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TOPIC: MENSTRUAL CYCLE

The menstrual cycle is divided as the following phases:

	Days
Menstruation	1-4
Follicular phase (proliferative phase)	5-13
Ovulation	14
Luteal phase (secretory phase)	15-28

	Follicular phase (proliferative phase)	Luteal phase (secretory phase)
Ovarian histology	A number of follicles develop. One follicle will become dominant around the mid-follicular phase	Corpus luteum
Endometrial histology	Proliferation of endometrium	Endometrium changes to secretory lining under influence of progesterone
Hormones	A rise in FSH results in the development of follicles which in turn secrete oestradiol When the egg has matured, it secretes enough estradiol to trigger the acute release of LH. This in turn leads to ovulation	Progesterone secreted by corpus luteum rises through the luteal phase. If fertilization does not occur, the corpus luteum will degenerate and progesterone levels fall Oestradiol levels also rise again during the luteal phase
Cervical mucus	Following menstruation, the mucus is thick and forms a plug across the external os Just prior to ovulation the mucus becomes clear, acellular, and low viscosity. It also becomes 'stretchy' - a quality termed spinnbarkeit	Under the influence of progesterone, it becomes thick, scant, and tacky
Basal body temperature	Falls prior to ovulation due to the influence of estradiol	Rises following ovulation in response to higher progesterone levels

TOPIC: PREMENSTRUAL SYNDROME (PMS)

BACKGROUND INFORMATION

- PMS is a condition characterized by **psychological, physical, and behavioural symptoms** occurring in the luteal phase of the normal menstrual cycle (the time between ovulation and onset of menstruation).
- A diagnosis of PMS is supported by the timing (rather than the types) of symptoms and the degree of impact on daily activity.
    - PMS symptoms start during the 2nd half of the cycle and are repetitive and cyclical. They also resolve shortly after the onset of menses.
    - Symptoms occurs only during ovulatory menstrual cycles; it doesn't occur prior to puberty, during pregnancy or post-menopause.
        - it resolves completely post-menopause and transiently during pregnancy
    - To differentiate PMS from physiological premenstrual symptoms, it must be demonstrated that symptoms cause significant impairment to the woman during the luteal phase of the menstrual cycle.
- **Premenstrual dysphoric disorder (PMDD)** is a severe form of PMS defined in the DSM-V as occurring when a woman suffers from at least 5 out of 11 distinct psychological premenstrual symptoms, one of which must include mood.

DIFFERENTIAL DIAGNOSIS

- Depression
- Anxiety and panic disorders
- Menopausal transition
- Thyroid disease (mainly hypothyroidism)
- Anaemia
- Dysmenorrhoea
- Irritable bowel syndrome
- Interstitial cystitis
- Endometriosis
- Chronic fatigue syndrome
- Fibromyalgia
- Systemic lupus erythematosus
- Migraine
- Substance abuse

RISK FACTORS

- Presence of ovulatory menstrual cycles
    - This is supported by the absence of PMS prior to puberty, during pregnancy, and after menopause
- Other possible risk factors include:
    - Family history of PMS
    - Mood disorders
    - Cigarette smoking
    - Alcohol intake
    - Sexual abuse and/or trauma.

- Weight gain
- Stress

DIAGNOSIS

- **HISTORY TAKING:**
  - **TYPES** of symptoms experienced, including:
    - **Psychological symptoms**, like mood swings, irritability, depressed mood, anxiety, loss of confidence, feeling out of control, poor concentration, change in libido, and food cravings.
    - **Physical symptoms**, such as breast tenderness, bloating, headaches, backache, weight gain, acne, and gastrointestinal disturbance.
    - **Behavioural symptoms**, such as reduced visio-spatial and cognitive ability, aggression, and increase in accidents.
  - **TIMING** of symptoms in relation to the menstrual cycle.
    - If appropriate, also ask about the presence of symptoms before, during, and after pregnancy.
  - **SEVERITY** of the symptoms and the degree of impact on daily activity.
    - ask about the effect of symptoms on the woman's work, school, family life, and interpersonal relationships.
  - **MEDICATION HISTORY**:
    - Any previous treatment tried and whether symptoms improved
    - Other current medications, including contraceptives.
  - **MEDICAL HISTORY:**
    - Underlying chronic illness (such as diabetes, depression, epilepsy, asthma, and migraine) and whether symptoms of the illness significantly worsen premenstrually.
    - Mood disorders.
    - Surgical procedures (such as a hysterectomy or endometrial ablation).
  - **PSYCHOSOCIAL HISTORY:** including smoking and alcohol use.
- **PHYSICAL EXAMINATION:**
  - as indicated by the woman's age and routine gynaecological and medical recommendations.
  - If the woman has pelvic pain or abdominal swelling, perform a full physical examination, including an abdominal and pelvic examination, to rule out any other underlying cause of their symptoms.
- **DAILY SYMPTOM DIARY:**
  - Ask the woman to record a daily symptom diary for two or three cycles, and review the woman with the diary.
  - The **Daily Record of Severity of Problems (DRSP)** is a validated questionnaire that is widely used for tracking symptoms related to PMS. Other questionnaires are available.
- **BLOOD TESTS:**
  - Consider doing a CBC/thyroid function test or other investigations to rule out potential differential diagnoses
- **DIAGNOSIS OF PMS SHOULD BE CONFIRMED IF:**
  - The diary shows a prominence of symptoms during the luteal phase of the menstrual cycle, which resolve with the onset of menses or soon after, followed by a symptom-free week.

- Symptoms are severe enough to affect daily functioning or interfere with the woman's work, school, performance, or interpersonal relationships.
- There is an absence of other conditions that could explain the symptoms (such as the differential diagnoses mentioned previously)

- If the completed symptom diary alone is inconclusive, refer the woman to secondary care.
  - **Gonadotrophin-releasing hormone (GnRH) agonists** are widely used by gynaecologists to establish a definitive diagnosis of PMS by inhibiting cyclical ovarian function. They are used for 3 months (a month for the agonist to generate a complete hormonal suppressive effect plus 2 months' worth of symptom diaries). Resolution of symptoms after ovarian suppression is considered diagnostic for PMS.

MANAGEMENT

- Management of PMS should be tailored to the severity and type of symptoms, the woman's treatment preferences, and any desire to become pregnant.
  - There are no formal criteria available for defining mild, moderate, or severe PMS; severity is usually based on clinical judgement after considering:
    - the woman's perception of symptom severity,
    - the impact of symptoms on the woman's quality of life,
    - the presence or absence of distress or impairment of socioeconomic function.
- For all women with PMS:
  - Offer lifestyle advice that includes:
    - Regular, frequent (2–3 hourly), small, balanced meals rich in complex carbohydrates.
    - Regular exercise.
    - Regular sleep.
    - Stress reduction.
    - Smoking cessation
    - Alcohol restriction
- If the predominant symptom is **PAIN** (ex: headache/ generalized aches and pains):
  - prescribe a simple analgesic, such as paracetamol or a nonsteroidal anti-inflammatory drug (NSAID).
- If the predominant symptom is **BREAST PAIN:**
  - Advice on wearing a well-fitted bra during the day, more supportive bra during exercise and/or a soft support bra at night.
  - Oral paracetamol and/or ibuprofen, or a topical nonsteroidal anti-inflammatory preparation, as required, are first-line options for pain relief.
  - A pain diary may be useful to assess the severity and timing of the pain, and its response to treatment.
  - If breast pain is severe enough to affect quality of life and sleep and does not respond to first-line treatment after 3 months, referral to a specialist should be considered for other treatment options.
- Advise that there is limited evidence to support the use of complementary treatments and dietary supplements
  - including reflexology, acupuncture, calcium and vitamin D, ginkgo biloba, evening primrose oil, vitamin B6, and magnesium.
- **MODERATE PMS:**
  - Consider prescribing a new-generation **combined oral contraceptive (COC)**, especially if the woman requires contraception (off-label use if used solely to treat PMS symptoms).

    - There is more evidence to support the use of Yasmin® (drospirenone 3 mg and ethinylestradiol 0.030 mg).
    - Current data suggest use of the COC continuously rather than cyclically.
  - **Cognitive behavioural therapy (CBT)** if the woman would benefit from psychological intervention.
- **SEVERE PMS:**
  - Consider prescribing a **selective serotonin reuptake inhibitor (SSRI, off-label use):**
    - To be taken continuously or just during the luteal phase (for example days 15–28 of the menstrual cycle, depending on its length).
    - Give an initial trial of 3 months' treatment; if there is benefit, continue treatment for 6 months to 1 year.
    - Suggested doses are fluoxetine 20 mg a day, sertraline 50 mg to 100 mg a day, citalopram 20 mg a day, escitalopram 20 mg a day, or paroxetine 20 mg a day.
    - If the woman wishes to become pregnant, advise that PMS symptoms will abate during pregnancy and SSRIs should therefore be discontinued prior to and during pregnancy.
      - Advise the woman on how to safely stop SSRI treatment: women taking luteal phase SSRIs can discontinue the treatment safely at any time, whereas women using a continuous regimen should taper the dose over a period of time.
  - **Cognitive behavioural therapy (CBT)** if the woman would benefit from psychological intervention.

FOLLOW UP

- The woman should be reviewed after 2 months (or earlier if indicated) to assess the effectiveness of the treatment.
  - If primary care management fails to control symptoms satisfactorily, referral to a clinic with a specific interest in PMS (or a general gynaecology clinic if this is not available) should be considered.

TOPIC: ABNORMAL UTERINE BLEEDING (AUB)

DEFINITION:

- AUB is a term which refers to uterine bleeding of abnormal quantity, duration, or schedule and is a common gynecologic concern in reproductive-age females.

Normal menstruation parameters:

Parameter	Normal	Abnormal
Frequency	≥24 and ≤38 days	Absent (no bleeding): amenorrhea
		Frequent (<24 days)
		Infrequent (>38 days)
Duration	≤8 days	Prolonged (>8 days)
Regularity	Regular: shortest to longest cycle variation: ≤7 to 9 days*	Irregular: shortest to longest cycle variation: ≥10 days
Flow volume (patient determined)	Patient considers normal	Patient considers light
		Patient considers heavy
Intermenstrual bleeding (bleeding between cyclically regular onset of menses)	None	Random
		Cyclic (predictable): ▪ Early cycle ▪ Mid cycle ▪ Late cycle
Unscheduled bleeding on progestin± estrogen gonadal steroids (contraceptive pills, rings, patches, IUDs, or injections)	Not applicable for patients not on gonadal steroid medication	Present
	None (for patients on gonadal steroid medication)	

IUDs: intrauterine contraceptive devices.

DIAGNOSIS

HISTORY

- Recent or current pregnancy
- First day of the last menstrual period and several previous menstrual periods
- For how many days does bleeding continue? How many days of full bleeding and how many days of light bleeding or brown staining does this include?
- Does bleeding occur between menstrual periods?
- If bleeding is irregular, how many bleeding episodes have there been in the past 6 to 12 months? What is the average time from the first day of one bleeding episode to the next?
- Is the bleeding associated with dysmenorrhea or dyspareunia (ie, suggestive of endometriosis and/or adenomyosis)?
- How heavy is the bleeding?
- Is the patient certain that the bleeding is from the vagina?
  - -Does the patient see the blood in the toilet only during or after either urination or defecation?
  - -Does the patient see the bleeding only when wiping with toilet tissue? If so, has the patient tried to separately dab the urethra, vagina, and anus with toilet tissue to check the source of the bleeding?
  - -If the patient uses a pad, on what part of the pad is blood visible? If the patient uses a tampon, is bleeding visible while a tampon is in the vagina?
- Were there precipitating factors, such as trauma, intercourse, or a procedure? Bleeding related to trauma or intercourse suggests a vaginal or cervical source of bleeding (eg, cervical dysplasia, cervicitis, vulvovaginal atrophy, cervical polyp), while bleeding after a procedure (eg, pregnancy termination, intrauterine device [IUD] placement) may suggest a uterine source of bleeding.
- Note: In general, if the bleeding occurs solely with urination or defecation and the pattern of bleeding or findings on physical examination are consistent with a urinary or gastrointestinal tract source, this should be the focus of further evaluation. Changes in bladder or bowel function may also suggest a mass effect from an enlarged fibroid uterus or a neoplasm
- **Sexual history** – A sexual history helps to determine whether the patient might be pregnant, as pregnancy is a common cause of uterine bleeding; however, a pregnancy test is generally also performed, Or risk of STI.
- **History of obstetric or gynecologic surgery**
  - •A prior cesarean birth can result in a cesarean scar defect (particularly if a patient has had multiple cesarean births)
  - •Past myomectomy suggests the possibility of recurrent or persistent uterine fibroids
  - •Past excisional cervical procedures performed for diagnosis or management of cervical intraepithelial neoplasia suggest the possibility of recurrent or persistent cervical neoplasia
- **Risk factors for endometrial cancer** – Endometrial hyperplasia and carcinoma often present with AUB (Increase age , diabetis mellitis , obesity , early menarche , late menopause , tamoxifen , Unopposed estrogen therapy)

Physical exam and investigations:

A complete pelvic examination should be performed, with a particular focus on:

- Potential sites of bleeding from the vulva, vagina, cervix, urethra, anus, or perineum

- size and contour of the uterus – An enlarged uterus may be due to pregnancy, uterine leiomyomas, adenomyosis, or uterine malignancy.
- Presence of an adnexal mass or tenderness

Other investigations include:

- **Pregnancy test** — Pregnancy should be excluded in all reproductive-age patients with AUB. Pregnancy testing should be performed even in patients with recent vaginal bleeding since this may represent bleeding during pregnancy rather than menses. It should also be performed in patients who report no sexual activity and in those who report use of contraception.
- **Pelvic ultrasound** – Pelvic ultrasound is the first-line imaging study in patients with AUB. Transvaginal examination should be performed.
  - While transvaginal ultrasound can provide useful information regarding structural causes of AUB (eg, fibroids, adenomyosis, polyps), measurement of endometrial thickness is not used as an alternative to endometrial sampling for the evaluation of endometrial neoplasia in reproductive-age patients as major variation in endometrial thickness occurs during the normal menstrual cycle.

Evaluation and differential diagnosis of abnormal uterine bleeding (AUB) in non-pregnant reproductive-age women:

Bleeding pattern	Other associated clinical features	Differential diagnosis		Evaluation
		Common etiologies	Less common etiologies	
Regular menses that are heavy or prolonged	Enlarged uterus on examination, discrete masses may be noted	Uterine leiomyoma		- Pelvic ultrasound - Saline infusion sonography or hysteroscopy (if intracavitary pathology is suspected)
	- Dysmenorrhea - Enlarged, boggy uterus on examination	Adenomyosis		Pelvic ultrasound
	- Family history of bleeding disorder - Symptoms of bleeding diathesis	Bleeding disorder		Testing for bleeding disorder

	- Anticoagulant therapy			
	Risk factors for uterine malignancy		Endometrial carcinoma or uterine sarcoma	Endometrial sampling
Regular menses with intermenstrual bleeding		Endometrial polyp		- Pelvic ultrasound - Saline infusion sonography or hysteroscopy (if available)
	Risk factors for uterine malignancy		Endometrial carcinoma or uterine sarcoma	Endometrial sampling
	Recent history of uterine or cervical procedure or childbirth, particularly if infection was present		Chronic endometritis	Endometrial sampling
Irregular bleeding, may be more or less frequent than normal menses and volume and duration may vary		Ovulatory dysfunction:		
	Hirsutism, acne, and/or obesity	PCOS		Total testosterone and/or other androgens (may not be increased in all women with PCOS)
	Galactorrhea	Hyperprolactinemia		Prolactin
	- Recent weight gain or loss - Heat or cold intolerance - Family history of thyroid dysfunction	Thyroid disease		Thyroid function tests
	Risk factors for uterine malignancy		Endometrial carcinoma	Endometrial sampling

			or uterine sarcoma	
Secondary amenorrhea	History of irregular bleeding	Ovulatory dysfunction		Refer to ovulatory dysfunction above
	Poor nutrition or intense exercise	Hypothalamic amenorrhea		- Follicle-stimulating hormone - Luteinizing hormone - Estradiol
	Hot flushes	Premature ovarian insufficiency		Follicle-stimulating hormone
	Recent history of uterine or cervical procedure or childbirth, particularly if infection was present (menses may present, but abnormally light or brief)		Cervical stenosis	On pelvic examination, instrument cannot be passed through internal cervical os
			Intrauterine adhesions (Asherman syndrome)	Hysteroscopy
Irregular or heavy bleeding in a patient using hormonal contraceptives or with an intrauterine device		Iatrogenic AUB		

Causes of abnormal genital tract bleeding in females:

Genital tract disorders
Uterus
Benign conditions:
Endometrial polyps
Endometrial hyperplasia
Adenomyosis
Leiomyomas (fibroids)
Arteriovenous malformation (also referred to as enhanced myometrial vascularity)
Cancer:
Endometrial adenocarcinoma
Sarcoma
Infection:
Pelvic inflammatory disease
Endometritis
Ovulatory dysfunction
Cervix
Benign conditions:
Cervical polyps
Ectropion
Endometriosis
Cancer:
Invasive carcinoma

Trauma
Sexual intercourse
Sexual abuse
Foreign bodies (including intrauterine device)
Pelvic trauma (eg, motor vehicle accident)
Straddle injuries
Cesarean scar defect (prior cesarean delivery)
Drugs
Contraception:
Hormonal contraceptives
Intrauterine devices
Postmenopausal hormone therapy
Anticoagulants
Tamoxifen
Corticosteroids
Chemotherapy
Phenytoin
Antipsychotic drugs

Metastatic (uterus, choriocarcinoma)	Antibiotics (eg, due to toxic epidermal necrolysis or Stevens-Johnson syndrome)
Infection:	**Systemic disease**
Cervicitis	Diseases involving the vulva:
Vulva	Crohn disease
Benign conditions:	Behçet syndrome
Skin tags	Pemphigoid
Sebaceous cysts	Pemphigus
Condylomata	Erosive lichen planus
Angiokeratoma	Lymphoma
Cancer	Bleeding disorders:
	von Willebrand disease
Vagina	Thrombocytopenia or platelet dysfunction
Benign conditions:	Acute leukemia
Gartner duct cysts	Some coagulation factor deficiencies
Polyps	Advanced liver disease
Adenosis (aberrant glandular tissue)	Thyroid disease
Cancer	Polycystic ovary syndrome
Vaginitis/infection:	Cushing syndrome
Bacterial vaginosis	Hormone-secreting adrenal and ovarian tumors
Sexually transmitted infections	
Atrophic vaginitis	Renal disease
Upper genital tract disease	Emotional or physical stress
Pelvic inflammatory disease	Smoking
Fallopian tube cancer	Excessive exercise

Ovarian cancer	**Diseases not affecting the genital tract**
Pregnancy complications	Urethritis
	Bladder cancer
	Urinary tract infection
	Colorectal cancer
	Inflammatory bowel disease
	Hemorrhoids
	Other
	Endometriosis

PALM-COEIN

Classification system for abnormal uterine bleeding in nongravid reproductive-age women

- Polyp
- Adenomyosis
- Leiomyoma (fibroids)
- Malignancy
- Coagulopathy
- Ovulatory dysfunction
- Endometrial
- Iatrogenic (surgery, ocp)
- Not yet classified

TOPIC: MENORRHAGIA

DEFINITION

- Menorrhagia has been defined, in research and clinical settings, as blood loss of more than 80 ml and/or a duration of more than 7 days however, this definition is no longer considered useful because measurement of menstrual blood loss is rarely undertaken in routine clinical practice and women's self-assessment of blood loss has been widely demonstrated to differ from measured blood loss.

COMPLICATIONS

- Negative effects on a woman's physical, social, emotional, and/or material quality of life.
- Other complications include:
  - Iron deficiency anaemia.
  - Complications of secondary care interventions, such as:
    - Endometrial ablation — vaginal discharge, increased period pain or cramping, infection, and perforation.
    - Hysterectomy — infection, thrombosis, urinary dysfunction, and damage to other abdominal organs (such as the urinary tract or bowel).
    - Uterine artery embolization — infection, persistent vaginal discharge, premature ovarian failure (particularly in women over 45 years old), haematoma, and haemorrhage.

DIAGNOSIS

Assessment:

- **History:**
  - Ask about the nature of the bleeding, including:
    - Onset (for example, peri-menarcheal, sudden, or gradual).
    - Frequency/pattern (cyclic or non-cyclic).
    - Duration (acute or chronic).
    - Temporal association (for example, postcoital or postpartum).
    - Related symptoms (such as persistent intermenstrual bleeding, pelvic pain, and/or pressure symptoms) — may suggest uterine cavity abnormality, histological abnormality, adenomyosis, or fibroids.
  - Ask about her menstrual history, including:
    - Age of menarche.
    - Gravidity and parity.
    - Normal menstrual cycle and any variation of this pattern (including regularity, duration, and number of pads/tampons per day).
    - Last menstrual period (the start date of the most recent menstrual period).
  - Ask about:
    - Medical history, including recent medical procedures and comorbidities (such as endometriosis or history of coagulation disorders [for example, von Willebrand disease]).
    - Drug history, including current medications and previous treatment for menorrhagia.

- Sexual history, including current contraceptive use (hormonal contraceptives may cause irregular bleeding), contraceptive plans, and future plans for a family (as this may impact the choice of treatment).
- Cervical screening history — to confirm she has attended as scheduled and results were normal.
- Signs and symptoms of anaemia, such as dyspnoea, fatigue, headache, and sore tongue.
- The impact of the bleeding on the woman's quality of life.

❖ **If the woman has a history of menorrhagia without other related symptoms**,
  - ➢ consider starting pharmacological treatment without carrying out a physical examination (unless the treatment chosen is the levonorgestrel-releasing intrauterine system [LNG-IUS]).

❖ **If the woman has a history of menorrhagia with other related symptoms,**
  - ➢ offer a physical examination (as appropriate) to exclude an underlying cause.
    - Perform an abdominal examination — to assess for large fibroids and other masses.
    - Perform a bimanual pelvic examination, including a speculum examination of the cervix (except in young girls who are not sexually active) — to exclude an underlying cause, such as ascites, fibroids, or gynaecological cancer.
    - Look for features of an underlying systemic disease, for example:
      - Hypothyroidism — goitre.
      - Coagulation disorders (for example, von Willebrand disease) — bruises or petechiae.
      - Polycystic ovary syndrome — acne, hirsutism.

❖ **Arrange investigations as appropriate to exclude an underlying cause.**
  - ➢ Arrange a full blood count in all women — to rule out iron deficiency anaemia (which is a strong indicator of excessive menstrual bleeding).
  - ➢ Arrange other investigations as suggested by history and clinical findings, for example:
    - A pregnancy test — in all women of reproductive age who have any deviation from their normal pattern of menstrual bleeding.
    - A vaginal or cervical swab — if an infection is suspected.
    - Thyroid function tests — if there are features of hypothyroidism.
    - Tests for coagulation disorders (for example, von Willebrand disease) — in women who have had heavy menstrual bleeding since menarche and a personal or a family history of a coagulation disorder.

❖ **DO NOT:**
  - ➢ Routinely carry out a serum ferritin test for women with menorrhagia.
  - ➢ Carry out female hormone testing for women with menorrhagia.
  - ➢ Measure blood loss to diagnose menorrhagia.

❖ **If the woman's history and/or examination suggests a low risk of fibroids, uterine cavity abnormality, histological abnormality, or adenomyosis,**
  - ➢ consider starting pharmacological treatment for menorrhagia without investigating the cause.

❖ **If the woman's history and/or examination suggests a high risk of fibroids, uterine cavity abnormality, histological abnormality, or adenomyosis,**
  - ➢ arrange further investigations to determine the cause of menorrhagia.

Investigating the cause of menorrhagia:

- **Suspected submucosal fibroids, polyps, or endometrial pathology**
  - Offer referral for outpatient hysteroscopy to women with menorrhagia if their history suggests submucosal fibroids, polyps, or endometrial pathology because they have:
    - Symptoms such as persistent intermenstrual bleeding or
    - Risk factors for endometrial pathology.
  - Explain what the procedure involves, and discuss the possible alternatives.
    - If the woman declines outpatient hysteroscopy, offer a referral for hysteroscopy under general or regional anaesthesia.
    - If the woman declines hysteroscopy under general or regional anaesthesia, consider referral for a pelvic ultrasound, explaining the limitations of this technique for detecting uterine cavity causes of menorrhagia.
- **Endometrial biopsy** should be considered at the time of hysteroscopy for women who are at high risk of endometrial pathology, such as:
    - Women with persistent intermenstrual or persistent irregular bleeding, and women with infrequent heavy bleeding who are obese or have polycystic ovary syndrome.
    - Women taking tamoxifen.
    - Women for whom treatment for menorrhagia has been unsuccessful.
- **Possible larger fibroids**
  - Offer a pelvic ultrasound to women with menorrhagia if any of the following apply:
    - Their uterus is palpable abdominally.
    - History or examination suggests a pelvic mass.
    - Examination is inconclusive or difficult, for example, in women who are obese.
- **Suspected adenomyosis**
  - Offer a transvaginal ultrasound (in preference to a transabdominal ultrasound or MRI [magnetic resonance imaging]) to women with menorrhagia who have significant dysmenorrhoea (period pain) or a bulky, tender uterus on examination that suggests adenomyosis.
    - If a transvaginal ultrasound is declined or unsuitable, consider a transabdominal ultrasound or MRI, explaining the limitations of these techniques.
    - Be aware that pain associated with menorrhagia may be caused by endometriosis rather than adenomyosis.

MANAGEMENT

When should I refer a woman with menorrhagia?

- Refer using a **suspected cancer pathway referral (for an appointment within 2 weeks)** if:
  - The woman has a pelvic mass associated with any other features of cancer (such as unexplained bleeding or weight loss).
  - The woman is aged 55 years and over with post-menopausal bleeding (unexplained vaginal bleeding more than 12 months after menstruation has stopped because of the menopause).
  - On examination, the appearance of their cervix is consistent with cervical cancer.
- **Refer urgently to gynaecology** if physical examination identifies ascites and/or a pelvic or abdominal mass (which is not obviously due to uterine fibroids).

- If the woman's history and/or examination suggests a high risk of fibroids, uterine cavity abnormality, histological abnormality, or adenomyosis, arrange further investigations to determine the cause of menorrhagia.

- **Refer for specialist management** (urgency depending on clinical judgement) if:
  - The woman has iron deficiency anaemia which has failed to respond to treatment, and other causes have been excluded.
  - Menorrhagia has not improved despite initial treatments.
- **Consider specialist referral** for women with fibroids of 3 cm or more in diameter.
- **If pharmacological treatment is needed** while the woman is awaiting treatment or referral appointment, offer tranexamic acid and/or a non-steroidal anti-inflammatory drug.

How should I manage women with menorrhagia in primary care?

If referral is not indicated:

- **Provide appropriate information and advice** on menorrhagia.
- **Discuss the natural variability and range of menstrual blood loss with the woman.**
  - For some women, reassurance may be all that is required, and treatment may not be needed.
  - If the woman feels she does not fall within the normal range, offer appropriate treatment taking into account:
    - The woman's priorities and preferences.
    - Any comorbidities.
    - The presence or absence of fibroids (including size, number, and location), polyps, endometrial pathology, or adenomyosis.
    - Associated symptoms, such as pressure and pain.
  - Discussions should cover the benefits and risks of the various options, suitable treatments if she is trying to conceive, and whether she wants to retain her fertility and/or her uterus.

PASSMEDICINE MANAGEMENT SUMMARY:

- Does not require contraception
  - either mefenamic acid 500 mg tds (particularly if there is dysmenorrhoea as well) or tranexamic acid 1 g tds. Both are started on the first day of the period
  - if no improvement then, try other drug whilst awaiting referral
- Requires contraception, options include
  - intrauterine system (Mirena) should be considered first-line
  - combined oral contraceptive pill
  - long-acting progestogens
- Norethisterone 5 mg tds should be stared at least 3 days before the expected period as a short-term option to rapidly stop heavy menstrual bleeding.

- If treatment is unsuccessful, the woman declines pharmacological treatment, or symptoms are severe, consider specialist referral for:
  - Additional investigations, taking into account any investigations the woman has already had.
  - Consideration of alternative treatments, such as pharmacological options not already tried or surgical treatment (second-generation endometrial ablation or hysterectomy). Hysteroscopic removal may be considered for women with submucosal fibroids.

- Secondary care treatment will depend on the size, location, and number of fibroids as well as the severity of symptoms. Treatment options include:
    - Non-hormonal treatment: NSAIDs (off-label use) or tranexamic acid.
    - Hormonal treatment: LNG-IUS (off-label use for some brands), CHC, or cyclical oral progestogens. Ulipristal acetate (Esmya®) may be considered for some premenopausal women in line with advice issued by the Medicines and Healthcare products Regulatory Agency (MHRA).
    - Uterine artery embolization.
    - Surgical treatment: myomectomy, hysterectomy, or second-generation endometrial ablation.

ADDITIONAL information on treatment options:

Surgical treatments

- Endometrial ablation
    - is a procedure that surgically destroys the lining of the uterus. Second-generation endometrial ablation techniques include:
        - Microwave endometrial ablation — a probe is inserted into the uterus through the vagina and cervix. This probe produces microwaves that destroy the lining of the womb.
        - Thermal balloon endometrial ablation — a deflated balloon is moved into the uterus through the vagina and cervix. Once in place, the balloon is filled with a heated liquid. The heat destroys the lining of the womb.
- Uterine artery embolization
    - is a procedure to remove uterine fibroids. Small particles are injected into the blood vessels that take blood to the uterus. This impairs the blood supply to fibroids, causing them to shrink.
- Myomectomy
    - is a procedure to remove uterine fibroids while preserving the uterus. Fibroids are removed via a number of routes: abdominal, vaginal, laparoscopic, and hysteroscopic. The choice of route is determined by the size and position of the fibroids.
- Hysterectomy
    - is the partial or total surgical removal of the uterus. It may also involve removal of the cervix, ovaries, and other surrounding structures.

TOPIC: AMENORRHOEA

DEFINITION

- Amenorrhoea is the absence or cessation of menstruation.
- It can be divided into two types: primary and secondary amenorrhoea.
  - **Primary amenorrhoea**
    - is the failure to establish menstruation by the time of the expected menarche.
    - Some experts define it as the failure to establish menstruation by 15 years of age in those with normal secondary sexual characteristics (such as breast development), or by 13 years of age in those with no secondary sexual characteristics.
  - **Secondary amenorrhoea**
    - is the cessation of menstruation in women with previous menses. Definitions vary as to how long and include:
      - The cessation of menses for 3 months in women with previously normal menstruation, or for 6 months in women with previous oligomenorrhoea
- **Oligomenorrhoea** is defined as menses occurring less frequently than every 35 days.

CAUSES

Primary amenorrhoea	Secondary amenorrhoea (after excluding pregnancy)
• gonadal dysgenesis (e.g. Turner's syndrome) - the most common causes • testicular feminisation • congenital malformations of the genital tract • functional hypothalamic amenorrhoea (e.g. secondary to anorexia) • congenital adrenal hyperplasia • imperforate hymen	• hypothalamic amenorrhoea (e.g. secondary stress, excessive exercise) • polycystic ovarian syndrome (PCOS) • hyperprolactinaemia • premature ovarian failure • thyrotoxicosis* • Sheehan's syndrome • Asherman's syndrome (intrauterine adhesions)

PRIMARY AMENORRHOEA

- **Causes of primary amenorrhoea in those with normal secondary sexual characteristics (such as breast development) include:**
  - Physiological causes:
    - Constitutional delay — no anatomical abnormality, and maturation usually occurs spontaneously by 18 years of age. Secondary sexual characteristics area also likely to be delayed. The condition is frequently familial.
    - Pregnancy.
  - Genito-urinary malformations, such as:
    - Imperforate hymen.
    - Transverse septum.

        - Absent vagina or uterus.
    - Endocrine disorders, such as:
        - Hypothyroidism.
        - Hyperthyroidism.
        - Hyperprolactinaemia (for example, caused by prolactinoma or drugs).
        - Cushing's syndrome.
        - Polycystic ovary syndrome (a rare cause of primary amenorrhoea).
    - Androgen insensitivity syndrome (rare, previously known as 'testicular feminization').
- **Causes of primary amenorrhoea in those with no secondary sexual characteristics (such as breast development) include:**
    - Primary ovarian insufficiency (POI) due to:
        - Chromosomal irregularities (for example Turner's syndrome [46XO] and gonadal agenesis [46XX or 46XY]).
        - Other causes, such as chemotherapy, pelvic radiation, and autoimmune disease.
    - Hypothalamic dysfunction due to:
        - Stress, excessive exercise, and/or weight loss (functional hypothalamic amenorrhea). The female athlete triad is a medical condition observed in physically active females (such as distance runners, gymnasts, and dancers) involving low energy availability (with or without eating disorders), menstrual dysfunction, and low bone density.
        - Chronic systemic illness (such as uncontrolled diabetes, severe renal and cardiac disorders, coeliac disease, cancer, and infections [for example tuberculosis]).
        - Other causes of dysfunction of the hypothalamic-pituitary axis, including hypothalamic or pituitary tumours, cranial irradiation, infection or head injury; Kallman's syndrome (congenital gonadotrophin deficiency characterized by anosmia and other cranial anomalies); empty sella syndrome; Laurence–Moon–Biedl syndrome; and Prader–Willi syndrome.
    - Causes of ambiguous genitalia, such as:
        - 5-alpha-reductase deficiency.
        - Androgen-secreting tumours.
        - Congenital adrenal hyperplasia.
- Most of the causes of secondary amenorrhoea can also cause primary amenorrhoea, if they occur before the menarche.

SECONDARY AMENORROEA

- **Causes of secondary amenorrhoea in those with no features of androgen excess include:**
    - Physiological causes:
        - Pregnancy.
        - Lactation.
        - Menopause.
    - Premature ovarian insufficiency, for example, due to:
        - Chemotherapy.
        - Radiotherapy.
        - Autoimmune disease.
    - Hypothalamic dysfunction
        - due to stress, excessive exercise, and/or weight loss (functional hypothalamic amenorrhea). The female athlete triad is a medical condition observed in

physically active females (such as distance runners, gymnasts, and dancers) involving low energy availability (with or without eating disorders), menstrual dysfunction, and low bone density.

- Other hypothalamic causes, including:
  - Chronic systemic illness (such as severe cardiac, renal, or liver disease; inflammatory bowel disease; coeliac disease; AIDS; or cancer).
  - Cranial irradiation, infection or head injury.
  - Central nervous system tumours (such as craniopharyngiomas or metastases).
- Pituitary causes, including:
  - Prolactinoma and other hormone-secreting pituitary tumours.
  - Head injury and cranial irradiation.
  - Hypopituitarism (for example, after traumatic brain injury).
  - Sheehan's syndrome (pituitary infarction after major obstetric haemorrhage).
  - Sarcoidosis.
  - Tuberculosis.
- Uterine causes, including:
  - Cervical stenosis.
  - Asherman's syndrome (intrauterine adhesions).
- Thyroid disease (hypothyroidism or hyperthyroidism).
- Iatrogenic causes, including:
  - Contraceptives — extended-cycle combined oral contraceptives, injectable progesterone, implantable etonogestrel (Nexplanon®), and levonorgestrel intrauterine system (Mirena®) may cause amenorrhea
  - Drugs (such as antipsychotics, which can cause increased prolactin levels) and illicit drug use (in particular cocaine and opiates, which can cause hypogonadism).
  - Surgery (hysterectomy, endometrial ablation, and ovarian surgery).

- Causes of secondary amenorrhoea in those with features of androgen excess (such as hirsutism, acne and virilization) include:

  Polycystic ovary syndrome.

  Cushing's syndrome.

  Late-onset congenital adrenal hyperplasia.

  Androgen-secreting tumours of the ovary or adrenal gland (rare).

COMPLICATIONS

- Osteoporosis
- Infertility
- Cardiovascular disease (CVD)
- Psychological distress
  - often causes considerable anxiety, altered self-image, and loss of self-esteem. Many women have concerns about loss of fertility, loss of femininity, or unwanted pregnancy. The diagnosis of Turner's syndrome, androgen insensitivity syndrome, or developmental anomaly can be traumatic for both girls and their parents.

DIAGNOSIS

PRIMARY AMENORRHOEA

- **Take a history:**
  - Sexual history and contraception (to exclude pregnancy or a contraceptive cause of amenorrhoea).
  - Cyclical lower abdominal pain (suggesting haematocolpos, caused by a genital tract malformation).
  - Stress, depression, weight loss, disturbance of perception of weight or shape, level of exercise, and chronic systemic illness (suggesting hypothalamic dysfunction).
  - Headache, visual disturbance, or galactorrhoea (suggesting prolactinoma).
  - Age at menarche of mother and sisters (family history of late menarche suggests constitutional delay of puberty).
  - Family history of genetic anomalies (for example, androgen insensitivity [46XY female]).
- **Examine the patient**
  - Measure height and body weight, and calculate body mass index (BMI) for weight-related causes of amenorrhoea (a high BMI may suggest polycystic ovary syndrome [PCOS]; a low BMI may suggest an eating disorder or the female athlete triad).
  - Examine for:
    - Features of Turner's syndrome (short stature, web neck, shield chest with widely spaced nipples, wide carrying angle, and scoliosis).
    - Features of Cushing's syndrome (striae, buffalo hump, significant central obesity, easy bruising, hypertension, and proximal muscle weakness).
    - Hirsutism and acne (suggesting PCOS, especially in those with a high BMI).
    - Features of thyroid and other endocrine disease.
    - If appropriate, examine for:
      - ✓ Clitoromegaly if hirsutism is present (indicating virilization due to possible androgen-secreting tumour).
      - ✓ Galactorrhoea (suggesting raised prolactin).
      - ✓ Haematocolpos if there is a history of cyclical lower abdominal pain (separation of the labia reveals a bulging, blue-coloured membrane, and a pelvic mass may be palpable).
      - ✓ Features of decreased endogenous oestrogen (reddened or thin vaginal mucosa; breast development is a good marker for ovarian oestrogen production).
      - ✓ Features of androgen insensitivity (absence of axillary and pubic hair with normal breast development; testes may be palpable in the inguinal canal or labia).
  - A pelvic examination is inappropriate in young girls who are not sexually active; an ultrasound can be done to assess pelvic anatomy. In older women presenting with primary amenorrhoea, it may be appropriate to do a pelvic examination, for example, to look for an absent uterus.
- Although investigations for primary amenorrhoea are usually done by a specialist, consider the following preliminary investigations to aid diagnosis and/or guide referral:
  - Pelvic ultrasound (if the presence of a vagina and uterus cannot be confirmed by physical examination, or in place of a pelvic examination in young girls who are not sexually active).
  - Serum prolactin.

        - Do not examine the breasts before taking blood for prolactin levels, as this may then be falsely elevated. If the breasts have been examined, delay the blood test for at least 48 hours.
    - Thyroid-stimulating hormone.
    - Follicle-stimulating hormone and luteinizing hormone.
    - Total testosterone (if there are features of androgen excess).
    - Other investigations as suggested by history and clinical findings.

Interpretation of investigation findings:

- Referral to a specialist is likely to be needed for people with primary amenorrhoea regardless of the result of investigations, but the following information may guide referral, speed up diagnosis, and aid explanation to the person about the possible cause of amenorrhoea (pending specialist opinion).
- **Pelvic ultrasound**
    - Uterus present.
        - In girls with normal secondary sexual characteristics, causes include outflow obstruction (for example, imperforate hymen or transverse vaginal septum) and polycystic ovary syndrome (PCOS).
        - In girls with no secondary sexual characteristics, causes include Turner's syndrome (46XO; 'streak' ovaries only) and gonadal agenesis (46XX or 46XY).
    - Absent or abnormal uterus could be caused by androgen insensitivity syndrome.
- **Prolactin levels**
    - Prolactin levels greater than 1000 mIU/L usually warrant further investigation by an endocrinologist (usually magnetic resonance imaging [MRI] of the pituitary fossa is required). Causes include pituitary adenoma, empty sella syndrome, hypothyroidism, and drugs (including antipsychotics [particularly risperidone], antidepressants [particularly selective serotonin reuptake inhibitors], and antiemetics [such as metoclopramide or domperidone]).
    - If prolactin levels are mildly elevated (for example 500–1000 mIU/L, normal range less than 500 mIU/L), it should be repeated before referral.
    - The most likely causes of mild hyperprolactinaemia in primary care are stress and drugs (such as antipsychotics).
    - Other causes include:
        - Pregnancy and breastfeeding.
        - Recent breast examination (if the breasts have been examined, the blood test should be delayed for at least 48 hours).
        - Needle phobia or traumatic venesection (can double or occasionally quadruple basal prolactin concentrations).
        - PCOS (in 10–20% of people; rarely above 1000 mIU/L).
        - Renal impairment (occasionally associated with mild hyperprolactinaemia; typically, less than 2000 mIU/L).
        - Hypothyroidism (uncommon, often still within the reference range and only rarely above 1200 mIU/L).
    - Persistent moderate elevations may be due to pituitary adenomas.
- **Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)**
    - If secondary sexual characteristics are absent, karyotyping in secondary care may be necessary.
    - Short stature and high FSH and LH levels suggest Turner's syndrome.

- Short stature and low FSH and LH levels suggest an intracranial lesion for example, hydrocephalus.
- Normal height and high FSH and LH levels suggest ovarian failure (normal karyotype) or 46XY (abnormal karyotype).
- Normal height and low FSH and LH levels suggest constitutional delay, weight loss, anorexia nervosa, or excessive exercise.

- **Total testosterone levels**
  - High levels of total testosterone (5.0 nanomol/L or greater) warrant investigation to exclude androgen insensitivity (46XY genotype, female phenotype), late-onset congenital adrenal hyperplasia, Cushing's syndrome, or an androgen-secreting tumour.
  - A moderately increased testosterone level (2.5–5.0 nanomol/L) may be seen in PCOS.

SECONDARY AMENORRHOEA

- **Take a history.**
  - To exclude physiological causes, including pregnancy, lactation, and menopause (in women 40 years of age or older). Ask about the following:
  - Contraceptive use (extended-cycle combined oral contraceptives, injectable progesterone, implantable etonogestrel [Nexplanon®], and levonorgestrel intrauterine system [Mirena®] may cause amenorrhea).
  - Hot flushes and vaginal dryness (suggesting premature ovarian insufficiency [POI] or natural menopause).
  - Headaches, visual disturbances, or galactorrhoea (suggesting a pituitary tumour).
  - Acne, hirsutism, and weight gain (suggesting polycystic ovary syndrome [PCOS]).
  - Stress, depression, weight loss, disturbance of perception of weight or shape, level of exercise, and chronic systemic illness (suggesting hypothalamic dysfunction).
  - Symptoms of thyroid and other endocrine disease.
  - A history of obstetric or surgical procedures (such as endometrial curettage) that may have resulted in intrauterine adhesions.
  - A history of chemotherapy and pelvic radiotherapy (which can cause POI); and cranial radiotherapy, head injury, or major obstetric haemorrhage (which can cause hypopituitarism).
  - Drugs (such as antipsychotics, which can cause increased prolactin levels) and illicit drug use (in particular cocaine and opiates, which can cause hypogonadism).
  - A family history of cessation of menses before 40 years of age (suggesting POI).
- **Examine the woman.**
  - Measure height and body weight, and calculate body mass index (BMI) for weight-related causes of amenorrhoea (a high BMI may suggest PCOS; a low BMI may suggest an eating disorder or the female athlete triad).
  - Examine for features of:
    - Cushing's syndrome (striae, buffalo hump, significant central obesity, easy bruising, hypertension, and proximal muscle weakness).
    - Thyroid disease. See the CKS topics on Hyperthyroidism and Hypothyroidism for more information.
    - Excess androgens (hirsutism, acne) or virilization (hirsutism, acne, deep voice, temporal balding, increase in muscle bulk, breast atrophy, and clitoromegaly).
    - Decreased endogenous estrogen (reddened or thin vaginal mucosa).
  - If appropriate, examine for galactorrhoea (suggesting raised prolactin levels).
  - Assess visual fields (if a pituitary tumour is suspected).

- **Consider carrying out the following preliminary investigations in primary care to aid diagnosis and/or guide referral:**
  - Full blood count, urea & electrolytes, coeliac screen, thyroid function tests
  - Follicle-stimulating hormone and luteinizing hormone.
  - Prolactin level.
  - Total testosterone/androgens, gonadotropins and oestradiol
  - Exclude pregnancy with urinary or serum bHCG
  - Ultrasound scan (if PCOS is suspected).
  - Other investigations as suggested by clinical findings.

Interpretation of investigation findings:

- **Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)**
  - High FSH and LH levels on two occasions suggest premature ovarian insufficiency (in women younger than 40 years of age).
  - Normal or low FSH levels and normal or low LH levels suggest hypothalamic causes (weight loss, excessive exercise, stress, or rarely, a hypothalamic or pituitary tumour).
  - Normal FSH levels and normal or moderately increased LH levels may be found in polycystic ovary syndrome (PCOS).
- **Ultrasound scan**
  - Polycystic ovaries on ultrasound are defined as the presence of 12 or more follicles (measuring 2–9 mm in diameter) in one or both ovaries and/or increased ovarian volume (greater than 10 cm3).

The table below shows laboratory findings in common causes of secondary amenorrhoea.

STATE	FSH	LH	PROLACTIN	TESTOSTERONE
Hyperprolactinaemia	Normal/Low	Normal/Low	High	Normal
PCOS	Normal	Normal/slightly increased in 40%	Normal/slightly increased in 5–30%	Normal/moderately increased Free Androgen index increased
Ovarian Failure	High	High	Normal	Normal
Hypothalamic (e.g. weight loss, excessive exercise, or stress)	Low/Normal	Low/Normal	Normal	Normal

MANAGEMENT

PRIMARY AMENORRHOEA

- **Refer those with primary amenorrhoea to secondary care for specialist investigation and, where necessary, management, of the underlying cause.**
- **Refer at an earlier age if the girl or her parents are concerned or if an abnormality is suspected, for example in those with:**
  - Growth retardation.
  - Symptoms and signs of androgen excess (such as hirsutism) or thyroid disease.
  - Galactorrhoea.

- Suspected genital tract malformation, intracranial tumour (for example prolactinoma), chromosomal anomaly (for example Turner's syndrome or androgen insensitivity), or anorexia nervosa.
- Puberty lasting 5 years without menarche (for example presenting at 15 years of age when pubic hair and breast development started at 10 years of age).

- **Referral to a gynaecologist** (preferably with a special interest in adolescent gynaecology) is appropriate for most people.

- **Referral to an endocrinologist** is recommended for those with hyperprolactinaemia, thyroid disease, or features of androgen excess (such as hirsutism, acne, and virilization).
- **Manage amenorrhoea caused by weight loss, excessive exercise, stress, or chronic illness after an endocrinologist has assessed and excluded other hypothalamic or pituitary causes (such as a tumour).**
  - For weight-related amenorrhoea, encourage weight gain and refer to a dietician if necessary. If an eating disorder is suspected, consider referral to a psychiatrist.
  - For exercise-related amenorrhoea, advise reducing exercise, increasing calorie intake, and weight gain. Consider referral to, or liaison with, a sports physician if available.
  - For stress-related amenorrhoea, consider measures to manage stress and improve coping strategies, such as cognitive behavioural therapy.
- **If amenorrhoea persists for more than 12 months, consider whether osteoporosis prophylaxis is required.**

Managing osteoporosis risk

- **For girls and women with hypothalamic amenorrhoea, or hyperprolactinaemia** (women with amenorrhoea associated with low oestrogen levels who are at increased risk of developing osteoporosis):
  - Treat the underlying cause, if possible.
  - Assess their fragility fracture risk.
  - Advise maintaining a healthy lifestyle to optimize bone health. This involves doing weight-bearing exercises, avoidance of smoking, eating a balanced diet, and maintenance of normal body weight.
  - Correct vitamin D deficiency, and ensure an adequate calcium intake.
  - Consider offering hormone replacement therapy (HRT) (off-label use) if amenorrhoea persists for more than 12 months.
    - Offer cyclical combined HRT (at the doses used for menopause).
    - Review treatment at least annually. For women with amenorrhoea due to reversible causes (such as weight loss or excessive exercise), stop oestrogen replacement periodically (for example after 12 months of treatment, to see whether menses resumes off treatment.
  - If other pharmacological treatments, including bisphosphonates, are being considered, seek specialist advice.

SECONDARY AMENORRHOEA

- **Manage the following causes for secondary amenorrhoea in primary care:**
  - Polycystic ovary syndrome (PCOS), when appropriate.
  - Hypothyroidism (menses may take several months to resume with treatment).
  - Menopause (women 40 years of age or older).
  - Pregnancy.

- **<u>Refer all other people with secondary amenorrhoea for specialist investigation and, where appropriate, management of the cause:</u>**
- **Refer to a gynaecologist** if she has any of the following:
  - Persistently elevated follicle-stimulating hormone (FSH) and luteinizing hormone (LH) levels, which suggests premature ovarian insufficiency in women younger than 40 years of age.
  - Recent history of uterine or cervical surgery (such as endometrial curettage, Caesarean section, or myomectomy) or severe pelvic infection (endometritis), which suggests Asherman's syndrome or cervical stenosis.
  - Infertility.
  - Suspected PCOS, if diagnosis and management are not feasible in primary care.
- **Refer to an endocrinologist** if she has any of the following:
  - Hyperprolactinaemia: serum prolactin level greater than 1000 mIU/L, or 500–1000 mIU/L on two occasions. This includes those taking drugs that are known to increase prolactin levels.
  - Low FSH and LH levels (to exclude hypopituitarism or a pituitary tumour, although stress, excessive exercise, or weight loss are more likely causes).
  - An increased testosterone level that is not explained by PCOS (suggesting an androgen-secreting tumour, late-onset congenital adrenal hyperplasia, or Cushing's syndrome).
  - Other features of Cushing's syndrome or late-onset congenital adrenal hyperplasia (besides an increased testosterone level).
- **Manage amenorrhoea caused by weight loss, excessive exercise, stress, or chronic illness after an endocrinologist has assessed and excluded other hypothalamic or pituitary causes (such as a tumour).**
  - For weight-related amenorrhoea, encourage weight gain and refer to a dietician if necessary. If an eating disorder is suspected, consider referral to a psychiatrist.
  - For exercise-related amenorrhoea, advise reducing exercise, increasing calorie intake, and weight gain. Consider referral to, or liaison with, a sports physician if available.
  - For stress-related amenorrhoea, consider measures to manage stress and improve coping strategies, such as cognitive behavioural therapy.
- Offer contraceptive advice to women who do not wish to become pregnant, as a small number of women with secondary amenorrhoea will become pregnant.

TOPIC: DYSMENORRHOEA

DEFINITION

- Dysmenorrhoea is painful cramping, usually in the lower abdomen, which occurs shortly before or during menstruation, or both.
- There are two types of dysmenorrhoea: primary and secondary.
  - **Primary dysmenorrhoea** occurs in the absence of any identifiable underlying pelvic pathology. It usually begins 6–12 months after the menarche, once cycles are regular, and is thought to be caused by the production of uterine prostaglandins during menstruation.
    - Features
      - ✓ pain typically starts just before or within a few hours of the period starting
      - ✓ suprapubic cramping pains which may radiate to the back or down the thigh
    - Pathophysiology:
      - ✓ Before menstruation begins, progesterone levels drop, causing endometrial cells to release prostaglandins. Prostaglandins stimulate uterine myometrial contractions, leading to decreased blood flow, uterine hypoxia, and pain.
      - ✓ Further effects of prostaglandins on smooth muscles may manifest as gastrointestinal symptoms (such as vomiting, nausea, and diarrhoea), which frequently coexist with primary dysmenorrhoea.
      - ✓ Leukotrienes and vasopressin may also play a role in the aetiology of primary dysmenorrhoea.
  - **Secondary dysmenorrhoea** often starts after several years of painless periods and is caused by an underlying pelvic pathology
    - **Causes:** endometriosis, adenomyosis, fibroids (myomas), endometrial polyps, pelvic inflammatory disease, ectopic pregnancy, ovarian/cervical cancer, intrauterine device (IUD) insertion.
    - In contrast to primary dysmenorrhoea the pain usually starts 3-4 days before the onset of the period.

RISK FACTORS

- Primary dysmenorrhoea (or more severe episodes)
  - Risk factors: earlier menarche, heavy menstrual flow, nulliparity, family history of dysmenorrhoea, and emotional stress.
  - Primary dysmenorrhoea has also been associated with a BMI of less than 20 kg/m2, cigarette smoking, and a history of sexual abuse.
  - Primary dysmenorrhoea tends to improve with increased age, parity, and use of oral contraceptives.
  - There is a lack of data on an association between primary dysmenorrhoea and nutritional deficiency. However, treatment with iron supplements can cure or improve dysmenorrhoea in women with iron deficiency.
- Risk factors for secondary dysmenorrhoea will depend on the underlying cause.

COMPLICATIONS

- Dysmenorrhea adversely affects the woman's quality of life and well-being and can lead to restriction of daily activities and absence from school or work.

- Women with dysmenorrhoea have poorer mood and sleep quality during menstruation compared with their pain-free follicular phase and compared with the menstruation phase of pain-free control women.
- Depression and anxiety are more common in women with dysmenorrhoea.

DIAGNOSIS

- Secondary causes of dysmenorrhoea must be excluded before considering a diagnosis of primary dysmenorrhoea.
- To assess for a secondary cause of dysmenorrhoea:
  - **Take a history**. Ask about:
    - When the symptoms started in relation to the menarche.
    - Characteristics of the pain, including type of pain, timing and duration, severity, and exacerbating and alleviating factors.
    - Any associated symptoms, including other gynaecological symptoms and non-gynaecological symptoms.
    - Menstrual history, including length of menstrual cycle, regularity, and duration, and the volume of menstrual flow.Risk factors for primary dysmenorrhoea, such as family history of dysmenorrhoea.
    - Medical history, past and present. Several conditions (for example, irritable bowel syndrome and lactose intolerance) can mimic dysmenorrhoea.
    - Obstetric history, including plans for pregnancy.
    - Drug history, including treatment(s), tried and the effect.
  - **Examine the patient**
    - Perform an abdominal examination in all women — to assess for large fibroids and other masses.
    - Perform a pelvic examination (including a speculum examination of the cervix), except in young women who are not sexually active if the symptoms are suggestive of primary dysmenorrhoea.
  - **Consider arranging the following investigations:**
    - An ultrasound scan — to rule out fibroids, adnexal pathology, and endometriosis or to assess an intrauterine contraceptive device.
    - High vaginal and endocervical swabs — if the woman is at risk of a sexually transmitted infection, especially if pain is associated with vaginal discharge and abnormal vaginal bleeding.
    - Pregnancy test — to exclude an ectopic pregnancy.
- **Primary dysmenorrhoea is the most likely diagnosis when:**
  - Menstrual pain starts 6–12 months after the menarche, once cycles are regular.
  - Pain, usually cramping in nature, occurs in the lower abdomen but may radiate to the back and inner thigh.
  - Pain starts shortly before the onset of menstruation and lasts for up to 72 hours, improving as the menses progresses.
  - Non-gynaecological symptoms, such as nausea, vomiting, diarrhoea, fatigue, irritability, dizziness, bloating, headache, lower back pain, and emotional symptoms, are present.
  - Other gynaecological symptoms are not present.
  - Pelvic examination is normal.
- **Secondary dysmenorrhoea is the most likely diagnosis when:**
  - Pain starts after several years of painless periods.

- Pain is not consistently related to menstruation alone and may persist after menstruation finishes or may be present throughout the menstrual cycle but is exacerbated by menstruation.
- Other symptoms are present, including:
  - Other gynaecological symptoms, such as dyspareunia, vaginal discharge, menorrhagia, intermenstrual bleeding, and postcoital bleeding.
  - Non-gynaecological symptoms, such as rectal pain and bleeding (which may be associated with endometriosis).
  - Pelvic examination is abnormal, although the absence of abnormal findings does not exclude secondary dysmenorrhoea.

MANAGEMENT

PRIMARY DYSMENORRHOEA

- Offer a **nonsteroidal anti-inflammatory drug (NSAID)** unless contraindicated.
- Offer **paracetamol** if NSAIDs are contraindicated or not tolerated, or in addition to an NSAID if the response is insufficient.
- **If the woman does not wish to conceive**, consider prescribing a 3–6 months trial of a hormonal contraceptive as an alternative first-line treatment.
  - Monophasic combined oral contraceptive (COC) preparations containing 30–35 micrograms of ethinylestradiol and norethisterone, norgestimate, or levonorgestrel are usually the first choice.
  - Oral (desogestrel 75 micrograms), parenteral (Depo-Provera® or Sayana Press®, and Nexplanon®), and intrauterine progestogen-only (Mirena®) contraceptives may also be considered after a full discussion of the advantages and disadvantages.
- **If the response to individual treatments is insufficient,** a combination of an NSAID (or paracetamol) and hormonal contraception may be considered.
- **Consider recommending the following non-drug measures** (in addition to drug treatments) to help reduce pain:
  - Local application of heat (for example, a hot water bottle or heat patch).
  - Transcutaneous electrical nerve stimulation (TENS) — set to a high frequency.
- **If symptoms are severe and do not respond to initial treatment within 3–6 months, or if there is doubt about the diagnosis, refer to a gynaecologist.**

SECONDARY DYSMENORRHOEA

- Management of secondary dysmenorrhoea will depend on the underlying cause.
- **Suspect a serious secondary cause of dysmenorrhoea and refer urgently if any of the following 'RED FLAGS' are present:**
  - A positive pregnancy test with pelvic pain and tenderness and vaginal bleeding.
  - Ascites and/or a pelvic or abdominal mass (where it is clear that this is not due to uterine fibroids).
  - An abnormal cervix on examination — this may indicate the presence of cervical cancer.
  - Persistent intermenstrual or postcoital bleeding without associated features of pelvic inflammatory disease (PID), such as pelvic pain, deep dyspareunia, and abnormal vaginal or cervical discharge — this may indicate the presence of cervical cancer, endometrial cancer, or endometrial polyps.
  - An ultrasound suggestive of cancer.

TOPIC: DYSPAREUNIA

DEFINITION

- **Female sexual pain (FSP) or Dyspareunia**
  - Is pain during or after sexual intercourse.
- It has a significant negative impact on a woman's health, self-esteem, relationships, quality of life, and work productivity.
- It is unclear if sexual pain is a sexual disorder, pain disorder, or both.
- Etiologies range from simple anatomic problems to complex biopsychosocial issues. In addition, a woman can have more than one etiology of her pain.
- It is useful to classify dyspareunia according to where the pain is felt, although there may be a degree of overlap.

Superficial dyspareunia	Deep dyspareunia
Lack of sexual arousal Vaginal atrophy (e.g. post-menopausal) Vaginitis secondary to infection e.g. *Candida, Trichomonas* Painful episiotomy scar Vaginismus	Pelvic inflammatory disease Endometriosis Cervicitis secondary to infection e.g. *Chlamydia* Prolapsed ovaries in the pouch of Douglas Adenomyosis Fixed retroverted uterus

ETIOLOGY

- Dyspareunia can be grouped into identifiable or idiopathic categories; patients can have either or both types.
- Women with FSP and no identifiable cause appear to have a type of chronic pain syndrome.
- The most common identifiable causes of premenopausal FSP are provoked vulvar pain syndrome, myofascial pelvic pain syndrome, vulvar dermatologic conditions, endometriosis, and interstitial cystitis/painful bladder syndrome.
- Genitourinary syndrome of menopause is the most common cause of postmenopausal FSP.

FEATURES

- Pain with genital sexual contact is the hallmark symptom of women with FSP.
  - Common sites of pain include the vulva, vaginal vestibule, introitus, vagina, pelvis, or pelvic floor.
- Women with FSP can also report the following pain characteristics:
- Location:
  - The pain can be isolated to one location or appreciated at multiple sites.
- Onset:
  - Sexual pain may be lifelong (i.e., pain present since the individual became sexually active) or acquired (i.e., the pain began after a relatively normal period of sexual function).
- Frequency:
  - Pain may be constant (with each sexual event and activity) or situational (with some experiences or partners but not others).

HISTORY

Pain time, onset and association:

- Does the pain occur with foreplay but before genital touch? If so, the pain may be related to generalized vulvodynia or anxiety with sexual contact.
- Does the pain occur at the very beginning of penetration? If so, the pain may be related to provoked vestibulodynia or an active vaginal infection.
- Does the pain occur with deep pelvic thrusting? If so, the pain may be related to pelvic floor muscle dysfunction, pelvic inflammatory disease, painful bladder syndrome/interstitial cystitis, endometriosis, or inflammatory bowel syndromes.
- Does the pain occur after sex? If so, it may be related to vestibulodynia, vulvar fissures, or other dermatoses.
- Does the pain happen at other nonsexual times? If so, the pain may be related to generalized vulvodynia, pelvic floor dysfunction, or pudendal neuralgia.

Location of the pain?

- Is the pain on the vulva (labia majora, labia minora, interlabial sulci, clitoral or periclitoral, or in the vestibule)? Pain in these locations may reflect lichen dermatoses, contact dermatitis, or generalized vulvodynia.
- Is the pain perineal or perianal? Pain in these locations may be related to nerve entrapment, fissures, deep pelvic floor muscle hypertonus, or dermatoses in addition to vulvodynia.
- Is the pain at the entry (introital) or other parts of the vagina, clitoris, or vulva? Pain can be caused by vulvar pain syndromes, fissures, dermatoses, superficial pelvic floor muscle hypertonus, or infection.
- Is the pain in the pelvis (i.e., deep dyspareunia)?

Was the onset of pain associated with a specific event?

- Childbirth (including episiotomy, lacerations, lactation)
- Pelvic or vaginal surgery (particularly placement of synthetic vaginal mesh)
- Back or hip injury
- Physical trauma (e.g., motor vehicle accident)
- Reconstructive surgery for incontinence or pelvic organ prolapses
- Radiation therapy and chemotherapy treatment
- Undesired sexual event or contact

What is the character and pattern of the pain?

- Is it a sharp or burning or tingling sensation? For how long does each episode last? How severe is it? How the pain changes during the menstrual cycle.
- Cyclic symptoms of deep dyspareunia may indicate the presence of endometriosis.
- Does the pain occur or worsen when you are sitting or exercising?
- This may reflect pudendal nerve dysfunction or pudendal neuralgia.
- What have you tried to relieve the pain?
- Inquire about evaluations and treatments, including self-initiated (eg, vaginal lubricant, position changes), that the patient has tried. Ask what has helped and what has not.

Cultural factors:

By asking open-ended questions, such as "Tell me a little about your family, cultural, and religious background so that I can best take care of your sexual concerns,"
Inquire about cultural and religious factors that may impact the woman's sexual function or expectations.

DIFFERENTIAL DIAGNOSES

Urological:

- Infectious cystitis and interstitial cystitis/bladder pain syndrome can cause deep midline dyspareunia.
- Urethral disorders (diverticulum, urethritis, cyst) can cause pain during vaginal penetration or thrusting.

Gastrointestinal:

- Any inflammatory bowel process (e.g., diverticulitis, Crohn disease, ulcerative colitis) can cause sexual pain. Crohn disease can also present with knife-slit vulvar lesions that can be an independent source of pain.
- Other intestinal dysfunction, such as irritable bowel syndrome or chronic constipation, can impair the pelvic floor muscle function and result in pain.

Musculoskeletal:

- Inquire about joint or muscle pain, limited mobility, injuries, falls, chronic low back pain, degenerative disc or joint disease, hip pain, and altered gait.
- All of these can contribute to pelvic floor muscle hypertonus or dysfunction which in turn contribute to FSP.

Dermatologic:

- Inquire about a history of dermatitis, including skin conditions that can affect the vulva such as eczema, allergies, and contact dermatitis.
- Ask about use of over-the-counter feminine products, soaps, cleansing wipes, and similar products that can result in drying and irritation of the vulva. Some women develop a contact dermatitis.

Vascular:

- Sexual dysfunction appears to occur more frequently in hypertensive women.
- It is unclear whether this is associated with the hypertension itself or with antihypertensive medications.

PHYSICAL EXAMINATION

- During physical examination, the pain is mapped and the anatomy is inspected for possible causes of the symptoms.
- GAD7 and PHQ9 score

INVESTIGATIONS

- Women with vaginal or cervical discharge undergo measurement of vaginal pH, microscopy, and testing for sexually transmitted infections, if appropriate.
    - This test combination can distinguish among bacterial vaginosis, vaginal candidiasis, gonorrhea, chlamydia, and trichomonas.
- Women with genital ulcers are evaluated for herpes simplex virus, syphilis, chancroid, lymphogranuloma venereum, granuloma inguinale, and erosive vulvar dermatoses.
- Women with suprapubic or bladder tenderness suggestive of infection of painful bladder syndrome undergo urinalysis, and urine culture if indicated.
- Women suspected of having hyperprolactinemia, genital herpes (without visible ulcers), or Sjogren's disease undergo serum testing for the respective hormone levels, antibodies, or autoimmune abnormalities.

MANAGEMENT

Reassurance:

Dyspareunia is pain that happens just before, during, or after sex. It can happen in males and females, but is more common in females.

In females, pain can affect the vulva, which is the area around the opening of the vagina. Or the pain can be inside the vagina or in the lower belly.

Advice:

Vaginal lubricants, which are used during sex

Physical therapy to loosen the muscles around the vagina.

Prescription:

Management managed based on the etiology.

Indications for referral:

- Interstitial cystitis/painful bladder syndrome – Refer to a female pelvic medicine specialist.
- Graft versus host disease of the genital tissue
- Female genital cutting or circumcision, endometriosis
- Vulvar dermatoses, if the clinician is not familiar with the diagnosis or treatment.

TOPIC: ENDOMETRIOSIS

DEFINITION

- Endometriosis is characterized by the growth of endometrium-like tissue outside the uterus.
  - Endometriotic deposits are most commonly distributed in the pelvis; on the ovaries, uterosacral ligaments, pouch of Douglas, rectum and sigmoid colon, bladder, and distal ureter. Extra-pelvic deposits, such as in the bowel, diaphragm, umbilicus, and pleural cavity, are rare.
  - Endometriosis is hormone mediated and is associated with menstruation. The hormonal changes in the menstrual cycle induce bleeding, chronic inflammation, and scar tissue formation.

PREVALENCE

- Endometriosis is one of the most common gynaecological disorders in women of reproductive age.
- The prevalence is difficult to determine because of variability in clinical presentation. Also, some women with endometriosis may be asymptomatic.
- The only reliable diagnostic test is laparoscopy.
- On average it takes 7.5 years from onset of symptoms to get a diagnosis of endometriosis.

CAUSES

- The exact cause of endometriosis is unknown. Several theories have been proposed to explain the underlying mechanisms for the development and progression of the disease, but no single one can account for every case or manifestation.
- It is thought that endometriosis develops as a result of a **combination of the following factors**:
  - **Retrograde menstruation** — this is where the endometrial cells flow backwards from the uterine cavity, through the fallopian tubes, and implant on pelvic organs, where they can seed and grow. This theory does not explain why endometriosis occurs in some women after hysterectomy or, rarely, in some men following exposure to oestrogen through drug treatments. It has been suggested that most women (about 90%) experience some form of retrograde menstruation. This theory also does not explain the low rate of the disease compared with how common retrograde menstruation is thought to be.
  - **Lymphatic or circulatory dissemination** — it has also been suggested that endometriotic tissue may also be able to travel to distant sites (such as the lungs, eyes, and brain) through the lymphatic system or in the bloodstream.
  - **Genetic predisposition** — genetic predisposition to endometriosis has been well documented in family and twin-based studies, but no specific gene has been identified. There are currently no genetic tests available
  - **Metaplasia** — this described a process by which cells in the pelvic and abdominal area change into endometrial-type cells of the germinal epithelium. This theory could explain how endometritic cells appear spontaneously in the body and how they appear in areas such as the lung and skin. It could also explain why endometriosis occurs in some women after hysterectomy and in men taking hormone treatment.

- **Environmental factors** — this theory suggests that certain environmental toxins can affect the body, immune system, and reproductive system and cause endometriosis. Animal studies have shown the development of endometriosis in animals following exposure to high levels of dioxin.
- **Immune dysfunction** — many women with endometriosis appear to have reduced immunity to other conditions. It is not known whether this contributes to endometriosis or whether it is as a result of endometriosis.

RISK FACTORS

- Early menarche.
- Late menopause.
- Delayed childbearing.
- Nulliparity.
- Family history.
- Vaginal outflow obstruction.
- White ethnicity.
- Low body mass index (BMI).
- Autoimmune disease (an increased prevalence of autoimmune diseases has been noted in women with surgically confirmed endometriosis)
- Late first sexual encounter.
- Smoking.

COMPLICATIONS

- Endometriomas
  - (ovarian cysts containing blood and endometriosis-like tissue) — if the ovaries are affected, endometriomas may develop. These may rupture and can affect fertility by causing distortion of pelvic anatomy.
- Fertility problems
  - endometriosis is commonly associated with infertility, with a prevalence of 25–40% in infertile women compared with 0.5–5% in fertile women.
  - Severe disease can lead to tubal adhesions (resulting in marked distortion of pelvic anatomy and pelvic pain), reduced ovarian reserve and oocyte and embryo quality, and poor implantation.
  - Endometriosis can also impair fertility by disturbing the function of the fallopian tube, embryo transport, and the eutopic endometrium. Even mild endometriosis can impair fertility.
- Adhesion formation
  - may occur due to the endometriosis or secondary to surgery or infection, and may cause chronic pelvic pain.
- Bowel obstruction
  - partial or complete bowel obstruction can occur due to adhesion formation or a circumferential endometriotic deposit.
- Chronic pain
  - in some women, pain from endometriosis may become chronic even when visible disease has been removed.
- Reduced quality of life
  - symptomatic endometriosis can have a significant and sometimes severe impact on the woman's quality of life and activities of daily living, including relationships and sexuality, work productivity, fitness, and mental health

PROGNOSIS

- Endometriosis can be a chronic disease affecting women throughout their reproductive lives (and sometimes beyond).
- For the majority of women, symptoms can be controlled with hormonal treatment however, some women may have complex needs and require long-term support.
- It is unclear whether endometriosis is always progressive, remains stable, or improves with time.
- Prospective data indicate that recurrence after surgery ranges from 10–50% at one year and increases over time.
- **An association between endometriosis and ovarian cancer has been suggested in previous studies**.
  - The Committee concluded that no recommendations should be made based on the available evidence.
  - The Committee agreed that for other types of cancer of reproductive organs, the evidence was negative or inconclusive.

DIAGNOSIS

- **Suspect endometriosis in women (including young women aged 17 years and younger) presenting with 1 or more of the following symptoms or signs:**
  - Chronic pelvic pain (defined as a minimum of 6 months of cyclical or continuous pain).
  - Period-related pain (dysmenorrhoea) affecting daily activities and quality of life.
  - Deep pain during or after sexual intercourse.
  - Period-related or cyclical gastrointestinal symptoms, in particular painful bowel movements.
  - Period-related or cyclical urinary symptoms, in particular blood in the urine or pain passing urine.
  - Infertility in association with 1 or more of the above.
- **If endometriosis is suspected:**
  - Take a detailed history. Ask questions about the symptoms and have a full discussion with women to:
    - Identify risk factors for endometriosis (such as early menarche or nulliparity).
    - Exclude differential diagnoses (such as irritable bowel syndrome or pelvic inflammatory disease).
    - Identify complications of endometriosis (such as fertility problems or depression).
  - Inform the woman that keeping a pain and symptom diary can aid discussions.
  - Examine the patient:
    - Offer an abdominal and pelvic examination to identify abdominal masses and pelvic signs, such as reduced organ mobility and enlargement, tender nodularity in the posterior vaginal fornix, and visible vaginal endometriotic lesions. An abdominal and pelvic examination will also help to exclude differential diagnoses (such as uterine fibroids).
    - If a pelvic examination is not appropriate, offer an abdominal examination to exclude abdominal masses.
- **Be aware that some women with endometriosis may be asymptomatic.**
  - Do not exclude the possibility of endometriosis if the abdominal or pelvic examination is normal.

- If clinical suspicion remains or symptoms persist, consider referral for further assessment and investigation.

DIFFERENTIAL DIAGNOSES

- **Uterine conditions, such as:**
  - Adenomyosis or uterine fibroids
    - typically causes lower abdominal pain and heavy menstrual bleeding, and an enlarged uterus may be identified on pelvic examination.
  - Primary dysmenorrhoea
    - painful cramping, usually in the lower abdomen, which occurs shortly before or during menstruation, or both. Primary dysmenorrhoea occurs in young females in the absence of any identifiable underlying pelvic pathology.
  - Uterine myoma
    - usually asymptomatic but often present with heavy and/or irregular menstrual bleeding. Pelvic examination may show an enlarged, nodular pelvic mass that can vary in size and shape.
- **Urological conditions, such as:**
  - Interstitial cystitis
    - diffuse chronic pain and dyspareunia are common and often indistinguishable from endometriosis. Symptoms are primarily localized to the bladder, such as urinary frequency and urgency. The woman may complain of pain with a full bladder that is relieved upon voiding.
  - Recurrent urinary tract infections (UTIs)
    - typical features of a UTI include dysuria, frequency, urgency, changes in urine appearance or consistency, nocturia, and suprapubic discomfort/tenderness.
- **Gynaecological conditions, such as:**
  - Pelvic inflammatory disease (PID)
    - typically causes lower abdominal pain, dyspareunia, abnormal vaginal bleeding, and malodorous vaginal discharge, with fever and general malaise in acute infection. Chronic PID may be indistinguishable from endometriosis.
  - Benign ovarian cyst
    - may be asymptomatic with an incidental pelvic mass or present with acute rather than chronic pain, such as with haemorrhagic cysts.
  - Ovarian cancer
    - typical symptoms represent advanced-stage cancer, and include weight gain despite lack of appetite, increased abdominal girth, and altered bowel habits.
  - Pregnancy, including ectopic pregnancy
    - common symptoms and signs of ectopic pregnancy include abdominal or pelvic pain, amenorrhoea or missed period, vaginal bleeding, and pelvic, adnexal, and abdominal tenderness.
- **Gastrointestinal conditions, such as:**
  - Irritable bowel syndrome, inflammatory bowel disease, or chronic constipation
    - typically cause lower abdominal or chronic pelvic pain associated with a range of bowel symptoms which may be cyclical.
  - Appendicitis
    - classic symptoms are abdominal pain, anorexia, nausea, vomiting, and constipation.
  - Gastroenteritis

        - diarrhoea (loose or watery stools, usually at least three times in 24 hours) is the main symptom. Other symptoms may include nausea, sudden onset of vomiting, blood or mucus in stools, and systemic features (such as fever or malaise).
    - Coeliac disease
        - characterized by persistent, unexplained abdominal or gastrointestinal symptoms, such as indigestion, diarrhoea, abdominal bloating, and constipation.
- **Other conditions, such as:**
    - Referred pain from degenerative disc disease
        - nerve entrapment pain is typically sharp, localised, or stabbing, and exacerbated by movement.
    - Other cancers, for example cervical, uterine, rectal, or bladder.
    - Congenital anomalies of the reproductive tract, such as imperforate hemivagina or vaginal septum — a rare cause of dysmenorrhoea in adolescents.
    - Musculoskeletal pain.

MANAGEMENT

For women with suspected or confirmed endometriosis:

- **Consider the need for referral.**
    - Consider referring to a gynaecology service for ultrasound or gynaecology opinion if the woman has:
        - Severe, persistent, or recurrent symptoms of endometriosis.
        - Pelvic signs of endometriosis.
- **Manage endometriosis-related pain.**
    - Consider a short trial (for example 3 months) of paracetamol and/or a non-steroidal anti-inflammatory drug (NSAID) for first-line management of pain.
        - Discuss the benefits and risks of analgesics, taking into account any comorbidities and the woman's preferences.
    - Offer hormonal treatment, for example, the combined oral contraceptive pill or a progestogen, such as the progestogen-only pill, implant (Nexplanon®), injectable (Depot-provera® or Sayana Press®), or levonorgestrel intrauterine system (Mirena®).
        - Explain that hormonal treatment can reduce endometriosis-related pain and has no permanent negative effect on subsequent fertility.
        - Advise the woman on the available options so that she can make an informed choice. The National Institute for Health and Care Excellence (NICE) has produced a patient decision aid on hormonal treatments for endometriosis.
        - Do not offer hormonal treatment to women with endometriosis who are trying to conceive, because it does not improve spontaneous pregnancy rates.
    - Give advice on non-pharmacological management.
        - Advise that the available evidence does not support the use of traditional Chinese medicine or other Chinese herbal medicines or supplements for treating endometriosis.
- **Assess her individual information and support needs,** taking into account her circumstances, symptoms, priorities, desire for fertility, aspects of daily living, work and study, cultural background as well as her physical, psychosexual, and emotional needs.
    - Assess for complications of endometriosis.

        - Assess the impact of the condition on the woman's quality of life, including anxiety or depression, and if necessary, offer treatment.
    - Provide information on endometriosis, including the symptoms and signs, how it is diagnosed, and treatment options (including laparoscopy).
- **Review the woman after 3–6 months, or earlier if symptoms are troublesome.**
    - If initial hormonal treatment for endometriosis is ineffective, not tolerated, or contraindicated, refer the woman to a gynaecology service, specialist endometriosis service, or paediatric and adolescent gynaecology service (as appropriate) for investigation and consideration of further management options, including surgery (laparoscopy).
- **For women with confirmed endometriosis, particularly women who choose not to have surgery, ensure that they are followed up in secondary care if they have:**
    - Deep endometriosis involving the bowel, bladder or ureter or
    - One or more endometrioma that is larger than 3 cm.

SECONDARY CARE ASSESSMENT & INVESTIGATIONS:

- **Diagnosis of endometriosis can only be made definitively by laparoscopic visualization of the pelvis.** However, other less invasive methods (including ultrasound) may be useful in assisting diagnosis.
    - A transvaginal ultrasound should be considered:
        - To investigate suspected endometriosis even if the pelvic and/or abdominal examination is normal.
        - To identify endometriomas and deep endometriosis involving the bowel, bladder, or ureter.
    - A transabdominal ultrasound scan of the pelvis should be considered, if a transvaginal scan is not appropriate.
    - Laparoscopy should be considered in women with suspected endometriosis, even if the ultrasound is normal.
    - If a full, systematic laparoscopy is performed and is normal, the woman should be advised that she does not have endometriosis, and alternative management offered.
- **The following are not recommended as primary investigations to diagnose endometriosis:**
    - Pelvic MRI (magnetic resonance imaging) — but it may be considered to assess the extent of deep endometriosis involving the bowel, bladder, or ureter.
    - Serum CA (cancer antigen) — if a coincidentally reported serum CA125 level is available, it is worth noting that:
        - A raised serum CA125 (that is, 35 IU/ml or more) may be consistent with having endometriosis.
        - Endometriosis may be present despite a normal serum CA125 (less than 35 IU/ml).

SECONDARY CARE MANAGEMENT

- **Surgical management**
    - Women with suspected or confirmed endometriosis should be asked about their symptoms, preferences, and priorities with respect to pain and fertility, to guide surgical decision-making.
    - Surgical treatment for endometriosis should be performed laparoscopically unless there are contraindications.

- During a laparoscopy to diagnose endometriosis, laparoscopic treatment of the following, if present, should be considered:
    - Peritoneal endometriosis not involving the bowel, bladder, or ureter.
    - Uncomplicated ovarian endometriomas.
- As an adjunct to surgery for deep endometriosis involving the bowel, bladder, or ureter, 3 months of treatment with a gonadotrophin-releasing hormone agonists should be considered before surgery.
- Excision rather than ablation should be considered to treat endometriomas, taking into account the woman's desire for fertility and her ovarian reserve.

❖ **Combination treatments**

- After laparoscopic excision or ablation of endometriosis, hormonal treatment (with, for example, the combined oral contraceptive pill) should be considered, to prolong the benefits of surgery and manage symptoms.

❖ **Hysterectomy in combination with surgical management**

- If hysterectomy is indicated (for example if the woman has adenomyosis or heavy menstrual bleeding that has not responded to other treatments), all visible endometriotic lesions should be excised at the time of the hysterectomy.
- Hysterectomy should be performed (with or without oophorectomy) laparoscopically when combined with surgical treatment of endometriosis, unless there are contraindications.

❖ **Surgical management if fertility is a priority**

- NICE states that these recommendations should be interpreted within the context of the NICE guideline Fertility problems: assessment and treatment. The management of endometriosis-related subfertility should have multidisciplinary team involvement with input from a fertility specialist. This should include the recommended diagnostic fertility tests or preoperative tests, as well as other recommended fertility treatments, such as assisted reproduction, that are included in the NICE guideline on fertility problems.
    - Excision or ablation of endometriosis plus adhesiolysis should be offered for endometriosis not involving the bowel, bladder, or ureter, because this improves the chance of spontaneous pregnancy.
    - Laparoscopic ovarian cystectomy with excision of the cyst wall should be offered to women with endometriomas, because this improves the chance of spontaneous pregnancy and reduces recurrence. The woman's ovarian reserve should be taken into account.
    - For women who have deep endometriosis involving the bowel, bladder, or ureter and who are trying to conceive (working with a fertility specialist), the benefits and risks of laparoscopic surgery as a treatment option should be discussed. Topics to discuss may include: whether laparoscopic surgery may alter the chance of future pregnancy; the possible impact on ovarian reserve; the possible impact on fertility if complications arise; alternatives to surgery; and other fertility factors.

TOPIC: PELVIC INFLAMMATORY DISEASE (PID)

DEFINITION

- Pelvic inflammatory disease (PID) is a general term for infection of the upper genital tract, which typically affects sexually active young women. Infection spreads upwards from the vagina and endocervix due to damage to the genital tract epithelium, causing one or more of the following:
  - Endometritis — inflammation and infection of the uterus.
  - Salpingitis — inflammation of the fallopian tubes.
  - Parametritis — inflammation of the parametrium, the connective tissue of the pelvic floor.
  - Oophoritis — inflammation of the content of one or both ovaries.
  - Tubo-ovarian abscess — due to complex infection of the adnexae.
  - Pelvic peritonitis — inflammation of the peritoneum.

CAUSES

- **Sexually transmitted infections (STIs)**
  - Chlamydia trachomatis
    - is the most commonly identified cause (14–35% of cases)
  - Neisseria gonorrhoeae (found in 2–3% of cases).
    - There is an increased risk of gonococcal infection if there is severe PID infection or there has been sexual contact abroad
  - Mycoplasma genitalium
    - may be associated with 10–25% of cases of PID and cervicitis in women
- **Other infections**
  - Micro-organisms that are part of the normal vaginal microbiome
    - such as Gardnerella vaginalis and anaerobes such as Prevotella, Atopobium, and Leptotrichia)
  - Other respiratory or enteric pathogens
    - such as streptococci, staphylococci, Escherichia coli, Bacteroides spp and Haemophilus influenzae may be transmitted from the oropharynx or rectum to the vagina during sex and then colonize the lower genital tract. These may then ascend into the upper genital tract causing infection
  - Actinomyces spp
    - is a commensal anaerobic bacteria found in the genital tract. Pelvic actinomycosis is a very rare, chronic bacterial pelvic infection that is associated with long-term intrauterine contraceptive (IUC) use
  - Rarely, infection can also reach the upper genital tract from the parametrium through the lymphatic system or through the haematogenous route in women with tuberculosis
  - Note: pathogen-negative PID is also common

RISK FACTORS

- Factors related to sexual behaviour, such as:
  - Age younger than 25 years.
  - Not using a barrier method of contraception.
  - Younger age at onset of sexual activity (less than 15 years).

- Multiple sexual partners.
- Recent new sexual partner (within the previous 3 months).
- Previous history of PID.
- History of sexually transmitted infection (STI) in the woman or her sexual partner.

- Recent instrumentation of the uterus or interruption of the cervical barrier, due to:
  - Termination of pregnancy (TOP).
  - Insertion of an intrauterine device (IUD) within the past 3 weeks, especially in women with pre-existing gonorrhoea or chlamydia infection. Overall, the risk is very low (less than 1% of all IUD users).
  - Hysterosalpingography or hysteroscopy.
  - In vitro fertilization (IVF) assisted reproductive techniques.

COMPLICATIONS

- The risk of complications of pelvic inflammatory disease (PID) is increased with severe or repeated infection and delayed treatment. Complications may include:
  - Ectopic pregnancy
    - PID increases the relative risk of a subsequent pregnancy being ectopic, but the absolute risk remains low, at around 1%
  - Pelvic peritonitis and sepsis
  - Tubo-ovarian abscess
  - Perihepatitis (Fitz-Hugh-Curtis syndrome)
  - Tubal factor infertility
    - may result from salpingitis with subsequent scarring and adhesions causing partial or total obstruction of the fallopian tubes and/or selective loss of ciliated epithelial cells along the fallopian tube, impairing ovum transport
  - Chronic pelvic pain
    - May be due to scarring and adhesions following acute infection

DIAGNOSIS

- The diagnosis of pelvic inflammatory disease (PID) should be made on the basis of clinical features. Do not delay making a diagnosis and starting antibiotic drug treatment while waiting for laboratory test results. If a diagnosis of PID is suspected:
- **Ask about:**
  - Any typical symptoms of PID, such as recent-onset:
    - Pelvic or lower abdominal pain (usually bilateral but can be unilateral).
    - Deep dyspareunia, particularly if recent onset.
    - Secondary dysmenorrhoea.
    - Abnormal vaginal bleeding (intermenstrual, postcoital, or heavy menstrual bleeding) which may be secondary to associated cervicitis and endometritis.
    - Abnormal vaginal or cervical mucopurulent discharge which may be due to associated cervicitis, endometritis, or vaginal infection.
    - Right upper quadrant pain or right shoulder pain due to possible perihepatitis.
    - Systemic symptoms such as fever (may be absent), nausea or vomiting (may be present in 50% of women) or malaise, which may suggest severe infection or a complication.
    - Note: PID may be asymptomatic in some women.
  - Any symptoms of an alternative underlying cause.

- What contraception is being used (if any), including recent intrauterine device (IUD) insertion, and whether the woman is (or might be) pregnant.
- Sexual history, including number of recent partners, any symptoms in sexual partner(s), and any other risk factors for developing PID.
    - Note: have a low threshold for suspecting a diagnosis of PID if the woman is sexually active, even if they report being in a monogamous relationship and their sexual partner is at low risk of sexually transmitted infection (STI).
- Smear history, including recent results.

❖ **Examination:**

- Perform a general, abdominal, and pelvic examination (with a chaperone) to assess for:
    - Fever and signs of systemic illness (which may suggest severe infection or a complication).
    - Lower abdominal tenderness (usually bilateral) or right upper quadrant pain if associated perihepatitis.
    - Adnexal tenderness (with or without a palpable mass), cervical motion tenderness, or uterine tenderness on bimanual vaginal examination.
    - Abnormal cervical or vaginal mucopurulent discharge and/or cervical friability on speculum examination.

❖ **Investigations:**

- Arrange investigations in primary care depending on clinical judgement. Consider signposting the woman to attend a local specialist sexual health service for STI screening and contact tracing, depending on clinical judgement and local availability.
    - Perform a pregnancy test in any sexually active woman to exclude ectopic pregnancy.
    - Offer to take vaginal swabs to assess for infection, including chlamydia and gonorrhoea. Be aware that a negative vaginal swab result does not rule out a diagnosis of PID.
        - In addition, consider testing for Mycoplasma genitalium, depending on local testing availability.
    - Assess for endocervical or vaginal pus cells under a microscope on a wet-mount vaginal smear if there is equipment available and appropriate clinician experience and expertise.
        - If pus cells are absent, a diagnosis of PID is unlikely. If present, this may be associated with PID or lower genital tract infection.
    - Consider arranging blood tests, depending on clinical judgement.
        - Leucocyte count — if elevated may support the diagnosis.
        - Inflammatory markers such as erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP) — if elevated may support the diagnosis.
        - HIV, hepatitis serology, and syphilis serology — as part of an STI screen.

DIFFERENTIAL DIAGNOSES

❖ Ectopic pregnancy or rupture.
❖ Threatened miscarriage.
❖ Acute appendicitis.
❖ Endometriosis.

- Gastrointestinal disorders, such as acute gastroenteritis, irritable bowel syndrome, diverticulitis, bowel obstruction, perforation, or necrosis.
- Ovarian cyst complications, such as rupture, torsion, or haemorrhage — symptoms are often of sudden onset and unilateral.
- Urinary tract infection, pyelonephritis, or renal tract stone.
- Mittelschmerz pain — benign mid-cycle ovulation pain.
- Ruptured corpus luteal cyst — a type of ovarian cyst which may rupture around the time of menstruation.
- Functional pelvic pain — may be associated with other chronic symptoms.

MANAGEMENT

- Arrange **urgent hospital admission** if:
  - The woman is pregnant, or an ectopic pregnancy is suspected.
  - There are severe systemic symptoms and/or an adnexal mass suggesting a possible complication, such as a tubo-ovarian abscess or pelvic peritonitis.
  - The woman is systemically unwell, and there is an uncertain diagnosis.
  - The woman is unable to follow or tolerate management in primary care.
- Consider **seeking advice from an appropriate specialist** if:
  - A woman has HIV — liaise with her infectious diseases specialist as needed.
  - A woman has an intrauterine contraceptive (IUC) and actinomyces-like organisms (ALOs) detected on cervical cytology — liaise with the local microbiology and gynaecology teams regarding the need for further investigation and management on a case-by-case basis.
    - Note: if a woman is asymptomatic but has positive ALO on cervical screening, there is no need for further investigation or management.
  - A complication such as perihepatitis is suspected.
  - There is any uncertainty about clinical management.
- **If hospital admission or referral is not needed**, advise the woman to attend a local specialist sexual health service for sexually transmitted infection (STI) screening, treatment, and contact tracing of current and recent sexual partners (within the last 6 months), depending on clinical judgement and local availability.
- Advise on the use of **over-the-counter medication for symptom relief**.
  - Options include paracetamol and/or ibuprofen if there are no contraindications.
- Advise to start **empirical antibiotic drug treatment** as soon as a working diagnosis of PID is made.
  - Advise the woman to ideally have STI screening before starting antibiotic treatment so that an accurate diagnosis can be made. Do not, however, delay starting drug treatment while awaiting tests to be arranged or the results of laboratory tests.
  - Advise to complete the course of antibiotic treatment, even if vaginal swabs are negative, in order to reduce the risk of long-term complications.
- **If a woman has a copper or levonorgestrel intrauterine device (IUD) in situ**:
  - If she has mild-to-moderate symptoms, advise that the IUD can remain in situ, provided she is clinically improving within 48–72 hours of starting antibiotic treatment. If symptoms are not improving, the IUD should be removed.
  - If she has severe symptoms at presentation, the IUD should be removed.
  - If the IUD is removed, consider the need for emergency hormonal contraception, follow-up pregnancy testing, and ongoing contraception.

- **Advise to abstain from all sexual activity** until both the woman and any sexual partner(s) have completed antibiotic treatment, are symptom-free, and have had a 'test of cure' if needed, depending on clinical judgement.
    - Advise that a barrier method of contraception should be used if sexual contact cannot be avoided. Advise that future use of a barrier method of contraception will greatly reduce the risk of reinfection and protect against other STIs.
- **Arrange to review the woman in primary care,** the timescale depends on clinical judgement.

ANTIBIOTIC REGIMEN:

- Advise the woman to attend a local specialist sexual health service for sexually transmitted infection (STI) screening, treatment, and contact tracing of current and recent sexual partners, depending on clinical judgement and local availability. If the woman is unable or unwilling to attend, consider prescribing antibiotic treatment in primary care.
- **If the risk of gonococcal infection is high:**
    - Prescribe ceftriaxone 1 g as a single intramuscular (IM) dose, followed by oral doxycycline 100 mg twice daily plus oral metronidazole 400 mg twice daily for 14 days.
- **If the risk of gonococcal infection is low**, prescribe one of the following antibiotic regimens, taking into account drug availability, any contraindications, and local antimicrobial sensitivity patterns.
    - First-line, prescribe ceftriaxone 1 g as a single IM dose, followed by oral doxycycline 100 mg twice daily plus oral metronidazole 400 mg twice daily for 14 days.
    - Second-line, consider prescribing one of the following antibiotic regimens:
    - Oral ofloxacin 400 mg twice daily plus oral metronidazole 400 mg twice daily for 14 days, or
    - Oral levofloxacin 500 mg once daily plus oral metronidazole 400 mg twice daily for 14 days, or
    - Oral moxifloxacin 400mg once daily for 14 days.
- **If the initial test result for Mycoplasma genitalium is positive**:
    - Prescribe oral moxifloxacin 400mg once daily for 14 days.
- **If all first- and second-line antibiotic regimens are contraindicated or not tolerated, consider prescribing:**
    - Ceftriaxone 1 g as a single IM dose, followed by oral azithromycin 1 g per week for 2 weeks.
- **If there is a risk of very early pregnancy (too early for a pregnancy test to be positive):**
    - Consider starting empirical antibiotic treatment, as the benefits of treatment outweigh the low risks of antibiotic treatment in very early pregnancy.

FOLLOW UP:

- Review the woman within 72 hours, depending on clinical judgement.
    - Ask about current symptoms to assess the response to treatment.
    - If she is not clinically improving, check compliance with treatment, reassess for an alternative underlying cause, and consider arranging urgent hospital admission.
- Check the antibiotic sensitivities from vaginal swab results (if available).
    - Adjust antibiotic drug treatment if necessary.
    - Continue antibiotic treatment even if vaginal swabs are negative.

    - If there are mild-to-moderate PID symptoms and the woman is unable to tolerate metronidazole due to adverse effects, advise her to stop metronidazole but continue the other antibiotics in the regimen.
- Consider arranging a further review 2–4 weeks after completion of antibiotic treatment.
    - Check compliance with and response to antibiotic treatment.
    - Check that any sexual partner(s) have been screened and treated.
    - Advise a repeat pregnancy test and test of cure if clinically indicated.
- Advise on the future use of a barrier method of contraception to reduce the risk of reinfection and protect against other STIs, and advise on options for additional methods of contraception.
- Ensure a 'test of cure' is arranged by the local specialist sexual health service or in primary care if the woman has:
    - A positive initial test result for gonorrhoea.
        - Repeat testing should be routinely arranged 2–4 weeks after completion of treatment.
    - A positive initial test result for chlamydia.
        - Repeat testing should be arranged 3–5 weeks after completion of treatment if there are persisting symptoms or if compliance with oral antibiotics and/or tracing of sexual contacts indicates the possibility of persisting or recurrent infection.
    - A positive initial test result for Mycoplasma genitalium.
        - Repeat testing should be arranged 4 weeks after completion of treatment.
- Persistent symptoms after completing antibiotic treatment.
- An initial test result showing unknown antibiotic sensitivity or antibiotic resistance (in cases of gonorrhoea or Mycoplasma genitalium).
- Suspected poor compliance with antibiotic treatment or treatment has not been tolerated.
- Possible persisting or recurrent infection, for example, due to repeated sexual contact with untreated partners.

MANAGING SEXUAL PARTNERS:

- Advise the woman that any current and recent sexual partners (within the last 6 months) should attend a local specialist sexual health service for sexually transmitted infection (STI) screening, treatment, and contact tracing. If a sexual partner is unable or unwilling to attend a specialist clinic:
- Offer screening for chlamydia and gonorrhoea.
    - Whilst awaiting the test results, offer any male partner(s) empirical treatment with a broad-spectrum antibiotic, such as doxycycline 100 mg twice daily for one week.
    - If chlamydia or gonorrhoea infection is confirmed in the partner(s), treat both the partner(s) and the woman appropriately.
- Offer screening for Mycoplasma genitalium if the woman has confirmed M. genitalium infection (depending on local testing availability).
    - If the test result for M. genitalium is positive, arrange antibiotic treatment for both the partner(s) and the woman.
- Advise to abstain from all sexual activity until both the woman and sexual partner(s) have completed a course of antibiotic treatment, are symptom-free, and have had a 'test of cure' if needed, depending on clinical judgement.
- Advise on the future use of a barrier method of contraception to reduce the risk of reinfection and protect against other STIs.

TOPIC: UTERINE FIBROIDS

DEFINITION

- **Uterine fibroids (leiomyomas)** are benign tumours which are caused by proliferation of a mixture of smooth muscle cells and fibroblasts, which form hard, round, whorled tumours in the myometrium.
  - They can be single, or multiple, and their size varies from a few millimetres to 30 cm or larger
  - They typically develop in women of reproductive age, as their growth and maintenance are thought to be oestrogen- and progesterone-dependent.
- Fibroids may develop anywhere within the myometrium. They may be defined as:
  - **Subserosal**
    - These develop near the outer serosal surface of the uterus and extend outside the uterus into the peritoneal cavity. They are commonly asymptomatic or minimally symptomatic even when relatively large. When they are sufficiently large, they may cause symptoms due to pressure on adjacent structures.
  - **Intramural**
    - These develop within the myometrium without extending predominately into the uterine cavity or peritoneal cavity. They may cause heavy menstrual bleeding and dysmenorrhea by interfering with the constriction of blood vessels during menstruation.
  - **Submucosal**
    - These develop near the inner mucosal surface of the uterus and extend into the uterine cavity. Even relatively small submucosal fibroids may cause significant heavy menstrual bleeding, dysmenorrhea, or reduce fertility.

RISK FACTORS

- Increasing age
  - risk increases with age during reproductive years until the menopause.
- Early menarche
  - risk increased if menarche before age of 11 years.
- Nulliparity.
- Older age at first pregnancy
  - fibroids may enlarge during the first trimester of pregnancy, and tend to shrink post-partum.
- Comorbidities
  - obesity (weight gain and central distribution of body fat); diabetes; hypertension.
- Ethnicity
  - higher risk in black and Asian women compared with white women. In addition, fibroids are more likely to be symptomatic, occur at an earlier age, be larger, and multiple in these ethnic groups.
- Family history
  - risk is higher in women who have an affected first-degree relative.

COMPLICATIONS

- Iron deficiency anaemia.

    - Heavy or abnormal uterine bleeding may occur due to distortion of the endometrial lining by fibroids, leading to endometrial and vascular dysfunction.
- Compression of adjacent organs by large fibroids, which may cause:
    - Recurrent urinary tract infection (UTI) — if pressure on the bladder.
- Urinary retention
    - if pressure on the bladder.
- Hydronephrosis
    - rare, due to a large fibroid causing ureteric compression which may affect renal function.
- Subfertility or infertility (rare).
    - Submucosal or deep intramural fibroids may distort the uterine cavity and interfere with implantation and live birth rates.
        - The risk increases with age.
- Complications during pregnancy (rare).
    - Miscarriage
        - submucosal or deep intramural fibroids may distort the uterine cavity and interfere with pregnancy outcome.
    - Fibroid vascular infarction ('red degeneration')
        - may cause acute pain due to degenerative changes when rapid growth of a fibroid, promoted by high levels of sex hormones, causes it to outgrow its blood supply.
    - Possible higher rates of caesarean instrumental delivery.
    - Fetal malpresentation.
    - Pre-term delivery.
- Torsion of a pedunculated fibroid (rare).
    - This can affect subserosal and submucosal fibroids, if they are attached to the myometrium by a pedicle containing their blood supply, and may present with acute pelvic or abdominal pain.
    - Occasionally, fibroid regression may occur when the blood supply is cut off due to torsion.
- Haemoperitoneum (rare).
    - This may result from spontaneous rupture or avulsion of fibroids, or rupture of an overlying blood vessel.

PROGNOSIS

- The natural history of fibroids is unpredictable and varies depending on the woman's age and reproductive status.
    - Expert opinion in a review article states that about 3–7% of untreated fibroids in premenopausal women regress over six months to three years.
- Once formed, fibroids tend to persist until the menopause when they usually shrink.
    - Shrinkage may be limited by calcification or delayed by taking hormone replacement therapy (HRT).
    - Occasionally, fibroids may degenerate before the onset of menopause

DIAGNOSIS

- Fibroids are commonly asymptomatic and may be identified incidentally during a routine pregnancy assessment or gynaecology assessment for a different condition.
- **Typical symptoms may include:**
    - Heavy menstrual bleeding and/or dysmenorrhoea.

- Pelvic pain, pressure, or discomfort; dyspareunia.
- Crampy abdominal discomfort or bloating (mainly during menstruation and felt in the lower abdomen)
- Back pain.
- Urinary symptoms, such as frequency, urgency, urinary incontinence, or retention; and increased risk of urinary tract infections (particularly if a large fibroid is causing pressure on the bladder).
- Bowel symptoms, such as bloating, constipation, and/or painful defecation (particularly if a large fibroid is causing pressure on the bowel).
- Subfertility or infertility (particularly if there are submucosal fibroids).
- rare features:
  - polycythaemia secondary to autonomous production of erythropoietin

- **On abdominal and pelvic examination there may be:**
  - Typically, a firm, enlarged, and irregularly shaped non-tender uterus on pelvic examination.
  - A central irregular abdominal mass (if a large fibroid).
- **Pelvic or renal tract ultrasound may show:**
  - Classical appearance of uterine fibroid(s).
  - Evidence of a complication such as hydronephrosis if a large fibroid causes ureteric compression.

ASSESSMENT

- If a woman has suspected or confirmed fibroids (for example on pelvic ultrasound scan):
- **Ask about:**
  - Any symptoms, their severity, duration, and impact on daily functioning and quality of life.
  - Any clinical features which may suggest an additional or alternative diagnosis.
  - Any previous fertility issues and hopes for future fertility.
  - Any risk factors, including family history.
  - History of cervical screening and results.
  - Any previous treatments for heavy menstrual bleeding and/or fibroids.
- **Examine the patient**
  - Perform an abdominal and bimanual pelvic examination to assess for the presence of pelvic tenderness and any mass(es).
- **Arrange a routine pelvic ultrasound scan (transabdominal and transvaginal, if needed)** to determine the number, size, and location of fibroid(s), particularly if there are any of the following:
  - The uterus is palpable abdominally.
  - History or examination suggests a pelvic mass.
  - Examination is inconclusive or difficult, for example, if a woman is obese.
- **Consider arranging additional investigations,** depending on clinical judgement.
  - Check a full blood count to assess for iron deficiency anaemia if there is a history of heavy menstrual bleeding and/or symptoms of anaemia.

DIFFERENTIAL DIAGNOSES

- **Malignant causes**
  - Ovarian cancer.
  - Endometrial cancer.

- Very rarely a pelvic mass may be due to a leiomyosarcoma which typically presents with rapid or unexpected growth after menopause. It can only be reliably diagnosed by histopathology. It is uncertain whether leiomyosarcoma is due to de novo growth or malignant transformation of a benign uterine fibroid.

- Gastrointestinal malignancy
- Urinary tract malignancy

- **Benign causes**
  - Endometrial polyp or hyperplasia.
  - Adenomyosis or endometriosis.
  - Urinary retention.
  - Pregnancy including ectopic pregnancy.

MANAGEMENT

If a woman has suspected or confirmed fibroids:

- **Arrange an urgent referral to an appropriate specialist using a two-week cancer pathway if the woman has:**
  - Any clinical or radiological features suggesting a gynaecological or other malignancy.
- **Arrange a referral to an appropriate specialist, the urgency depending on clinical judgement, if the woman has:**
  - An uncertain diagnosis.
  - Severe heavy menstrual bleeding or compressive symptoms.
  - Symptoms which cannot be successfully managed in primary care.
  - Confirmed fibroids measuring 3 cm or more in diameter or suspected submucosal fibroids (for example on pelvic ultrasound scan).
  - Suspected fertility or obstetric issues associated with fibroids.
  - Rapid or unexpected growth of fibroids after the menopause.
- **If a woman has heavy menstrual bleeding** associated with fibroid, manage the menorrhagia
- **If a woman has confirmed fibroids which are currently asymptomatic:**
  - Advise that no treatment or follow-up is needed routinely.
  - Advise her to arrange medical review if there are new symptoms, clinical features suggesting a gynaecological or other malignancy, and/or rapid growth of a pelvic mass.
- **Provide information and advice on management options for contraception, if required.**
  - Advise that options for women with fibroids vary depending on whether there is distortion of the uterine cavity or not.
- **Provide information and advice on management options for menopausal symptoms, if required.**
  - Advise that in some women, hormone replacement therapy (HRT) may increase the size of fibroids and possibly cause symptoms.
    - If HRT is started in a woman with fibroids, assess for fibroid symptoms at each review and advise her to return for review if symptoms develop at any time.
    - If a woman has symptomatic fibroids and requests HRT, consider seeking specialist advice before prescribing.

SPECIALIST ASSESSMENT AND MONITORING

- **Specialist assessment and monitoring**
  - Pelvic ultrasound scan or MRI may be needed to check fibroid position, size, number, vascularity, and growth, for example before planned surgery. Hysteroscopy may be arranged to identify submucosal fibroids.

- **Management options**
  - Management options may include drug treatments, interventional radiology procedures, or surgical procedures, depending on the number, size, and location of fibroid(s); the woman's age; symptoms; previous treatments; and her desire to retain fertility and/or uterus
  - Drug treatments
    - Options include nonsteroidal anti-inflammatory drugs (NSAIDs), the antifibrinolytic agent tranexamic acid, and hormonal contraception or cyclical oral progestogens.
    - Additional specialist drug treatments include gonadotrophin-releasing hormone analogues (GnRH analogues) which may be used pre-operatively to treat fibroids which are causing an enlarged or distorted uterus, or in women approaching the menopause.
    - The selective progesterone receptor modulator ulipristal acetate has been granted restricted use for intermittent treatment of moderate or severe symptoms before menopause when surgical procedures including uterine artery embolization are unsuitable or have failed, as there is a risk of severe liver injury and liver failure.
  - Interventional radiology procedures
    - Uterine artery embolization may be offered if there are symptomatic fibroids and a woman wishes to avoid surgery.
  - Surgical treatments
    - Options include myomectomy, hysterectomy (may be open, hysteroscopic, or laparoscopic approaches) or second-generation endometrial ablation. Surgery may be indicated if fibroids are large, rapidly growing, or symptoms are not responding to drug treatment(s)
    - hysteroscopic endometrial ablation
    - hysterectomy

FIBROID DEGENERATION

- Uterine fibroids are sensitive to oestrogen and can therefore grow during pregnancy.
  - If growth outstrips their blood supply, they can undergo red or 'carneous' degeneration.
- This usually presents with low-grade fever, pain and vomiting.
- The condition is usually managed conservatively with rest and analgesia and should resolve within 4-7 days.

TOPIC: POLYCYSTIC OVARIAN SYNDROME (PCOS)

DEFINITION

- **Polycystic ovary syndrome (PCOS) is a heterogeneous endocrine disorder that appears to emerge at puberty.**
- It is characterized by hyperandrogenism (with clinical features such as acne and hirsutism), ovulation disorder (usually manifested as infrequent or no menstruation), and polycystic ovarian morphology on ultrasound.
- The clinical features of PCOS vary widely, with symptoms of hyperandrogenism and severe menstrual disturbances at one end of the spectrum (previously known as Stein–Leventhal syndrome) and mild symptoms at the other. Symptoms may vary over time

CAUSES

- **The cause of polycystic ovary syndrome (PCOS) is unknown.** It is likely to be multifactorial, with both genetic and environmental factors playing a part.
- **Insulin resistance and the consequent compensatory hyperinsulinaemia are key factors in the pathogenesis of PCOS in many, but not all, women.**
  - Effects of hyperinsulinaemia include:
    - Reduced production of sex hormone-binding globulin (SHBG) in the liver — as testosterone is bound to SHBG, more testosterone is available in the blood in the biologically active unbound form, even though the total testosterone level may be normal or only modestly elevated.
    - Increased androgen production (via several mechanisms) — an excessive increase in androgen production stops follicular development and therefore causes anovulation (failure of the ovaries to produce eggs) and menstrual disturbance.
- **Hormonal imbalance is common in women with PCOS.**
  - Serum luteinizing hormone (LH) levels are elevated in approximately 40% of women with PCOS due to increased production from the anterior pituitary (increased amplitude and frequency of LH pulses).
  - The theca cells of the ovary produce excess androgens due to hyperinsulinaemia or increased serum levels of LH.
    - When the concentration of LH increases relative to that of follicle-stimulating hormone, the ovaries preferentially synthesize androgens from androgen precursors rather than oestrogens
    - The theca cells in women with PCOS seem to be more efficient at converting androgen precursors to testosterone than normal theca cells
- **Women with PCOS may have increased serum oestrogen levels.**
  - Follicular development is arrested at some stage short of full maturation of an ovulatory follicle. Therefore, although there is no ovulation, oestrogen production continues and there is no oestrogen deficiency. However, as a result of continued exposure to oestrogen unopposed by progestogen, the endometrium may become hyperplastic
  - In addition, testosterone is converted to oestrogen in peripheral fat
- **PCOS appears to have a genetic link.** However, specific genes associated with the condition have not been identified

COMPLICATIONS

- **Infertility**
  - PCOS is the single most common cause of infertility in young women. It is the underlying cause in 75% of women who have infertility due to anovulation
  - Women with PCOS may have a pregnancy rate significantly less than the rate of successful ovulation induction
- **Cardiovascular disease (CVD)**
  - The cardiometabolic profile is adversely altered in PCOS, and many CVD risk factors (such as body mass index, dyslipidaemia, hypertension, insulin resistance, metabolic syndrome, and deficiencies in insulin secretion) are increased
  - CVD risk factors beyond standard recognized risks (identified as 'novel risk factors') and early onset cardiovascular dysfunction (including endothelial dysfunction, arterial stiffness, plaques, and coronary artery calcification) have been noted in women with PCOS and are related to insulin resistance and obesity
  - High androgen levels and low sex hormone binding globulin levels have also been linked to increased CVD risk in both premenopausal and postmenopausal women with PCOS
  - Type 2 diabetes is a major CVD risk factor, and insulin resistance has been shown to worsen CVD risk in PCOS
- **Metabolic disorders**
  - Insulin resistance is present in around 65–80% of women with PCOS. This is independent of obesity but is further exacerbated by excess weight. Insulin resistance has been shown to worsen reproductive and metabolic features and type 2 diabetes in PCOS
- **Obstructive sleep apnoea syndrome**
  - About 20–45% of women with PCOS have OSAS or sleep-disordered breathing. Therefore, all women with PCOS should be screened for these conditions
- **Psychological disorders**
  - PCOS is associated with an increased risk of many mental health conditions, including mood and anxiety disorders
  - One meta-analysis found that the odds of being diagnosed with an eating disorder (such as binge-eating disorder and bulimia nervosa) were 3.9-fold higher in women with PCOS compared with controls
- **Pregnancy complications**
  - Women with PCOS have a clinically significant increased risk of pregnancy complications compared with women without PCOS
  - Characteristic features of PCOS, such as hyperandrogenism, obesity, insulin resistance, and metabolic abnormalities, may contribute to the increased risk of obstetric and neonatal complications
  - Evidence from systematic reviews and meta-analyses have shown increased rates of gestational diabetes, pregnancy-induced hypertension, pre-eclampsia, caesarean delivery, miscarriage, hypoglycaemia, preterm delivery, macrosomia, newborn admission to the neonatal intensive care unit, neonatal asphyxia, and perinatal mortality in PCOS
- **Cancer**
  - Women with PCOS appear to have an increased risk of endometrial cancer due to prolonged oligomenorrhoea or amenorrhoea and an increased prevalence of obesity

- Untreated anovulatory women with PCOS have chronic oestrogen exposure with no progesterone exposure. This could lead to abnormal uterine bleeding, endometrial hyperplasia, and cancer
- Intervals between menstruation of more than 3 months (equivalent to fewer than four periods each year) may be associated with endometrial hyperplasia and later cancer
- There does not appear to be an association with breast or ovarian cancer and no additional surveillance is required

- **Non-alcoholic fatty liver disease (NAFLD)**
  - NAFLD may be present in 40–55% of women with PCOS due to high androgen levels

PROGNOSIS

- Polycystic ovary syndrome (PCOS) is a chronic condition.
  - There is no cure, and treatment is usually continued throughout the woman's reproductive years.
- The aim of treatment is to:
  - Alleviate the signs and symptoms of PCOS.
  - Prevent the development of complications, such as type 2 diabetes and cardiovascular disease.
- Symptoms generally recur if treatments are stopped during the woman's reproductive years.
- As the woman gets older or reaches menopause, hyperandrogenic manifestations may improve as ovarian function declines allowing withdrawal of treatments aimed at the hyperandrogenic features of PCOS.

DIAGNOSIS

WHEN TO SUSPECT PCOS:

- **In adults, suspect polycystic ovary syndrome (PCOS) if a woman has one or more clinical features of:**
  - Hyperandrogenism, such as acne and hirsutism.
  - Ovulation disorders, such as oligomenorrhoea, amenorrhoea, and infertility.
- **In adolescents, suspect PCOS if the girl has the following:**
  - Clinical features of hyperandrogenism (such as acne and hirsutism).
  - Irregular menstrual cycles, defined as:
    - Normal in the first-year post-menarche as part of the pubertal transition.
    - More than 1 year to less than 3 years of irregular cycles (more than 45 days or less than 21 days) after the onset of menarche.
    - More than 3 years of irregular cycles (more than 35 days or less than 21 days, or less than 8 cycles every year) post menarche to perimenopause.
    - More than 1 year of irregular cycles (more than 90 days for any one cycle) post menarche.
  - Primary amenorrhea by age 15 years or more than 3 years of irregular cycles post thelarche (breast development).
- **Suspect PCOS if one or more of the following are present:**
  - A family history of PCOS.
  - Indirect evidence of insulin resistance, such as obesity (especially central obesity) and acanthosis nigricans.

- **<u>If PCOS is suspected,</u>** arrange investigations to help confirm the diagnosis and exclude differential diagnoses.
  - In adolescents, consider investigations after only 1 year of irregular cycles if the combined oral contraceptive (COC) pill is about to be started, as COCs will mask diagnosis of PCOS by suppressing hyperandrogenaemia.

Investigations & diagnosis:

- **<u>If polycystic ovary syndrome (PCOS) is suspected:</u>**
  - Measure total testosterone — this is normal to moderately elevated in women with PCOS.
  - Measure sex hormone-binding globulin (SHBG) — this is normal to low in women with PCOS and provides a surrogate measurement of the degree of hyperinsulinaemia.
  - Calculate free androgen index (100 multiplied by the total testosterone value divided by the SHBG value) to assess the amount of physiologically active testosterone present — this is normal or elevated in women with PCOS (the normal range is usually defined as lower than 5, but may vary within local laboratories).
    - Note that reliable assessment of biochemical hyperandrogenism is not possible in women on hormonal contraception due to effects on SHBG and altered gonadotrophin-dependent androgen production. If assessment of biochemical hyperandrogenism is indicated, hormonal contraception should be stopped for at least 3 months before the assessment. The need for a non-hormonal alternative contraceptive method should be considered.
- **<u>Measure the following to rule out other causes of oligomenorrhoea and amenorrhoea</u>** (such as premature ovarian failure, hypothyroidism, and hyperprolactinaemia):
  - Luteinizing hormone and follicle-stimulating hormone — may be increased in women with premature ovarian failure, and decreased in women with hypogonadotropic hypogonadism.
  - Prolactin (normal range is less than 500 mU/L) — may be mildly elevated in women with PCOS.
  - Thyroid-stimulating hormone (normal range 0.4–4.5 mU/L).
- **<u>Refer adults for ultrasound scan to look for the classic picture of polycystic ovaries</u>** (unless the diagnosis of PCOS is obvious on clinical and biochemical grounds).
  - Polycystic ovaries on ultrasound are defined as the presence of 12 or more follicles in at least one ovary (measuring 2–9 mm diameter) or increased ovarian volume (greater than 10 cm3).
  - Note that polycystic ovaries do not have to be present to make the diagnosis of PCOS, and the finding of polycystic ovaries does not alone establish the diagnosis.
  - Ultrasound scan should not be used for the diagnosis of PCOS in adolescents.
- **<u>Exclude other causes of hyperandrogenism (such as late-onset congenital adrenal hyperplasia, Cushing's syndrome, or an androgen-secreting tumour) if:</u>**
  - There are signs of virilization, for example, deep voice, reduced breast size, increased muscle bulk, and clitoral hypertrophy.
  - There is rapidly progressing hirsutism (less than 1 year between hirsutism being noticed and seeking medical advice).
  - The total testosterone level is significantly elevated (greater than 5 nmol/l or more than twice the upper limit of normal reference range).

After investigations and the exclusion of differential diagnoses:

- **Diagnose PCOS in adults if two of three of the following criteria are present:**
  - Infrequent or no ovulation (usually manifested as infrequent or no menstruation).
  - Clinical and/or biochemical signs of hyperandrogenism (such as hirsutism, acne, or elevated levels of total or free testosterone).
  - Polycystic ovaries on ultrasound scan, defined as the presence of 12 or more follicles (measuring 2–9 mm in diameter) in one or both ovaries and/or increased ovarian volume (greater than 10 cm3).
- **In adolescent girls, both hyperandrogenism and irregular menstrual cycles are required for a diagnosis of PCOS.**
  - Great caution should be taken before diagnosing PCOS if there are clinical features of androgen excess (such as hirsutism and biochemical hyperandrogenism) without irregular menstrual cycles.
  - Adolescents who have features of PCOS but do not meet the diagnostic criteria should be considered to be at 'increased risk' of PCOS and reassessed at or before full reproductive maturity (that is, 8 years post-menarche). This includes those with PCOS features before combined oral contraceptive pill commencement, those with persisting features, and those with significant weight gain in adolescence.

DIFFERENTIAL DIAGNOSES

The diagnosis of polycystic ovary syndrome involves the exclusion of all of the following disorders, which may have a similar clinical presentation:

- Simple obesity
  - Raised androgen levels with or without symptoms.
  - Oligomenorrhoea or amenorrhoea is not often present.
- Primary hypothyroidism
  - Normal or mildly raised androgen levels with or without symptoms.
  - Oligomenorrhoea or amenorrhoea may be present.
  - Goitre may be present.
  - There may be elevated thyroid-stimulating hormone levels and subnormal plasma thyroxine levels.
  - Prolactin levels may be raised.
- Premature ovarian failure
  - Normal androgen levels.
  - Oligomenorrhoea or amenorrhoea is present.
  - May be associated with other autoimmune endocrinopathies.
  - Elevated plasma follicle-stimulating hormone levels and normal or subnormal estradiol levels.
- Raised prolactin or prolactinoma
  - Normal or mildly raised androgen levels with or without symptoms.
  - Oligomenorrhoea or amenorrhoea is present.
  - Galactorrhoea.
  - Elevated plasma prolactin levels.
- Non-classic congenital adrenal hyperplasia due to deficiency of 21-hydroxylase
  - Raised androgen levels with or without symptoms.
  - Oligomenorrhoea or amenorrhoea is not often present.
  - There is a family history of infertility, hirsutism, or both, common in Ashkenazi Jewish people.

    - Elevated basal 17-hydroxyprogesterone levels in the morning or on stimulation.
- Cushing's syndrome
    - Raised androgen levels with or without symptoms.
    - Oligomenorrhoea or amenorrhoea is present.
    - Symptoms include hypertension, striae, and easy bruising.
    - Elevated 24-hour urinary free cortisol levels.
- Androgen-secreting tumour (virilizing adrenal or ovarian neoplasm)
    - Extremely high plasma androgen levels with or without symptoms.
    - Oligomenorrhoea or amenorrhoea is present.
    - Symptoms include clitoromegaly, extreme hirsutism, and male pattern alopecia.
- Acromegaly
    - Normal or mildly raised androgen levels with or without symptoms.
    - Oligomenorrhoea or amenorrhoea is often present.
    - Symptoms include enlargement of the extremities, coarse features, and prognathism.
    - Increased plasma insulin-like growth factor levels.

Also consider:

- Hypogonadotropic hypogonadism (that is, central origin of ovarian dysfunction).
- Hyperandrogenic-insulin resistant-acanthosis nigricans (HAIRAN) syndrome.
- High-dose exogenous androgens.
- Drug-related causes.
    - Androgenic drugs that may cause hirsutism include testosterone, danazol, gestrinone, adrenocorticotropic hormone, high-dose corticosteroids, androgenic progestogens in oral contraceptives, and anabolic steroids.
    - Non-androgenic drugs that may cause hirsutism include ciclosporin, diazoxide, minoxidil, phenytoin, and, rarely, carbamazepine, sodium valproate, and acetazolamide.
    - Drugs that may cause hypertrichosis include ciclosporin, diazoxide, minoxidil, and phenytoin.
- Other causes of amenorrhoea, such as physiological causes (including pregnancy, lactation, and menopause) and hypothalamic dysfunction (by weight loss, excessive exercise, or chronic systemic illness).

MANAGEMENT

ADULTS:

- **Inform the woman about the possible long-term complications of polycystic ovary syndrome (PCOS), including type 2 diabetes and cardiovascular disease (CVD).**
    - Encourage a healthy lifestyle and optimal weight management to reduce the risk of complications and help improve the clinical features of PCOS.
- **Manage the clinical features of PCOS, including acne, hirsutism, and menstrual irregularity (oligomenorrhoea or amenorrhoea).**
- **Assess for (and manage):**
    - Cardiovascular disease risk factors, such as hypertension, obesity, and type 2 diabetes. See the section on Managing CVD risk factors for more information.
    - Other possible complications of PCOS, such as infertility, obstructive sleep apnoea syndrome, and psychological disorders.

For women who are not pregnant or planning a pregnancy:

- **Encourage healthy lifestyle behaviours, including healthy eating and regular exercise.**
  - Advise that healthy lifestyle behaviours will:
    - Help achieve and/or maintain a healthy weight.
    - Optimize hormonal outcomes.
    - Improve general health and quality of life.
  - If the woman is overweight or obese, explain that weight loss may:
    - Reduce hyperinsulinism and hyperandrogenism.
    - Reduce the risk of type 2 diabetes and CVD.
    - Result in menstrual regularity.
- **Offer a combined oral contraceptive (COC) pill, provided there are no contraindications (off-label use).**
  - COCs with 35 micrograms of ethinyloestradiol plus cyproterone acetate preparations should not be considered first line in polycystic ovary syndrome (PCOS) due to adverse effects, including venous thromboembolic risks.
- **For women with acne,** consider adding a topical-retinoids, topical antibiotics, and/or oral antibiotics as appropriate.
- **For women with hirsutism,** discuss methods of hair reduction and removal (such as shaving and waxing), as these will remain an important part of management.
- **For women with prolonged amenorrhea** (less than one period every three months) or abnormal vaginal bleeding:
  - Prescribe a cyclical progestogen (such as medroxyprogesterone 10 mg daily for 14 days) to induce a withdrawal bleed, then refer for a transvaginal ultrasound to assess endometrial thickness.
  - If endometrial thickening is present (greater than 10 mm) or the endometrium has an unusual appearance, refer for endometrial sampling to exclude endometrial hyperplasia or cancer.
  - If the endometrium is of normal thickness and appearance, advise treatment to prevent endometrial hyperplasia. The choice of treatment depends on factors such as whether the woman wishes to have regular withdrawal bleeds (at least every 3–4 months), whether she has acne or hirsutism, and whether there are any contraindications to treatment.
  - Options include:
    - A cyclical progestogen, such as medroxyprogesterone 10 mg daily for 14 days every 1–3 months.
    - A low-dose COC.
    - The levonorgestrel-releasing intrauterine device (LNG-IUD).
  - If the woman is unwilling to take cyclical hormone treatment or use the LNG-IUD, seek specialist advice or refer.
    - Regular ultrasonography is likely to be required (for example every 6–12 months) to assess endometrial thickness and morphology.
- **Metformin has been used off label to treat PCOS**. However, there is uncertainty regarding the relative benefits compared with COC.
  - Consider seeking specialist advice before initiating metformin in primary care for women without diabetes.
  - The International PCOS Network makes the following recommendations on the use of metformin for non-fertility indications:
    - Metformin (alone or in combination with COC) may offer greater benefit in high metabolic risk groups, including those with diabetes risk factors, impaired glucose tolerance, or high-risk ethnic groups.

- In addition to lifestyle measures, metformin should be considered in women with a body mass index (BMI) of 25kg/m2 or more for the management of weight and metabolic outcomes.
- In addition to lifestyle measures, metformin could be recommended for the treatment of weight, hormonal, and metabolic outcomes.
- In combination with COC, metformin should be considered for management of metabolic features where COC and lifestyle changes do not achieve desired goals.

➢ If metformin is initiated in primary care:
- Explain that it is an off-label use.
- Advise that adverse effects, such as gastrointestinal symptoms and reduced vitamin B12 levels, may occur.
- Explain that metformin use appears safe long-term, based on use in other populations. However, ongoing requirement needs to be considered.

How should I manage cardiovascular disease risk factors in women with polycystic ovary syndrome?

❖ **Body mass index (BMI)**
- ➢ Offer all women with polycystic ovary syndrome (PCOS) regular monitoring for weight change and excess weight. Monitoring could be at each visit or at a minimum of 6–12 monthly, with frequency, planned and agreed with the woman.
- ➢ At each visit, measure the woman's weight, height, and waist circumference, and calculate her BMI.
- ➢ For women who are overweight or obese, explain that weight loss may:
  - Reduce hyperinsulinism and hyperandrogenism.
  - Reduce the risk of type 2 diabetes and CVD.
  - Result in menstrual regularity.
  - Improve the chance of pregnancy (if it is desired).

❖ **Diet and physical activity**
- ➢ Encourage healthy lifestyle behaviours, including healthy eating and regular exercise.
- ➢ Advise that healthy lifestyle behaviours will:
  - Help achieve and/or maintain a healthy weight.
  - Optimize hormonal outcomes.
  - Improve general health and quality of life.

❖ **Blood pressure**
- ➢ Offer routine blood pressure checks (annually, or more frequently based on global cardiovascular disease [CVD] risk).

❖ **Lipid levels**
- ➢ Offer all overweight and obese women with PCOS, regardless of age, a fasting lipid profile (cholesterol, low density lipoprotein cholesterol, high density lipoprotein cholesterol, and triglyceride level) at diagnosis. Thereafter, the frequency of measurement should be based on the presence of hyperlipidaemia and global CVD risk.
- ➢ Refer to a specialist if dyslipidaemia requires treatment.

❖ **Glycaemic status**
- ➢ Assess glycaemic status at baseline in all women with PCOS. Thereafter, assessment should be every 1–3 years, depending on the presence of other diabetes risk factors.

- Perform a 2-hour post 75 g oral glucose tolerance test (OGTT), fasting plasma glucose, or HbA1c to assess glycaemic status.
- In high-risk women with PCOS, an OGTT is recommended. This includes:
  - Women with BMI of 25 kg/m2 or more (or in Asians 23 kg/m2 or more).
  - Women with additional risk factors, such as hypertension, advanced age (older than 40 years of age), history of impaired fasting glucose, impaired glucose tolerance, gestational diabetes, or family history of type 2 diabetes.
- Offer an annual OGTT to women with one of the following:
  - Impaired fasting glucose (fasting plasma glucose level from 6.1 mmol/l to 6.9 mmol/l).
  - Impaired glucose tolerance (plasma glucose of 7.8 mmol/l or more but less than 11.1 mmol/l after a 2-hour OGTT).

- **Check the woman's smoking status.**
  - If appropriate, offer advice on smoking cessation.

How should I manage other possible complications of polycystic ovary syndrome?

- **Obstructive sleep apnoea syndrome (OSAS) as mentioned before**
- **Psychological disorders**
  - Screen for depression and anxiety, where appropriate.
  - Consider the possibility of the following psychological disorders:
    - Psychosexual problems — hirsutism, obesity, oligomenorrhoea, and infertility may lead to feelings of being unattractive and loss of feminine identity.
    - Negative body image — women with polycystic ovary syndrome (PCOS) may feel less physically attractive, physically fit or healthy and may be less satisfied with their body size than women without PCOS.
    - Eating disorders, such as anorexia nervosa, bulimia nervosa, binge-eating disorder, and atypical eating disorders.
    - Adverse impact of PCOS on quality of life.
- **Pregnancy complications**
  - If a woman with PCOS is pregnant or considering pregnancy, offer referral for a 75-g oral glucose tolerance test (OGTT).
    - If this is not performed preconception, it should be offered before 20 weeks, gestation.
    - All pregnant women with PCOS should be offered an OGTT at 24–28 weeks, gestation.
  - Consider whether any changes to drug treatment(s) should be made. For example, any hormonal treatment (such as medroxyprogesterone for inducing cyclical bleeding) should be stopped.
- **Cancer**
  - Have a low threshold for investigation of endometrial cancer in women with PCOS or a history of PCOS.
  - Transvaginal ultrasound and/or endometrial biopsy is recommended for women with persistent thickened endometrium and/or risk factors, including prolonged amenorrhea, abnormal vaginal bleeding, or excess weight.
  - Optimal prevention for endometrial hyperplasia and endometrial cancer is not known.
    - A pragmatic approach could include combined oral contraceptive (COC) or progestin treatment in women with cycles longer than 90 days.

Routine ultrasound screening of endometrial thickness in PCOS is not recommended.

TOPIC: URINARY INCONTINENCE

DEFINITION:

- Urinary incontinence is defined as any involuntary leakage of urine.
- The main **types** of urinary incontinence are:
  - **Stress urinary incontinence**
    - involuntary leakage on effort or exertion, or on sneezing or coughing.
  - **Urgency urinary incontinence (UUI)**
    - involuntary leakage accompanied by, or immediately preceded by, a sudden compelling desire to pass urine which is difficult to defer (urgency). UUI is part of a larger symptom complex known as overactive bladder (OAB) syndrome.
    - OAB is defined as urinary urgency (with or without urgency incontinence) which is usually associated with increased frequency and nocturia. OAB may be further described as either, 'OAB wet' where incontinence is present, or 'OAB dry' where incontinence is absent. Symptoms usually occur without a urinary tract infection, or other obvious pathology.
  - **Mixed urinary incontinence**
    - both stress and urgency incontinence; involuntary leakage is associated with both urgency and physical stress (exertion, effort, sneezing, or coughing)
  - **Overflow incontinence**
    - detrusor underactivity or bladder outlet obstruction results in urinary retention and leakage of urine. There may be straining to urinate or the person may feel the bladder has been incompletely emptied
- Continuous urinary incontinence describes constant loss of urine.
  - This may be because of the severity of the person's condition (for example overflow incontinence) or due to an underlying cause such as a fistula.
- Urinary incontinence may also be situational, for example urinary incontinence that occurs during sexual intercourse or giggling

CAUSES:

Stress incontinence

- There is no single cause for stress urinary incontinence, however, a number of risk factors predispose the person to developing stress urinary incontinence. These include:
  - Increasing age.
  - Pregnancy and vaginal delivery — muscles and connective tissue can be weakened during delivery, and damage may occur to pudendal and pelvic nerves.
  - Obesity — due to pressure on pelvic tissues and stretching and weakening of muscles and nerves from excess weight.
  - Constipation — straining may weaken pelvic floor muscles.
  - A deficiency in supporting tissues for example:
  - Prolapse — not a cause of stress urinary incontinence but may be caused by the same underlying deficiency of supporting tissues.
  - Hysterectomy — surgery may damage the pelvic floor muscles.
  - Lack of oestrogen at the menopause — oestrogens keep tissues that influence normal pressure transmission in the urethra healthy and maintain urethral secretions that help to create a 'seal'.

- Family history — women whose mother or sisters are incontinent are more likely to develop stress urinary incontinence.
- Smoking — smoking is associated with chronic cough which may contribute to stress urinary incontinence.
- Drugs — for example angiotensin-converting enzyme (ACE) inhibitors (can cause cough and worsen stress incontinence).

Urge incontinence

- Urgency urinary incontinence can be associated with overactive bladder syndrome.
  - The symptoms of an overactive bladder are thought to be caused by involuntary contractions of the detrusor muscle during the filling phase of the micturition cycle. This overactivity of the detrusor muscles can cause urgency and frequency, with or without incontinence
- Urgency incontinence is idiopathic in most women.
  - In some cases, it can be associated with systemic neurological conditions such as Parkinson's disease, multiple sclerosis, or injury to pelvic or spinal nerves
  - Comorbidities such as obesity, type 2 diabetes, and chronic urinary tract infection can increase urgency symptoms
- In addition, the adverse effects of some drugs may also cause detrusor overactivity, such as parasympathomimetics, antidepressants, and hormone replacement.
  - Diuretics can also increase urinary frequency [Tse et al, 2016].
- Urinary urgency can be exacerbated by caffeinated, acidic, or alcoholic drinks

Overflow incontinence

- Overflow incontinence occurs as a result of urinary retention which can be due to bladder outlet obstruction or detrusor underactivity.
  - Overflow incontinence is associated with incomplete bladder emptying. It is more frequently found in women with systemic neurological disease or abnormalities such as urethral obstruction
- Medications that can decrease bladder contractility, leading to retention and overflow incontinence include:
  - Angiotensin-converting enzyme (ACE) inhibitors.
  - Antidepressants.
  - Antihistamines.
  - Antimuscarinics.
  - Antiparkinsonian drugs
  - Beta-adrenergic agonists.
  - Calcium channel blockers.
  - Opioids.
  - Sedatives and hypnotics.

COMPLICATIONS

- Impairment in quality of life
  - impact on employment and leisure activities.
- Psychological problems
  - for example, depression, anxiety, embarrassment, feelings of shame, loss of self-confidence, low self-esteem, and fear that they smell of urine or might lose bladder control.

- Social isolation and avoidance of going away from home or to places where there may be difficulty accessing a bathroom.
- Sexual problems
  - urinary incontinence can result in reduced intimacy, affection, and physical proximity. Women may avoid sexual activity if they are concerned about incontinence.
- Loss of sleep
  - sleep is often affected, particularly in women with overactive bladder, for whom nocturia is a common symptom.
- Falls and fractures
  - particularly in older people.
- Financial problems
  - including cost of absorbent products and laundry.

DIAGNOSIS

1 – Assess the type of incontinence

- When taking the woman's **history:**
  - Ask:
    - If incontinence occurs when coughing, sneezing, or on effort or exertion (likely to be stress urinary incontinence), or
    - If there is sudden urgency, and if they have frequency and nocturia (likely to be urgency incontinence associated with overactive bladder syndrome).
    - If incontinence occurs about equally with physical activity and urgency (suggests mixed incontinence).
    - If incontinence occurs without physical activity or a sense of urgency (suggests a cause other than stress or urgency incontinence).
  - If the incontinence is not characterized by stress or urgency incontinence, ask about:
    - Voiding difficulty (for example straining to void, sensation of incomplete emptying) — may suggest chronic urinary retention (overflow incontinence).
    - Constant leakage of urine (may be intermittent if position dependent) — suggestive of a fistula (for example vesicovaginal).
    - Post-void dribbling, pain, urgency, frequency, recurrent urinary tract infection, vaginal discharge, and dyspareunia — consider a urethral diverticulum.
- **On examination:**
  - Perform a general examination, looking for features such as weight, abnormalities of gait, and indicators of neurological disease.
  - Examine the abdomen for a palpable bladder or a mass.
  - Perform a pelvic examination
    - During the pelvic examination:
      - ✓ Ask the woman to cough with a comfortably full bladder and observe the external urethral meatus for leakage (suggests that stress urinary incontinence is likely).
      - ✓ Assess pelvic muscle tone and contraction during bimanual examination by asking the woman to contract her pelvic floor muscles to squeeze the examining finger. For more information, see Assessing pelvic floor muscle contractions digitally.
- While performing the **pelvic examination**, also look for potential causes of urinary incontinence, for example:
  - Evidence of pelvic organ prolapse.

- Urethral diverticulum — a sac-like protrusion between the periurethral tissues and the anterior vaginal wall.
- Pelvic mass.
- Atrophic vaginitis.
- Assessing pelvic floor

- Assessing pelvic floor muscle contractions digitally:
  - The strength of pelvic floor muscle contractions may be assessed digitally. A modified Oxford grading system may be used to quantify the strength of the contraction.
    - 0 = no contraction. No discernible muscle contraction.
    - 1 = flicker. A flicker or pulsation is felt under the examiner's finger.
    - 2 = weak. An increase in tension is detected, without any discernible lift.
    - 3 = moderate. There is lifting of the muscle belly and also elevation of the posterior vaginal wall.
    - 4 = good. Increased tension and a good contraction elevate the posterior vaginal wall against resistance (pressure by the examining finger applied to the posterior vaginal wall).
    - 5 = strong. Strong resistance is applied to the elevation of the posterior vaginal wall. The examiner's finger is squeezed and drawn into the vagina.
- **Initial investigations**
  - bladder diaries should be completed for a minimum of 3 days
  - urine dipstick/routine and culture
  - urodynamic studies

2- Assess for underlying causes:

- Consider and look for the presence of any causes or conditions that may exacerbate urinary incontinence or overactive bladder.
  - Ask about fluid intake and the amount and type of fluids, for example, caffeine and alcohol.
  - Identify symptoms indicating an increased likelihood of a more serious diagnosis, for example, hematuria, persisting bladder or urethral pain, recurrent urinary tract infection, constant leakage (suggests a fistula).
  - Ask about any drugs the woman is taking that could be causing or exacerbating the incontinence.
    - Drugs that may be associated with urinary incontinence include alpha-1 adrenoceptor antagonists, antipsychotics, anticholinergics, anti-parkinsonism drugs, antidepressants, benzodiazepines, beta-adrenoceptor antagonists, diuretics, and hormone replacement therapy.
  - Ask about previous history of investigations and treatment, urinary tract disorders, low spinal surgery, prolapse or hysterectomy, and obstetric and gynaecological history.
  - Consider and look for neurological conditions, for example, multiple sclerosis, spinal cord injury, Parkinson's disease.
  - Consider and look for systemic disease, such as heart failure and diabetes mellitus.
  - Consider and look for cognitive impairment.
- Perform urine dipstick analysis in all women presenting with urinary incontinence to test for blood, glucose, protein, leucocytes, and nitrites.
  - If the woman has symptoms of a urinary tract infection (UTI) and dipstick analysis is positive for both leukocytes and nitrites, send a mid-stream urine (MSU) sample for culture and sensitivities and prescribe an antibiotic whilst waiting for the culture result.

3- Assess for complications

- Assess the woman's renal function
  - acute kidney injury (AKI) may be present if there is urinary obstruction and will require urgent management.
  - Assess for the presence of depression.
  - Assess the woman for risk of falls and fractures.
  - Assess the woman for insomnia.
  - Assess for the effect on the woman's quality of life:
    - Ask about the effect on her social life and sexual functioning.
  - Assess the impact of urinary incontinence on the person's family and/or carers.

4- Assess for the severity of the incontinence

- To determine how severe the urinary incontinence is:
  - Ask how often the woman is incontinent, at what times, and during which activities.
  - Ask about the use of pads (including pad size) or changing of clothing.
  - Ask the woman how often she passes urine, including at night.
  - Ask the woman to keep a bladder diary for a minimum of 3 days, making sure that variations in her usual activities (for example working and leisure days) are covered.
    - The diary should document the amount, type, and timing of fluids she drinks, voided volume, frequency of micturition, episodes of urgency, episodes of incontinence, activities causing leakage, and pad and clothing changes.

MANAGEMENT:

Stress incontinence

Referral

- **Refer urgently (within 2 weeks)** using a suspected cancer pathway referral for bladder cancer if the woman:
  - Is aged 45 years and over with:
  - Unexplained visible haematuria without urinary tract infection, or
  - Visible haematuria that is persistent or recurrent after successful treatment of urinary tract infection, or
  - Is aged 60 and over with unexplained non-visible haematuria and dysuria or a raised white cell count on a blood test.
- **Refer to an appropriate specialist** (urologist, urogynaecologist, or nephrologist), using clinical judgement to determine urgency, if there is:
  - A bladder that is palpable on abdominal or bimanual examination after voiding.
  - Voiding difficulty.
  - Persistent bladder or urethral pain (refer urgently if cancer is suspected).
  - A pelvic mass that is clinically benign.
  - Associated faecal incontinence.
  - Suspected neurological disease.
  - A history of previous incontinence surgery, pelvic cancer surgery, or previous radiation therapy.
  - Recurrent urinary tract infection — for recurrent or persistent unexplained urinary tract infection in people aged 60 years and over, consider non-urgent referral for bladder cancer.
  - Suspected urogenital fistulae.

Management:

- **Manage any reversible causes or contributing factors of stress urinary incontinence.**
- **Give lifestyle advice on:**
  - Reducing caffeine intake — this may improve symptoms of urgency and frequency but not incontinence.
  - Fluid intake — advise the woman to avoid drinking either excessive amounts, or reduced amounts, of fluid each day.
  - Weight loss if the woman's body mass index is 30 kg/m2 or greater.
  - Smoking if this is appropriate
- **Offer referral for a trial of at least 3 months' supervised pelvic floor muscle training (PFMT) to an appropriate practitioner** (such as a continence adviser, nurse specialist in urogynaecology, physiotherapist specialising in women's health).
  - A minimum of eight pelvic floor muscle contractions should be performed at least three times a day.
- **Absorbent containment products, hand-held urinals, and toileting aids should not be used as treatment for urinary incontinence, and should only be considered in the following circumstances:**
  - To cope with urinary leakage whilst awaiting assessment and treatment.
  - To contain leakage whilst awaiting response to ongoing treatment.
  - For long-term management only after treatment options have been explored — review at least once a year and assess continence, skin integrity, efficacy of the absorbent containment product, and suitability of other available treatments.
    - Review should be carried out or overseen by a healthcare professional trained in assessing continence and referring to specialist services.
- **If the above conservative treatments fail or if the woman wants further management, consider:**
  - Referring to a urogynaecologist, gynaecologist, or urologist (depending on local service provision) for assessment and further surgical management.
    - Treatment options in secondary care include colposuspension, autologous rectus fascial sling, and retropubic mid-urethral mesh sling, intramural urethral bulking agents.
  - Offering duloxetine as a second-line treatment, but only if the woman prefers drug to surgical treatment or is not suitable for surgical treatment, and is counselled about its adverse effects.

Urge incontinence

- The same referral indications as the ones underlying stress incontinence also apply to urge incontinence (mentioned above)

Management:

- Exclude or manage treatable causes of overactive bladder syndrome if possible.
- Advise the woman on **fluid intake and lifestyle measures.**
- **Do not routinely offer** absorbent containment products, hand-held urinals, and toileting aids as treatments for urinary incontinence, but they may be considered in certain circumstances.
- **Offer referral for bladder re-training** (lasting for at least 6 weeks). This may be available from the local continence nurse, continence physiotherapist, or urology clinic.
  - lasts for a minimum of 6 weeks, the idea is to gradually increase the intervals between voiding)

These methods are to help you regain bladder control. Think of it as "mind over bladder."
1. Start by going to the toilet and trying to urinate as often as your shortest voiding interval (the length of time between trips to the bathroom) based on your voiding diary. For example, go every hour if that is what your bladder diary indicates is the shortest interval of time between visits to urinate. Make these regular trips to the toilet while you are awake. You do not have to get up during the night!
2. You must try to urinate whether you feel the need or not. You must try to urinate even if you have just been incontinent.
3. If you get a strong urge to go to the bathroom before your scheduled time, use distraction or relaxation: ▪ Stop, do not run to the bathroom! ▪ Stand still or sit down if you can. ▪ RELAX. Take a deep breath and let it out slowly. ▪ Concentrate on making the urge decrease or even go away anyway you can (imagine the pressure becoming less and less). You can also try doing quick contractions of your pelvic floor muscles. ▪ DISTRACT yourself, for example by doing math problems in your head. ▪ When you feel IN CONTROL OF YOUR BLADDER, walk slowly to the bathroom, and then go.
4. Keep this schedule until you can go 1 day without urine leakage. Then, increase the time between scheduled trips to the toilet by 15 minutes. When you can go 1 day on this new schedule without urine leakage, extend the time between bathroom trips again by 15 minutes.
5. Keep this up until you can go 4 hours between trips to the toilet (which is NORMAL), or until you are comfortable with a shorter time interval. This may take several weeks.
6. DO NOT GET DISCOURAGED! Bladder training takes time and effort, but it is an effective way to get rid of incontinence without medication or surgery.

- **If symptoms persist despite bladder training and frequency is a troublesome symptom, encourage the woman to continue bladder training and consider adding in:**
  - An antimuscarinic (anticholinergic) drug with the lowest acquisition cost. Oxybutynin (immediate release), tolterodine (immediate release), or darifenacin (once daily preparation) can be used first-line. Do not offer flavoxate, propantheline, or imipramine.

        - Other conditions (for example poor bladder emptying, cognitive impairment, dementia) — do not offer immediate-release oxybutynin to older women who may have an increased risk of sudden physical or mental health deterioration. Offer a transdermal overactive bladder treatment for women who are unable to tolerate oral medication.
        - Other medications affecting anticholinergic load.
        - Risk of adverse effects (for example cognitive impairment).
    - Mirabegron — if an antimuscarinic drug is contraindicated (depending on local prescribing policy).
- **Before starting drug treatments, explain:**
    - The medicine may take time to work (at least 4 weeks) and symptoms may continue to improve.
    - Likely adverse effects — some adverse effects (such as dry mouth and constipation) may indicate that anticholinergic medicine is starting to work.
    - The long-term effects of anticholinergic drugs on cognitive function are uncertain for women using them for overactive bladder.
    - The likelihood of treatment success.
- **Review the woman after 4 weeks of drug treatment** (sooner if drug adverse effects are intolerable). If the first-line drug treatment is not effective or tolerated, treatment options include:
    - Referral to secondary care, if the woman does not want to try another drug but would like to consider further treatment.
    - Changing the dose, if possible (for more information, see the antimuscarinic prescribing information on Initiation and titration) and reviewing after a further 4 weeks.
    - Offering an alternative antimuscarinic drug and reviewing after a further 4 weeks (do not offer flavoxate, propantheline, or imipramine).
    - Considering another medication such as mirabegron (depending on local prescribing policy) and reviewing after a further 4 weeks.
- **If treatment is effective and the woman wishes to continue treatment, review every 12 months.**
    - More frequent review is required (every 6 months) if the woman is older than 75 years of age.
- In addition, **if the woman:**
    - Is post-menopausal and has vaginal atrophy, consider intravaginal oestrogen therapy.
        - Review at least annually to re-assess the need for continued treatment and to monitor for symptoms of endometrial hyperplasia or carcinoma in women with a uterus.
    - Has troublesome nocturia, consider prescribing desmopressin (off-label indication). Avoid in women aged 65 years or over with cardiovascular disease or hypertension.
    - If conservative treatment options fail, consider referral for specialist urological assessment and management.
        - Treatment options in secondary care include injection of botulinum toxin type A into the bladder wall, percutaneous sacral nerve stimulation, augmentation cystoplasty, and urinary diversion.
- **Lifestyle and diet advice:**
    - Lifestyle advice for women with urgency urinary incontinence includes:
        - Reducing caffeine intake — this may improve symptoms of urgency and frequency but not incontinence.

- Fluid intake — advise the woman to avoid drinking either excessive amounts, or reduced amounts, of fluid each day.
- Weight loss if the woman's body mass index is 30 kg/m2 or greater.
- Smoking if this is appropriate. For more information,

❖ **Alternative (second-line) antimuscarinics**

➢ The National Institute for Health and Care Excellence (NICE) recommends that if a first-line medicine for overactive bladder is not effective or well tolerated, an alternative medicine with a low acquisition cost may be considered (do not offer flavoxate, propantheline, or imipramine). Options include:

- An untried first-line antimuscarinic — oxybutynin (immediate release), tolterodine (immediate release), or darifenacin (once daily preparation), or
- One of the following:
- Fesoterodine.
- Oxybutynin extended release.
- Oxybutynin transdermal — suitable treatment option for women who are unable to tolerate oral medicines.
- Propiverine or propiverine (extended release).
- Solifenacin.(vesicare 5mg)
- Tolterodine (extended release).
- Trospium or trospium (extended release).

Mixed incontinence

❖ The same referral criteria indicated above also apply to mixed incontinence.

❖ If referral is not indicated, manage the woman according to the most predominant type of urinary incontinence:

Overflow incontinence

❖ Refer to an appropriate specialist (urologist, urogynaecologist, or nephrologist), using clinical judgement to determine urgency, if there is:

➢ A bladder that is palpable on abdominal or bimanual examination after voiding, and/or

➢ History of chronic urinary retention

Urogenital fistula

❖ Refer women with suspected urogenital fistulae to an appropriate specialist (urologist, urogynaecologist), using clinical judgement to determine urgency.

TOPIC: VAGINAL DISCHARGE

Vaginal discharge is a common presenting symptom and is not always pathological

Common causes:

- Physiological
- *Candida*
- *Trichomonas vaginalis*
- Bacterial vaginosis

Less common causes:

- *Gonorrhoea*
- *Chlamydia* can cause a vaginal discharge although this is rarely the presenting symptoms
- Ectropion
- Foreign body
- Cervical cancer

Key features of the common causes are listed below

Condition	Key features
Candida	'Cottage cheese' discharge Vulvitis Itch
Trichomonas vaginalis	Offensive, yellow/green, frothy discharge Vulvovaginitis Strawberry cervix
Bacterial vaginosis	Offensive, thin, white/grey, 'fishy' discharge

DEFINITION

- **Normal physiological vaginal discharge** is a white or clear, non-offensive discharge that can vary over time. For example:
  - It is thick and sticky for most of the menstrual cycle but becomes clearer, wetter, and stretchy for a short period around the time of ovulation.
  - It is heavier and more noticeable during pregnancy, with contraceptive use, and with sexual stimulation.
  - It decreases in volume at menopause due to a fall in oestrogen levels.
- **Abnormal vaginal discharge** is characterized by a change of colour, consistency, volume, and/or odour, and may be associated with symptoms such as itch, soreness, dysuria, pelvic pain, or intermenstrual or post-coital bleeding. It is most commonly caused by infection; however, there can be non-infective causes.

CAUSES

- Infections are the most common cause of abnormal vaginal discharge. This includes:

- Vaginal infections:
  - **Bacterial vaginosis** (the most common cause) — caused by an overgrowth of anaerobic bacteria, particularly Gardnerella vaginalis.
  - **Vaginal candidiasis** — caused by fungal infection with Candida albicans.
  - **Trichomoniasis** (less common) — a sexually transmitted infection (STI) caused by the protozoan Trichomonas vaginalis.
- Endocervical infections caused by sexual transmission of chlamydia (caused by the obligate intracellular bacterium Chlamydia trachomatis, common) or gonorrhoea (caused by Neisseria gonorrhoeae, less common).
  - The infection may remain localized causing cervicitis, or ascend to the upper genital tract causing pelvic inflammatory disease (PID).
  - Cervicitis or pelvic inflammatory disease (PID) is characterized by vaginal discharge associated with post-coital or intermenstrual bleeding, dysuria, deep dyspareunia, or lower abdominal pain.

- Non-infective causes:
  - A retained foreign body, such as a tampon, condom, or vaginal sponge.
  - Inflammation due to allergy or irritation caused by substances such as deodorants, lubricants, and disinfectants.
  - Tumours of the vulva, vagina, cervix, and endometrium.
  - Atrophic vaginitis in post-menopausal women.
  - Cervical ectopy or polyps.
  - Fistulae.
  - Recent childbirth (with perineal, vaginal lacerations, or episiotomy) or vaginal surgery — vaginal discharge may result from granulation tissue or from a surgical site infection.
  - Physiological causes — the nature of physiological discharge can vary over time. For example:
    - It is thick and sticky for most of the menstrual cycle but becomes clearer, wetter, and stretchy for a short period around the time of ovulation.
    - It is heavier and more noticeable during pregnancy, with contraceptive use, and with sexual stimulation.
    - It decreases in volume at menopause due to a fall in oestrogen levels.

DIAGNOSIS

Assessment

- Take a detailed clinical history to assess whether it is a normal physiological discharge or whether an infective or non-infective cause is likely. **Ask about:**
  - The characteristics of the discharge, including the onset, duration, colour, odour, consistency, and associated symptoms.
  - Any exacerbating factors (such as after intercourse).
  - Any treatments tried (prescription or over-the-counter) and their effects.
  - The use of vaginal products, such as douches, deodorant, and vaginal washes.
  - Cyclical symptoms.
  - Past medical history.
  - Drug history, including contraceptive use.
- Assess the woman's risk of sexually transmitted infection (STI). Women are considered at increased risk of an STI if they are:
  - Younger than 25 years of age, or
  - Have had a new sexual partner or more than 1 sexual parner in the last 12 months, or

    - Have had a previous STI.
- Offer examination and investigations to women with an abnormal vaginal discharge.
    - Consider referring women at high risk of an STI or with characteristic features of trichomoniasis, cervicitis, or PID to a genito-urinary medicine (GUM) clinic or other local specialist sexual health service to facilitate screening for infections and partner notification.
- Examination may be omitted and empirical treatment started for women with:
    - Characteristic symptoms of BV if all the following apply:
        - The woman is at low risk of an STI.
        - The woman does not have symptoms of other conditions causing vaginal discharge.
        - Symptoms have not developed pre or post a gynaecological procedure.
        - The woman is not postnatal or post miscarriage.
        - The woman is not pre or post termination.
        - This is a first episode of suspected BV, or if recurrent, a previous episode of recognizably similar symptoms was previously diagnosed to be BV following examination.
        - The woman is not pregnant.
    - Characteristic symptoms of vaginal candidiasis if all the following apply:
        - The woman is at low risk of an STI.
        - The woman does not have symptoms of other conditions causing vaginal discharge.
        - Symptoms have not developed following a gynaecological procedure.
        - This is a first episode of suspected vaginal candidiasis, or if recurrent, recurrence is infrequent (less than four times a year).

Examination & investigations

- If indicated, perform the following examination and investigations (obtain consent and offer a chaperone).
    - Palpate the abdomen to assess for tenderness or a mass (which may indicate malignancy).
    - Inspect the vulva for lesions, discharge, vulvitis, ulcers, and any other changes.
    - If the history is suggestive of pelvic inflammatory disease (PID), perform a bimanual pelvic examination to identify cervical motion tenderness, adnexal tenderness, and abnormal masses.
    - Perform a speculum examination to visualize the cervix and vagina to look for signs of cervicitis, vaginal discharge, and any possible foreign body (such as a tampon or condom).
- Look for characteristic signs that may indicate an infective cause of vaginal discharge.
    - **BV** is characterized by a thin, white/grey, homogeneous coating of the vaginal walls and vulva that has a fishy odour. There is usually no vaginal or vulval inflammation or soreness, unless there is associated candidiasis.
    - **Vaginal candidiasis** is characterized by an odourless, white, curdy discharge. Examination may reveal erythema (usually localized to the vagina and vulva, but may extend to the labia majora and perineum), vaginal fissuring and/or oedema, and excoriation of the vulva. Vulval satellite lesions of erythema may occur in addition to vaginal erythema.
    - **Trichomoniasis**, when symptomatic, is characterized by a yellow-green, frothy discharge with a fishy odour. Inflammation of the vulva and vagina, or more rarely a strawberry appearance of the cervix (cervicitis) may be observed on pelvic examination.

    - **Cervicitis** caused by chlamydia (or less commonly by gonorrhoea) is characterized by an inflamed cervix which bleeds easily and may be associated with a mucopurulent discharge.
    - **PID** caused by chlamydia (or less commonly by gonorrhoea) is characterized by lower abdominal pain, with or without fever. Cervicitis may be seen, and adnexal tenderness and cervical excitation found on bimanual palpation.
- Take a high vaginal swab (or use a self-taken low vaginal swab) for Gram staining and to exclude other causes of symptoms from all women of reproductive age with vaginal discharge if:
    - The woman is postnatal or post miscarriage, pre or post gynaecological surgery, or pre or post termination of pregnancy.
    - The woman has vaginitis without discharge.
    - Symptoms are not characteristic of BV.
    - Symptoms are recurrent (four or more cases a year).
    - It is within 3 weeks of intrauterine contraceptive insertion.
    - Previous treatments have failed.
        - To take a high vaginal swab, roll a swab anywhere on the vaginal wall to obtain discharge.
        - Place all swabs for bacterial culture in Amies transport medium with charcoal.
        - Transport samples to the laboratory as soon as possible. If there is a delay in transportation, the swab should be refrigerated at 4°C for no longer than 48 hours.
- If the woman is at high risk of an STI:
    - Test for the following:
        - Chlamydia
        - Gonorrhoea
        - Trichomoniasis
    - Offer blood tests for HIV and syphilis if testing for gonorrhoea and chlamydia.
    - Offer chlamydia screening to all sexually active people younger than 25 years of age, annually, or more frequently if they have changed their partner.
- Consider the following additional tests, depending on the history:
    - Urine pregnancy test — to exclude pregnancy.
    - Urine dipstick — to exclude a urinary tract infection.

MANAGEMENT

- Consider the need for admission or referral.
    - Consider a suspected cancer pathway referral (for an appointment within 2 weeks) for women with suspected gynaecological cancer.
    - Arrange referral (for same day assessment) to a genito-urinary medicine (GUM) clinic for investigation and management for women with suspected pelvic inflammatory disease (PID).
        - Arrange urgent admission for women with suspected PID who are pregnant or have severe symptoms and signs (such as nausea, vomiting, and a fever greater than 38°C).
    - Consider referral to GUM clinic for further investigation if:
        - Chlamydia, gonorrhoea, or trichomoniasis is suspected or confirmed (trichomoniasis should always be managed in GUM clinic).
        - Symptoms are recurrent or persistent.
        - There is doubt about the diagnosis.

    - Arrange referral to a GUM clinic for partner notification for women with microbiologically confirmed:
        - Gonorrhoea.
        - Chlamydia.
        - Trichomoniasis.
- If admission or referral is not indicated, manage the woman in primary care.
    - Manage infective causes of abnormal vaginal discharge.
        - For women with suspected bacterial vaginosis or vaginal candidiasis, prescribe empirical antibiotic treatment.
        - For women with cervicitis, treat for chlamydia while awaiting swab results.
        - For women with suspected PID if same day treatment in a GUM clinic is not possible, prescribe empirical antibiotic treatment.
        - Note that empirical antibiotics while awaiting swab results are not required for women at increased risk of a sexually transmitted infection (STI), if they do not have any clinical features suggestive of infection.
    - Manage non-infective causes of vaginal discharge, where possible, or refer appropriately.
    - For women with no identifiable infective or non-infective causes of vaginal discharge, and no clinical clues from history and examination, reassure that the discharge is probably physiological.
    - Advise women on:
        - Basic personal hygiene.
        - Avoidance of feminine hygiene products (such as douches and vaginal washes) and tight, synthetic clothing.
        - Safe sexual behaviours and practices.

TOPIC: TRICHOMONIASIS

- **Trichomoniasis is a sexually transmitted infection (STI) caused by the flagellated protozoan *Trichomonas vaginalis*.**
- In adults, transmission of *T. Vaginalis* is almost exclusively through sexual intercourse. Vertical transmission can occur from an infected mother to baby during vaginal delivery.
- In women, the organism is found in the vagina, urethra, and paraurethral glands.
- In men, the organism is usually located in the urethra but has also been isolated from the subpreputial sac and from lesions on the penis.

What are the complications?

- **Trichomoniasis may be associated with the following complications:**
  - **In women:**
    - Perinatal complications (preterm delivery and/or low birthweight infant).
    - Predisposition to maternal postpartum sepsis.
    - Facilitation of HIV transmission, and there is some evidence that there may be an increased risk of *Trichomonas vaginalis* infection in people who are HIV positive.
    - Pelvic inflammatory disease.
    - Alterations to the normal vaginal flora, increasing susceptibility to bacterial vaginosis.
    - Increased risk of cervical cancer, especially in women co-infected with human papilloma virus.
    - Infertility.
  - **In men:**
    - Acute and chronic prostatitis.
    - Facilitation of HIV transmission, and there is some evidence that there may be an increased risk of *T. vaginalis* infection in people who are HIV positive.
    - Increased risk of prostate cancer.
    - Infertility.

Diagnosis:

How should I diagnosis trichomoniasis in women?

- **Take a history.**
  - Ask about the symptoms experienced (if any), including the duration, severity, and any exacerbating factors (such as after intercourse or during the menstrual cycle).
    - Up to 50% of women with trichomoniasis are asymptomatic.
    - When present, common symptoms include vaginal discharge (in up to 70% of women), vulval itching, dysuria, or offensive odour, but these are not specific for trichomoniasis. Vaginal discharge varies in consistency from thin and scanty to profuse and thick. The classic discharge is frothy and yellow-green, and occurs in 10–30% of infected women.
    - Other symptoms include vulval soreness, vulval ulceration, lower abdominal pain, and dysuria.

- Ask about:
  - Any treatments tried (prescription or over-the-counter) and their effects.
  - Medical history (past and present).
  - Drug history, including the use of oral contraceptives.
- Assess the woman's risk of sexually transmitted infection (STI). Women are considered at increased risk of an STI if they:
  - Are younger than 25 years of age, *or*
  - Have had a new sexual partner or more than 1 sexual partner in the last 12 months, *or*
  - Have had a previous STI.

- **If trichomoniasis is suspected,** refer the woman to a genito-urinary medicine (GUM) clinic or other local specialist sexual health service for confirmation of the diagnosis. If this is declined or not possible, perform the following examination and investigations (obtain consent, and offer a chaperone).
  - **Palpate the abdomen (if appropriate)** to assess for tenderness or a mass (which may indicate malignancy).
  - **Inspect the vulva** for lesions, discharge, vulvitis, ulcers, and any other changes.
  - **Perform a speculum examination (except in a pregnant woman with a low-lying placenta)** to visualize the cervix and vagina to look for characteristic signs of trichomoniasis.
    - Trichomoniasis, when symptomatic, is characterized by a yellow-green, frothy discharge with a fishy odour.
    - Inflammation of the vulva and vagina, or more rarely a strawberry appearance of the cervix (cervicitis), may be observed on pelvic examination.
    - About 5–15% of women will have no abnormalities on examination.
  - **Test the pH of the vaginal discharge** to help distinguish between trichomoniasis and other causes for symptoms.
    - Collect discharge from the lateral wall of the vagina with a swab, and rub it onto narrow-range pH paper.
    - Measure the pH by comparing the colour of the moist test section of pH paper against the graded standard.
    - The normal vaginal pH in a woman of childbearing age is 3.5–4.5.
    - A pH greater than 4.5 is suggestive of trichomoniasis.
  - **Take a high vaginal swab from the posterior fornix (or use a self-taken low vaginal swab)** for Gram staining and to exclude other causes of symptoms.
    - Place all swabs for bacterial culture in Amies transport medium with charcoal.
    - Mark the request as 'suspected trichomoniasis infection' if the local laboratory does not routinely perform *Trichomonas vaginalis* wet microscopy or culture.
    - Transport samples to the laboratory as soon as possible. If there is a delay in transportation, the swab should be refrigerated at 4°C for no longer than 48 hours.
  - **Offer tests for:**
    - Chlamydia.
    - Gonorrhoea.
    - HIV.
    - Syphilis.
    - Hepatitis (if the person is at high risk).
- **In women with vaginal symptoms,** consider other causes of vaginal discharge.

How should I diagnose trichomoniasis in men?

- **Take a history.**
  - Ask about the symptoms experienced (if any), including the duration, severity, and any exacerbating factors (such as after intercourse).
    - About 15–50% of men are asymptomatic and usually present as sexual partners of infected women.
    - When present, the most common symptoms are urethral discharge and/or dysuria. Urethral discharge is present in about 20–60% of men and is usually in small or moderate amounts.
    - Other symptoms include urinary frequency and urethral irritation.
    - Rarely, the man may complain of a copious purulent urethral discharge, or complications such as prostatitis.
  - Ask about:
    - Any treatments tried (prescription or over-the-counter) and their effects.
    - Medical history (past and present).
    - Drug history.
- **If trichomoniasis is suspected,** refer the man to a genito-urinary medicine (GUM) clinic.
  - **Arrange a urethral swab and/or first-void urine** for culture and/or microscopy (depending on local availability).
    - In most men, there may be no signs, even in the presence of symptoms suggesting urethritis.
    - Rarely, the man may present with balanitis/balanoposthitis (inflammation of the glans penis and prepuce).
- **In men with urethral symptoms,** consider other causes of urethritis.

Management:

From age 13 years onwards

How should I manage a person with confirmed trichomoniasis?

Ideally, treatment of confirmed trichomoniasis should be provided by a genito-urinary medicine (GUM) clinic

- **Treat the infection.**
  - For men and women (not pregnant or breastfeeding):
    - Prescribe oral metronidazole 400–500 mg twice a day for 5–7 days, *or* metronidazole 2 g as a single oral dose, *or* tinidazole 2 g as a single oral dose.
    - Seek advice from a GUM specialist if the person has a confirmed metronidazole and tinidazole allergy.
  - For breastfeeding women and symptomatic pregnant women:
    - Prescribe oral metronidazole 400–500 mg twice a day for 5–7 days.
    - Where possible, inform other healthcare professionals involved in the woman's care (such as the woman's midwife and obstetrician) of the diagnosis and treatment.
    - Seek specialist advice from a GUM specialist if the woman is unable or unwilling to take metronidazole.
    - High-dose metronidazole (2 g single dose) and tinidazole are not recommended during pregnancy and breastfeeding.
  - For asymptomatic pregnant women:
    - Seek specialist advice from a GUM specialist.

    - For people with HIV:
        - Prescribe oral metronidazole 500 mg twice a day for 7 days.
        - Seek advice from a GUM specialist if the person has a confirmed metronidazole allergy.
    - **Treat current partner(s) simultaneously, and also treat any partner(s) from within the 4-week period prior to presentation.**
- **Offer screening for other STIs (if not already done).**
    - Also screen the person's current partner(s) and partner(s) from within the 4-week period prior to presentation.
- **Arrange follow up after treatment to:**
    - Review the person for persistent symptoms, which may indicate treatment failure.
    - Confirm that contact tracing has been carried out.
    - Discuss the results of the STI screen.
- **Advise sexual abstinence** for at least one week and until the person and partner(s) have completed treatment and follow up.
- **Do not routinely test to confirm cure.**

How should I manage treatment failure?

- **If symptoms persist or recur after the** first line treatment **is completed,** the person should ideally be treated by a genito-urinary medicine (GUM) clinic or other local specialist sexual health service. If this is declined or not possible:
    - **Check adherence to the first-line treatment**.
    - **Exclude the possibility of reinfection.**
        - Check that current partner(s) have been treated appropriately and simultaneously.
    - **Reconsider the diagnosis.**
        - Ensure that other causes of symptoms have been excluded.
        - Offer repeat testing.
    - **Manage the infection.**
        - For men and women (not pregnant or breastfeeding), repeat the 7-day course of metronidazole treatment (400–500 mg twice a day).
        - For pregnant or breastfeeding women, discuss with a GUM specialist or the woman's obstetrician before prescribing any further treatment.
    - **Seek advice from a GUM specialist:**
        - If the second-line treatment regimen fails.
        - For people with failed first-line single-dose treatment.

Managing people who do not respond to the first-line treatment

- Evidence identified by BASHH showed that 40% of non-responders to metronidazole 400–500 mg twice daily for 5–7 days responded to a repeat course.
- A 7-day (rather than 5-day) course is specified as second-line treatment.

When to seek specialist advice

- For people who do not respond to the second-line treatment, BASHH recommends higher dose treatment (metronidazole or tinidazole 2 g daily for 5–7 days, or metronidazole 800 mg three times daily for 7 days) based on evidence that 70% of people responded to higher doses of metronidazole.

TOPIC: BACTERIAL VAGINOSIS

DEFINITION

- **Bacterial vaginosis (BV)** is a dysbiosis of the vagina, characterized by an overgrowth of predominantly anaerobic organisms (such as Gardnerella vaginalis, Prevotella species, Mycoplasma hominis, and Mobiluncus species) and a loss of lactobacilli.
    - The vagina loses its normal acidity, and vaginal pH increases to greater than 4.5.
- BV is not generally regarded as a sexually transmitted infection; however, the prevalence is higher amongst sexually active women (especially women with concurrent STIs).

CONTRIBUTING FACTORS

- Factors that **INCREASE** BV risk:
    - Being sexually active
    - Having multiple male sexual partners.
    - Sexual relationships with more than one person.
    - Recent change in sexual partner.
    - Certain sexual behaviours may increase risk such as anal and oral intercourse followed by vaginal penetration, and in women who have sex with women, receptive oral sex, and the use and sharing of unwashed sex toys
    - Douching.
    - Seropositivity for herpes simplex virus 2.
    - Presence of a copper intrauterine contraceptive device.
    - Cigarette smoking.
    - Genital hygiene (not washing vaginal region, infrequent change of underwear)
    - Ethnicity — BV is more prevalent in Black women (45–55%) than in Caucasian women (5–15%)
- Factors which **REDUCE** BV risk:
    - Circumcised partner.
    - Consistent condom use.
    - Hormonal contraception

PREVALENCE

- BV is the most common cause of abnormal vaginal discharge in women of childbearing age, but may also be encountered in post-menopausal women

COMPLICATIONS

- Increased risk of acquiring sexually transmitted infections (STIs) and HIV
- Obstetric and gynecologic complications, including;
    - Late miscarriage.
    - Pre-term labour and delivery.
    - Pre-term premature rupture of membranes.
    - Spontaneous abortion.
    - Low birthweight baby.
    - Postpartum endometritis.
    - Post caesarean delivery wound infections.
    - Post-surgical infections.
    - Subclinical PID.

DIAGNOSIS

HISTORY

- Symptoms experienced:
    - BV is characterized by a fishy-smelling, thin, grey/white homogeneous discharge that is not associated with itching or soreness.
    - Check for associated symptoms which might indicate the presence of other pathology (such as dysuria, genital skin problems, abnormal vaginal bleeding, dyspareunia, and abdominal pain.)
    - Ask about symptom duration, severity and exacerbating factors.
    - Approximately 50% of women with BV are asymptomatic
- Other questions:
    - Contributing factors (including vaginal products like douches, deodorant, and vaginal washes, and the use of antiseptics, bubble baths, or shampoos in the bath.)
    - Medical history
    - Medication history (including the use of oral or intra-uterine contraceptives and any treatments trialed)
    - Possibility of pregnancy.
    - Possibility of a foreign body (retained tampon, any other items inserted into the vagina.)
- Sexual history:
    - Sexually activity status, the gender of their partner, whether they are in a committed or casual-relationship, the time interval since previous sexual partner, and about the use of condoms.
    - Ask about symptoms or risk factors for blood-borne viruses in the partner(s) (ex: known or suspected STIs, or injecting drug use)
- Assess the woman's risk of STI.
    - Anyone who has condomless sex with new or casual partners is at risk of STI, particularly those who have frequent changes of sexual partners.
    - Women are considered at increased risk of an STI if they are:
        - Younger than 25 years of age, or
        - Have had a new sexual partner or more than 1 sexual partner in the last 12 months, or
        - Have had a previous STI.

EXAMINATION

- Offer examination to all people presenting with genital symptoms.
- Before examining:
    - Consider referring women at high risk of a STI to a genito-urinary medicine (GUM) clinic or other local specialist sexual health service instead, to facilitate screening for infections and partner notification (if necessary).
- Examination:
    - Palpate the abdomen (if appropriate) to assess for tenderness/masses
    - Inspect the vulva for lesions, discharge, vulvitis, ulcers, and any other changes.
    - Bimanual examination
        - To be considered in order to assess for cervical or adnexal tenderness, masses and to exclude a foreign body as a cause of discharge.
    - Speculum examination

        - except in a pregnant woman with a low-lying placenta to visualize the cervix and vagina to look for characteristic signs of BV.
        - BV is characterized by a thin, white/grey, homogeneous coating of the vaginal walls and vulva that has a fishy odour.
        - Whilst examining, take a sample of the discharge for investigation.
- In some cases of suspected BV, examination and investigations may be omitted and empirical treatment started without further assessment if all the following apply:
    - Symptoms are characteristic of BV
    - The woman does not have symptoms of other conditions causing vaginal discharge (such as itching, soreness, abnormal vaginal bleeding, blood-stained discharge, or pelvic pain).
    - The woman is at low risk of an STI.
    - The woman is not pregnant.
    - Symptoms have not developed pre or post a gynaecological procedure.
    - The woman is not postnatal or post miscarriage.
    - The woman is not pre or post termination.
    - This is the first episode of suspected BV, or if recurrent, a previous episode of recognizably similar symptoms was previously diagnosed to be BV following examination.

INVESTIGATIONS

- Using a speculum, take a swab of the discharge from the lateral wall of the vagina.
- Test the pH of the discharge
    - A pH greater than 4.5 is suggestive of, but is not specific to, the diagnosis of BV; raised vaginal pH can also indicate other conditions, such as trichomoniasis.
- If the diagnosis cannot be made from history, examination and pH, send a sample of the discharge to the laboratory for Gram-staining and microscopy.

MANAGEMENT

NON-PREGNANT WOMEN:

- **Asymptomatic:**
    - treatment is not usually required, other than in the context of having some gynaecological procedures.
- **Symptomatic:**
    - Advise that she should reduce exposure to contributing factors, such as smoking, vaginal douching and the use of antiseptics, bubble baths, or shampoos in the bath.
    - Prescribe oral metronidazole 400 mg twice a day for 5 to 7 days.
        - If adherence to treatment is an issue, a single oral dose of 2 g may be used, if appropriate.
        - If the woman prefers topical treatment or cannot tolerate oral metronidazole
            - Prescribe intravaginal metronidazole gel 0.75% once a day for 5 days or intravaginal clindamycin cream 2% once a day for 7 days.
        - Oral clindamycin and oral tinidazole are alternatives but are less preferred.
    - A test of cure is not required if symptoms resolve.

PERSISTENT BV:

- Check that the initial treatment was adhered to.
- Reconsider the diagnosis. Perform a speculum examination and take swabs of the discharge for pH, microscopy and culture.
- Enquire about continued exposure to contributing factors, such as smoking, vaginal douching and the use of antiseptics, bubble baths, or shampoos in the bath.
    - For persistent bacterial vaginosis (BV) in women with an intrauterine contraceptive device, consider removing the device and advising the use of an alternative form of contraception.
- Ensure that the current episode is adequately managed.
    - Prescribe an alternative treatment, for example, if a single 2 g dose of metronidazole or an intravaginal preparation was used previously, prescribe oral metronidazole 400 mg twice a day for 7 days.
- In the unlikely event that a woman with confirmed BV has not responded to a 7-day course of oral metronidazole, consider discussing with a gynaecologist or genito-urinary medicine (GUM) specialist regarding further treatment.
- Routine screening and treatment of male partners is not indicated, but consider testing and treating the female partner in a same-sex relationship.

RECURRENT BV:

- Advise the woman that recurrence of symptoms is common.
- Reconsider the diagnosis. Perform a speculum examination and take swabs for pH, microscopy and culture. Further examination and investigations may not be necessary if a previous episode of recognizably similar symptoms was previously diagnosed to be BV and:
    - Symptoms and signs cleared after antibiotic treatment.
    - Symptoms, signs, and microbiological evidence from swabs of other conditions causing vaginal discharge were absent.
    - Characteristic symptoms and signs of BV were present.
- Enquire about continued exposure to contributing factors, such as smoking, vaginal douching and the use of antiseptics, bubble baths, or shampoos in the bath.
    - For recurring BV in women with an intrauterine contraceptive device, consider removing the device and advising the use of an alternative form of contraception.
- Treat the current episode with a 7-day course of oral metronidazole 400 mg twice a day.
- Routine screening and treatment of male partners is not indicated, but consider testing and treating the female partner in a same-sex relationship.
- If the diagnosis is confirmed and symptoms recur frequently despite adequate management in primary care, and symptoms are adversely affecting the woman, consider prescribing metronidazole vaginal gel as suppressive treatment (off-label use) if experienced in treating recurrent BV in primary care. Otherwise, discuss management with a gynaecologist or GUM specialist.

PREGNANT WOMEN:

- **Asymptomatic:**
    - Discuss with the woman's obstetrician whether treatment is appropriate.
- **Symptomatic:**

- Advise that, where possible, she should reduce exposure to contributing factors, such as smoking, vaginal douching and the use of antiseptics, bubble baths, or shampoos in the bath.
- Prescribe oral metronidazole 400 mg twice a day for 5 to 7 days.
    - High-dose regimens (single oral dose of 2 g) are not recommended during pregnancy.
- If the woman prefers topical treatment or cannot tolerate oral metronidazole:
    - Prescribe intravaginal metronidazole gel 0.75% once a day for 5 days (off-label for women aged younger than 18 years) or intravaginal clindamycin cream 2% once a day for 7 days (second or third trimester only for clindamycin).
- Oral clindamycin (in the second and third trimesters) is an alternative but is less preferred.
- Repeat testing after 1 month (and offer further treatment) if the woman is still symptomatic, or if treatment has been prescribed to reduce the risk of preterm birth.

TOPIC: VAGINAL CANDIDIASIS

DEFINITION

- Vulvovaginal candidiasis (genital thrush) is a symptomatic inflammation of the vagina and/or vulva caused by a superficial fungal infection (usually yeasts that belong to the genus Candida).
  - Candida yeasts are part of the normal flora of the mucous membranes of the female genital tract, but overgrowth can cause infection.
  - It typically causes symptoms of vulval or vaginal itch and irritation, and a non-offensive vaginal discharge.
  - Other yeasts are occasionally implicated:
    - Candida albicans (most common and accounts for 80–89% of cases).
    - C. glabrata (5% of cases).
    - C. tropicalis, C. parapsilosis, C. krusei, C. kefyr, C. guilliermondii, and Saccharomyces cerevisiaea (remaining cases).
- **Acute infection** describes a first or single isolated presentation of vulvovaginal candidasis, and Candida species is usually detected by microscopy and/or culture.
- **Recurrent infection** describes four or more symptomatic episodes in one year, with at least two episodes confirmed by microscopy or culture when symptomatic (at least one must be culture). Affected women may have a:
  - Good or complete response to therapy and are asymptomatic between episodes, or
  - Poor or partial response to therapy with persistence of symptoms between treatments.
- **Treatment failure** is defined as failure of symptoms to resolve within 7–14 days of treatment

PREVALENCE

- Vulvovaginal candidiasis is very common, and up to 20% of women of reproductive age may be colonized with asymptomatic Candida species, which does not require treatment.
  - Expert opinion in a review article notes that self-diagnosis of vulvovaginal candidiasis is unreliable, and it is frequently over-diagnosed due to its presentation with common, non-specific symptoms.

RISK FACTORS

- The risk factors for developing recurrent vulvovaginal candidiasis are uncertain, and women may have no obvious predisposing or underlying condition. Possible factors include:
  - Non-compliance with treatment.
  - Recent antibiotic use within three months, causing a change in the vaginal flora.
  - Local irritants such as soaps, shampoos, shower gels/douching can predispose or exacerbate symptomatic vulvovaginal candidiasis.
  - Persistent infection with Candida.
  - Infection with non-albicans Candida species.
  - Azole-resistant Candida species (such as Candida glabrata).
  - Uncontrolled diabetes mellitus and other causes of immunosuppression such as HIV infection and long-term corticosteroid use.
  - Endogenous and exogenous oestrogen

- Up to 20% of women of reproductive age may be colonized with asymptomatic Candida species. Candida species is typically found in pubertal/post pubertal and not pre-pubertal females.
- Increases in endogenous and exogenous oestrogen (such as pregnancy, the combined oral contraceptive pill [COCP], and hormone replacement therapy may cause overgrowth and symptomatic infection.
- Asymptomatic colonisation and symptomatic Candida infection is more common in pregnancy. There is no evidence of association with low birth weight or preterm delivery.
- There is inconsistent, mixed evidence of a link between COCPs, intrauterine contraceptive devices (IUCDs), and the intrauterine system (IUS) and the development of vulvovaginal candidiasis in the literature.

COMPLICATIONS

- Treatment failure
  - occurs in up to 10–20% of women receiving imidazole treatment for acute infection
- Recurrent infection
  - women may have a poor or partial response to therapy with persistence of symptoms between treatments
- Reduced quality of life and psychosexual difficulties
  - including embarrassment, reduced libido and arousal may affect women who have recurrent infection
- Candidal balanitis
  - may occur rarely in male partners of women with vulvovaginal candidiasis. This typically presents with erythematous areas on the glans of the penis, pruritus, and/or irritation

DIAGNOSIS

Assessment

- **Ask about:**
  - The duration and severity of symptoms, such as:
    - Vulval or vaginal itching (often the defining symptom).
    - Vulval or vaginal soreness and irritation.
    - Vaginal discharge (usually white, 'cheese-like', and non-malodorous).
    - Superficial dyspareunia.
    - Dysuria (pain or discomfort during urination).
  - The frequency of symptoms:
    - An isolated episode, or
    - Recurrent symtoms, or
    - Treatment failure.
  - Any risk factors for candidiasis.
  - Any risk factors for a sexually transmitted infection (STI), if appropriate.
  - Any treatments tried, including over-the-counter treatments.
  - Risk of pregnancy and contraceptive use.
  - Other symptoms which may indicate an alternative or additional diagnosis:
    - Foul smelling or purulent discharge could indicate a bacterial infection.
    - Urinary frequency and urgency could indicate a urinary tract infection (UTI) or STI.

        - Abnormal vaginal bleeding could indicate an STI or gynaecological cancer.
        - Other recurrent infections may indicate possible immunosuppression.
- **Examination:** Consider performing an examination of the external genitalia if there is suspected acute vulvovaginal candidiasis, and routinely offer an examination if there is treatment failure or suspected recurrent infection, to exclude an alternative or additional diagnosis.
    - Note: external genital examination may be normal.
    - Signs of severe vulvovaginal candidiasis include:
        - Erythema — usually localized to the vagina and vulva, but may extend to the labia majora and perineum.
        - Vaginal fissuring and/or oedema.
        - Satellite lesions (rare; may indicate other fungal conditions or herpes simplex virus [HSV] infection), or vulval excoriation.
- **Investigations** are not routinely needed if clinical features suggest acute vulvovaginal candidiasis. Consider investigations to confirm the diagnosis and/or exclude an alternative diagnosis if there is diagnostic uncertainty, or persistent or recurrent symptoms, depending on clinical judgement.
    - Note: a self-collected low vulvovaginal swab (LVS) may be appropriate if examination of the external genitalia is not possible or needed.
    - Consider a high vaginal swab (HVS) of vaginal secretions for microscopy first-line for suspected acute vulvovaginal candidiasis.
    - Arrange a HVS of vaginal secretions for culture for suspected recurrent vulvovaginal candidiasis.
        - This may identify Candida albicans, non-albicans Candida species, or an additional infection.
        - If there is a poor or partial response to maintenance therapy, request culture with full speciation and sensitivity testing.
        - Consider arranging a self-collected vaginal swab if initial results are negative.
    - Consider vaginal pH testing of secretions, but this is not essential to make a diagnosis of Candida infection.
    - Consider a midstream sample of urine (MSU) — if a UTI is suspected.
    - Consider an HbA1c test — to exclude diabetes mellitus in severe or recurrent infection.
    - Consider a full blood count and serum ferritin level — to exclude iron deficiency anaemia.
- Consider **STI screening** — if the woman is concerned or at risk, or if there are clinical features suggesting an STI.

MANAGEMENT

ACUTE INFECTION

- Advise on self-management measures to provide symptom relief:
    - Use simple emollients as a soap substitute to wash and/or moisturize the vulval area.
    - Avoid contact with potentially irritant soap, shampoo, bubblebath, or shower gels, wipes, and daily or intermenstrual 'feminine hygiene' pad products.
    - Avoid vaginal douching.
    - Avoid wearing tight-fitting and/or non-absorbent clothing, which may irritate the area.
    - Avoid use of complementary therapies such as application of yoghurt, topical or oral probiotics, and tea tree or other essential oils.

- If a woman has uncontrolled diabetes mellitus or is otherwise immuncompromised:
  - Optimize management of any underlying conditions causing immunocompromise if possible.
- Advise on antifungal drug treatment options and preparations, depending on the woman's age, co-morbidities, personal preference, and drug cautions and contraindications.
  - **Fluconazole** 150 mg oral capsule as a single dose first-line.
  - **Clotrimazole** 500 mg intravaginal pessary as a single dose if oral therapy is contraindicated.
- If there are vulval symptoms, consider advising on use of a topical imidazole in addition to an oral or intravaginal antifungal.
  - Options include clotrimazole 1% or 2% cream applied 2–3 times a day.
  - Advise that oral fluconazole, intravaginal clotrimazole, and topical clotrimazole can be bought over-the-counter.
    - Advise that topical imidazole preparations may damage latex condoms and diaphragms.
- If there are signs of severe infection, advise to repeat antifungal drug treatment after 72 hours:
  - Prescribe fluconazole 150 mg oral capsule on day 1 and 4 first-line.
- Arrange follow-up if symptoms have not resolved within 7–14 days or if symptoms are recurrent.
  - Advise that follow-up and test of cure are not necessary if symptoms have resolved.
  - Do not routinely treat an asymptomatic male sexual partner.
- **Alternative treatment regimens:**
  - If first-line oral azole or intravaginal imidazole therapy is contraindicated or not tolerated for acute infection, alternative treatment regimens are:
  - Intravaginal creams
    - Clotrimazole 10% cream — 5 g intravaginally as a single dose at night.
    - Miconazole 2% cream — 5 g intravaginally once at night for 7 nights.
  - Intravaginal pessaries
    - Clotrimazole 200 mg pessaries (3 pessaries) — insert 1 pessary intravaginally once at night for 3 nights.
    - Econazole nitrate 150 mg single-dose pessary (for adults and children aged over 16 years) — insert 1 pessary intravaginally once at night as a single dose.
    - Econazole nitrate 150 mg pessaries (3 pessaries) — insert 1 pessary intravaginally once at night for 3 nights.
    - Miconazole nitrate 1.2 g single-dose vaginal capsules (for adults aged 18 years and older) — insert 1 capsule intravaginally once at night as a single dose.
    - Miconazole nitrate 400 mg vaginal capsule (for adults aged 18 years and older) — insert 1 capsule intravaginally once at night for 3 nights.
    - Fenticonazole 200 mg vaginal capsules (3 capsules) — insert 1 capsule intravaginally once at night for 3 nights.
    - Fenticonazole 600 mg single-dose vaginal capsules — insert 1 pessary intravaginally once at night as a single dose.
  - Oral itraconazole
    - 200 mg twice a day for 1 day.
  - For breastfeeding women

        - Do not prescribe an oral antifungal. Topical imidazoles are a safe and equally effective alternative.
    - For girls aged 12–15 years
        - Consider prescribing topical clotrimazole 1% or 2% applied 2–3 times a day, or seek specialist advice. Do not prescribe an intravaginal or oral antifungal.

Failure of acute infection treatment:

- If a woman has acute vulvovaginal candidiasis which has not responded to initial treatment within 7–14 days:
    - Check that initial treatment was used as recommended.
        - Be aware that topical treatments can cause vulvovaginal irritation, which may be mistaken for treatment failure.
    - Reassess for risk factors for persistent or recurrent infection, and manage appropriately.
    - Arrange an examination of the external genitalia (if not already performed) with high vaginal swab (HVS), and arrange other investigations to assess for an alternative or additional diagnosis, depending on clinical judgement.
    - Treat any ongoing infection, depending on the HVS result.
        - If there has been poor compliance with initial treatment, prescribe an alternative preparation (for example, prescribe oral treatment instead of intravaginal treatment).
    - Reinforce self-management advice for symptom relief.
    - Arrange specialist referral or seek specialist advice, depending on clinical judgement, if:
        - An affected young person is aged 12–15 years.
        - There is uncertainty about the diagnosis.
        - Symptoms are not improving and treatment failure is unexplained.
        - Symptoms persist after a second course of antifungal treatment.
        - The woman has uncontrolled diabetes and treatment failure.

RECURRENT INFECTION

- Reinforce advice on self-management measures to provide symptom relief.
- Reassess for risk factors for persistent or recurrent infection, and manage appropriately.
- Arrange an examination of the external genitalia with high vaginal swab (HVS), and arrange other investigations to assess for an alternative or additional diagnosis, depending on clinical judgement.
- Offer an induction regimen immediately followed by maintenance treatment, the drug treatment and preparation depending on the woman's age, co-morbidities, and personal preference, and drug cautions and contraindications.
    - For induction, prescribe three doses of oral fluconazole 150 mg (to be taken every 72 hours) first-line.
    - For maintenance, prescribe oral fluconazole 150 mg once a week for six months first-line.
    - Note: routine use of a topical imidazole in addition to an oral or intravaginal antifungal for vulval symptoms is not recommended.
- **Alternative treatment regimens:**
    - If first-line oral azole therapy is contraindicated or not tolerated for recurrent infection:

- Induction therapy
  - Topical imidazole therapy (such as clotrimazole 500 mg intravaginal pessary) can be increased to 7–14 days, depending on symptom response.
- Maintenance therapy
  - Clotrimazole 500 mg pessary inserted intravaginally once a week for six months, OR:
  - Oral itraconazole 50–100 mg daily for six months.
- For breastfeeding women
  - Do not prescribe oral antifungal treatment. Intravaginal clotrimazole may be prescribed for induction and maintenance therapy as a safe and effective alternative.

❖ **Follow up:**

- If a woman has completed an induction and maintenance regimen of treatment for recurrent vulvovaginal candidiasis:
- If there are recurrent symptoms between maintenance treatment doses, options include:
  - Oral fluconazole 150 mg twice-weekly instead of once a week, or
  - Considering the use of cetirizine 10 mg once daily for six months, particularly if there is a history of allergy (off-label indication).
- If there is a poor or partial response to maintenance therapy:
  - Check concordance with therapy.
  - Arrange a high vaginal swab (HVS) and request culture with full speciation and sensitivity testing (if not already requested), and treat accordingly.
  - If HVS culture is negative and symptoms persist, consider assessment for an alternative diagnosis.
  - If a non-albicans Candida species or azole resistant Candida is identified, seek specialist advice.
- If there are infrequent further sporadic episodes of vulvovaginal candidiasis, treat each episode as an acute infection.
- If there is further recurrent vulvovaginal candidiasis, consider the use of a repeat induction and maintenance treatment regimen if clinically indicated.
- Arrange specialist referral or seek specialist advice, depending on clinical judgement, if:
  - An affected young person is aged 12–15 years.
  - There is uncertainty about the diagnosis.
  - A non-albicans Candida species or azole resistant Candida is identified.
  - A woman has uncontrolled diabetes and recurrent infection.

INFECTION DURING PREGNANCY

❖ Advise on self-management measures to provide symptom relief:

- Use simple emollients as a soap substitute to wash and/or moisturize the vulval area.
- Avoid contact with potentially irritant soap, shampoo, bubblebath, or shower gels, wipes, and daily or intermenstrual 'feminine hygiene' pad products.
- Avoid vaginal douching.
- Avoid wearing tight-fitting and/or non-absorbent clothing, which may irritate the area.
- Avoid use of complementary therapies such as application of yoghurt, topical or oral probiotics, and tea tree or other essential oils.

- Advise on antifungal drug treatment options and preparations for symptomatic acute infection, depending on the woman's age, co-morbidities, personal preference, and drug cautions and contraindications.
    - Prescribe clotrimazole pessary 500 mg intravaginally at night for up to 7 consecutive nights first-line (if aged 16 years and older).
- Offer an induction regimen immediately followed by maintenance treatment for recurrent infection.
    - For induction, prescribe topical imidazole therapy (such as clotrimazole pessary 500 mg intravaginally at night, if aged 16 years and older) for 10–14 days according to symptom response.
    - For maintenance, prescribe one clotrimazole pessary 500 mg intravaginally at night once a week for six months (if aged 16 years and older).
- If there are vulval symptoms, consider prescribing a topical imidazole in addition to an intravaginal antifungal.
    - Options include clotrimazole 1% or 2% cream applied 2–3 times a day.
- Arrange follow-up if symptoms have not resolved within 7–14 days for acute infection.
    - Advise that follow-up and test of cure are not necessary if symptoms have resolved.

TOPIC: PRUTITIS VULVAE

DEFINITION

- Pruritus vulvae is itching of the vulva
  - (which includes the mons pubis, labia majora, labia minora, clitoris, perineum and the external openings of the urethra and vagina).
- Pruritus vulvae is a symptom which is common to many conditions.
  - One or more underlying causes can usually be identified from the history and examination.
- Pruritus vulvae should be distinguished from vulval pain and from vulvodynia, although one or more of these symptoms may co-exist.

CAUSES

- Most women who present with pruritus vulvae have an identifiable cause. These include:
  - Dermatological conditions.
  - Infections and infestations.
  - Neoplastic conditions.
  - Hormonal changes (for example atrophic vaginitis).
  - Gastrointestinal disease and urinary incontinence.
  - Neurological causes.
  - Systemic conditions.
- Idiopathic pruritus vulvae is thought to be uncommon.
- There may be more than one cause. For example, dermatological conditions and infections may co-exist.

Which dermatological conditions can cause pruritus vulvae?

- **Eczema (dermatitis).** This may be:
  - Atopic dermatitis.
  - Allergic contact dermatitis.
    - Common allergens include fragrances, topical antibiotics and anaesthetics, and components of other topical treatments.
  - Irritant contact dermatitis
    - is the most common type of eczema causing pruritus vulvae, and one of the most common causes. it is an inflammatory reaction with itch as the main symptom. Lichenification and hyperpigmentation may occur when dermatitis is chronic. It can be caused by:
      - ✓ Proprietary creams (especially those containing local anaesthetics).
      - ✓ Topical antibiotic preparations (for example, neomycin).
      - ✓ Barrier contraceptives or lubricants.
      - ✓ Perfumes, soaps, bubble baths, or wet wipes.
      - ✓ Detergents, fabric conditioners, bleaches, or dyes.
      - ✓ Urine, particularly in those with urinary incontinence.
- **Psoriasis**
  - due to the moisture and friction of skin folds, the appearance may be different to classic psoriatis with scale being less common. There are well-defined erythematous plaques, but there may be no scale, or there may be fine silvery scales. Lesions on

the outer labia majora may have thicker scales. However, typical psoriasis lesions may be present elsewhere on the body (elbows, knees, scalp).

- Genital skin is affected in up to 63% of people with psoriasis during their lifetime.
- Genital areas may be the only region affected in 2 - 5% of patients.

❖ **Seborrhoeic dermatitis**

- ➢ ill-defined border, some scaling, with or without involvement of other sites, such as the axillae, face (eyebrows or nasolabial folds), anterior chest, or scalp. This rarely affects the vulva but may affect flexures or the mons pubis.

❖ **Lichen simplex**

- ➢ a localized plaque of chronic eczematous inflammation created by repeated rubbing or scratching of the skin. It is more common in people with a personal or family history of atopy, but can be the result of any factors causing repeated scratching or rubbing.
- ➢ It can affect the entire vulva and perianal area, or part of it. It can be localized, unilateral or bilateral, and the skin may appear leathery or lichenified, or may be pale and wrinkled.
- ➢ There may be labial swelling and erythema, and some hair loss due to scratching.
- ➢ There may be change in skin pigmentation (hyper-, hypo-, or depigmented areas).

❖ **Lichen planus**

- ➢ a chronic inflammatory condition which affects the skin and /or mucosal surfaces.
- ➢ Lesions may appear as erosive, glazed or glossy erythematous plaques; bluish-purple, shiny, flat-topped papules with small white dots or lines (Wickham's striae); or uniformly white hyperkeratotic plaques.
- ➢ There may be loss of vulvar architecture.
- ➢ There is a small risk of squamous cell carcinoma developing in women with lichen planus (less than 3%).

❖ **Lichen sclerosus**

- ➢ an inflammatory skin condition which affects the anogenital area more often than other cutaneous surfaces.
- ➢ It can occur at any age, but is most often diagnosed in women aged over 50 years.
- ➢ Any area from the clitoral hood to the perianal area can be affected, but the vaginal mucosa is not affected.
- ➢ Lesions are hypopigmented-to-white, crinkled, fragile plaques classically distributed in a figure eight pattern around the vulva, perineal body, and perianal skin, although it may also be localised to the vulva alone. Bruises, blood blisters, or ulcers may appear after scratching, or from minimal friction. Purpura (ecchymosis) is common.
- ➢ Scarring may cause loss of vulvar architecture, such as resorption of the labia minora, and mid-line fusing and stenosis of the introitus. The clitoral hood may be scarred, burying the clitoris. It mainly affects the inner, non-hair bearing areas of the vulva.
- ➢ There is a small risk of squamous cell carcinoma developing in women with lichen sclerosus (less than 5%).
- ➢ Early diagnosis is important to prevent scarring and malignant change.

Which infections and infestations can cause pruritus vulvae?

❖ Candidiasis
❖ Trichomoniasis
❖ Bacterial vaginosis

- Threadworm
- Genital herpes simplex
- Pubic lice (Pediculus pubis)
- Scabies (Sarcoptes scabiei)

Which pre-malignant and malignant conditions may cause pruritus vulvae?

Vulval intraepithelial neoplasia (VIN) is a pre-malignant skin lesion of the vulva.

- There are three distinct types of VIN, which may be related to human papilloma virus (HPV) infection, or follow chronic skin conditions such as lichen sclerosis. These have differing aetiology, malignant potential and treatment. Epidemiology varies between countries, and this may partly relate to differing HPV prevalence. Women of all ages may be affected.
- VIN may be completely symptom-free.however, women may present with:
  - Vulval itching.
  - Vulval burning.
  - Vulval pain.
  - Dyspareunia.
  - Lumps or erosions.
- Clinical appearance is highly variable. There may be one or more flat or slightly raised, well-defined or irregular skin lesions that may be pink, red, brown, or white.
- If left untreated, VIN may resolve spontaneously (especially the low-grade VIN), or it may develop into an invasive cancer. This may take on average between 2-7 years, depending on the type of VIN. There is an association with other genital intraepithelial neoplasias and cancers.

Malignant neoplasms of the vulva (uncommon). Squamous cell carcinoma (SCC) accounts for 90% of malignant disease of the vulva. Other less common neoplastic conditions include perianal intraepithelial neoplasia, basal cell carcinoma, melanoma, extramammary Paget's disease, and carcinoma of Bartholin's gland.

- Squamous cell carcinomas.
  - Certain types of HPV infection, VIN, vulval lichen sclerosus, and lichen planus predispose to vulval neoplasms. Risk is also increased by smoking and immunosuppression.
  - SCC of the vulva was historically a condition of postmenopausal women, and is still more common in older women, but the average age has fallen due to an increase in HPV infections worldwide, with the incidence in younger women increasing. Vulval cancer can affect women of all ages.
  - SCC of the vulva may be asymptomatic and be found incidentally on examination, but there may be itch, burning, pain, lumps, ulceration, weeping, or bleeding.
- Extramammary Paget disease (very rare)
  - a cutaneous neoplasm with a chronic eczema-like rash of the anogenital region and vulva. A common symptom is a mild to intense itching of a lesion found around the groin, genitalia, perineum, or perianal area. Pain and bleeding may occur from scratching lesions that have been around for a long time. Thickened plaques may form that can become red, scaly, weeping, and crusty. Although they may appear similar to eczema, they fail to clear up with topical steroid creams.

What hormonal changes can cause pruritus vulvae?

Atrophic vulvovaginitis

- In peri- and postmenopausal women, declining oestrogen levels may contribute to vaginal and vulval changes that result in vulvovaginal itching, dryness, and sometimes burning. The vulvovaginal area will be dry, pale, and thin.
  - Cessation of menstruation and other symptoms, such as hot flushes, may indicate that the cause is the menopause.
  - Physiological changes of menopause alone may cause vulval itching, but conditions such as lichen sclerosus are more common in postmenopausal women, and contact dermatitis may be more problematic in this age group.
- In breastfeeding women, elevated prolactin levels can have an antagonistic effect on oestrogen production, and may result in low oestrogen levels. This can lead to vaginal dryness, itching, burning, and irritation.

Pregnancy

- Increased hormone levels in pregnancy are associated with increased physiological vaginal discharge, and an increased risk of candidal vulvovaginitis, both of which cause vulvular pruritus.

How may gastrointestinal conditions and urinary incontinence cause pruritus vulvae?

- Gastrointestinal disease
  - irritable bowel syndrome, Crohn's disease, ulcerative colitis, other inflammatory bowel disease, and anal fissures may lead to prolonged contact of stool with the vulval skin due to faecal incontinence or poor perianal hygiene.
- Urinary and faecal incontinence
  - this can result in vulvar inflammation, which presents as pruritus.

What neurological conditions can cause pruritus vulvae?

Neurological causes of itch are due to neural dysfunction, either neuropathic (injury or damage to nerve fibres) or neurogenic (changes in the stimulation pathway within the central nervous system without structural damage). Some of these may affect the vulva. Causes include:

- Nerve or nerve root compression at L4 to S2 level.
- Varicella zoster (shingles) reactivation and post-herpetic neuralgia.
- Systemic diseases which affect nerve fibres (for example, diabetes mellitus, sarcoidosis, amyloidosis, Vitamin B12 deficiency, and viral infections).
- Central nervous system conditions (such as brain tumours, brain injury, and multiple sclerosis).

What systemic conditions may cause pruritus vulvae?

- Any cause of generalized pruritus may cause vulvar pruritus, including:
  - Drug reactions, for example tetracycline antibiotics, such as doxycycline, and nonsteroidal anti-inflammatory drugs (NSAIDs), such as ibuprofen.
  - Systemic diseases, such as renal or hepatic disease, diabetes, iron deficiency anaemia, lymphoma, other haematological abnormalities, viral infections, and thyroid dysfunction.
- Psychological problems
  - may occasionally present as pruritus vulvae.

- Stress
  - may be a cause of itch, or if not an initial cause, an exacerbating factor causing prolongation of symptoms or a flare-up.

COMPLICATIONS

Complications may result from:

- **The underlying cause of the pruritus vulvae.**
  - For example, complications of lichen sclerosus include:
    - Development of squamous cell carcinoma (less than 5% risk).
    - Development of clitoral pseudo-cyst.
    - Sexual dysfunction.
    - Urinary dysfunction.
    - Dysaesthesia.
  - Some conditions predispose to neoplasia. There is a risk of development of squamous cell carcinoma in lichen sclerosus, lichen planus and vulval intraepithelial neoplasia.
- **Persistent symptoms,** as chronic itch can result in:
  - Poorer quality of life.
  - Psychological problems, such as loss of self-esteem, relationship problems, depression, and anxiety.
  - Sexual dysfunction.
  - Sleep disturbance.
  - Severe and irreversible architectural damage due to chronic, deep scratching and gouging.
  - Lichen simplex — thickened plaques with exaggerated skin markings over the labia majora as a result of an itch-scratch cycle (regardless of the initial underlying cause of the itch).
- **Secondary bacterial infection** of excoriated lesions.

DIAGNOSIS

ASSESSMENT

- **Confirm that the woman is experiencing vulval itch, not vulval pain.**
- **Take a history to find an underlying cause for the vulval itch. Ask about:**
  - Location, duration and onset of symptoms.
    - Extragenital itch may indicate a more generalized problem.
    - Lichen sclerosus is normally confined to the vulva, typically does not affect the vagina and very rarely involves the oral mucosa, while lichen planus may affect the vagina, skin, hair, nails, as well as genital and oral mucous membranes.
    - Acute onset is associated with allergic contact dermatitis, or vulvovaginal candidiasis (which are the most common causes of acute vulval itch).
    - Nocturnal vulval itching may indicate threadworm infestation (particularly if combined with perianal itch).
  - Trigger factors:
    - Sexual activity.
    - Menses.
    - Exercise.

        - Friction, heat, moisture.
    - Associated symptoms.
        - Vaginal discharge may indicate infection, such as candidiasis, bacterial vaginosis, or trichomoniasis.

    - Hygiene practices and products.
        - Creams, perfumes, soaps, or deodorants can cause irritation.
        - Excessive washing may result in vulval irritation or even contact dermatitis.
    - Prescribed and over the counter medicines.
        - Topical antifungal creams, hormone replacement therapy creams, or pessaries may cause irritation.
        - Systemic medicines may cause fixed drug reactions.
        - Spermicides, or the latex in condoms or diaphragms may cause allergic reactions in some women.
    - Medical history.
        - A personal or family history of skin disorders or atopic conditions such as hayfever, asthma, or eczema — a positive family history is observed in about 10% of patients with vulval lichen sclerosus, and up to 75% of people with lichen simplex chronicus have a personal or family history of atopy.
        - Diabetes mellitus — the risk of candidal infection is increased.
        - Systemic illness, such as renal or hepatic impairment, or anaemia can cause generalized pruritus.
        - Faecal or urinary incontinence — these can damage the vulval skin either directly, or indirectly, by the use of sanitary products, or over washing.
        - Menopause — symptoms may be caused by atrophic vaginitis.
        - Breastfeeding can result in lowered oestrogen levels and consequent vulval symptoms.
    - Sexual history (if appropriate. Consider risk factors for sexually transmitted infections).
    - Assess the severity of symptoms and the impact this is having on the woman — in particular enquire about sexual problems, low mood, sleep disturbance, and anxiety.
- **Examine the anogenital region.**
    - Explain the examination before the person undresses and establish informed consent.
    - Offer a chaperone and document this information.
    - Maintain the person's dignity. Allow them privacy to undress, and cover their skin where possible.
    - Look at skin colour and texture, look for fissuring, excoriation, erosions, ulcerations, lichenification, lesions, tumours, atrophy, scarring, and vaginal discharge.
    - Consider using a speculum to visualize the vaginal mucosa for erythema, erosions, or lesions, and to take a high vaginal swab if there is a discharge.
- **Also examine the skin and other relevant extragenital sites.**
    - For example, the elbows, knees and nails for signs of psoriasis, skin elsewhere for eczema or contact dermatitis, the oral mucosa for lichen planus.

INVESTIGATIONS

Diagnosis is often clinical and further investigation may not be necessary. Investigations which may be relevant include:

- A high vaginal swab should be taken if there is vaginal discharge, signs of infection or suspicion of candidiasis.
- Blood tests if there is generalized pruritis. For example, full blood count, ferritin, thyroid function tests, renal function tests, liver function tests, blood glucose, and HbA1c.
- Further blood tests may be relevant in a person with lichen sclerosus, to exclude autoimmune diseases (such as thyroid disease and diabetes).
- In secondary care, further tests such as vulvoscopy, biopsy, and allergy tests may be performed.

MANAGEMENT

How should I manage dermatological conditions?

Manage the underlying dermatological cause of pruritus vulvae in primary care where appropriate.

Contact dermatitis

- Identify and remove exposure to irritants. Common irritants include soaps, fragrances, condoms, lubricants and topical treatments.
- Recommend a simple emollient as a soap substitute. Barrier preparations are helpful where urinary incontinence is the cause.
- For mild itching, consider prescribing a mild potency topical corticosteroid ointment, such as hydrocortisone 1% for 7–10 days.
- Stronger potency corticosteroids (such as betamethasone valerate or mometasone) for 7–10 days may be used if symptoms are moderate or severe, if the skin is lichenified, or to break the itch-scratch-cycle. Ointments are preferable to other preparations.
- Treat co-existing infection with a combination cream corticosteroid/antifungal, or corticosteroid/antibacterial, or oral antibiotic if required.
- A sedative antihistamine such as hydroxyzine may be helpful if there is sleep disturbance, or a low dose of a tricyclic antidepressant such as amitriptyline or doxepin.
- Consider referral to dermatology for skin patch testing and further advice if avoidance of irritants and topical corticosteroids have not improved the condition.

Seborrhoeic dermatitis

- Ketoconazole cream once or twice a day for at least 4 weeks, or an antifungal shampoo such as ketoconazole can be used as body wash for seborrhoeic dermatitis.

Lichen simplex

- Identify and avoid precipitating factors.
- Identify and treat secondary or associated infection.
- Recommend an emollient as a soap substitute.
- Consider prescribing a potent or very potent topical corticosteroid ointment (such as betamethasone 0.025%, mometasone or clobetasol), for 1–2 weeks to break the itch-scratch cycle and bring the condition under control. A gradual reduction in frequency and strength over 3–4 months may be required to prevent recurrence.
- Consider prescribing a mildly anxiolytic antihistamine (for example hydroxyzine) or doxepin for use at night.

- Cognitive behavioural therapy may be helpful in some cases where there are associated mental health issues.

Lichen sclerosus and lichen planus

- Refer to secondary care to confirm the diagnosis.
- Once the diagnosis is confirmed in secondary care, very potent corticosteroids are usually initiated by a specialist. Repeated, intermittent courses of topical steroids may be required for longer-term management.

How should I manage a possible vulval neoplasm?

- **Refer urgently (within 2 weeks) all women with:**
  - An unexplained vulval lump, ulceration, or bleeding.
  - An unexplained palpable mass in, or at the entrance to the vagina.
- **If vulval intraepithelial neoplasia (VIN) is suspected**, for any persistent area of abnormality that is thick, warty, or ulcerated, referral for a biopsy should be made. This should be on an urgent or two-week basis, depending on the clinical situation and suspicion.

How should I manage gastrointestinal disease or urinary incontinence which is causing pruritus vulvae?

Manage pruritus vulvae appropriately, depending on the underlying cause:

- Faecal incontinence
  - manage faecal incontinence (for example regular toileting, use of barrier creams). Refer to a specialist if required.
- Poor perianal hygiene
  - provide advice on appropriate hygiene measures.
- Urinary incontinence
  - Where incontinence has led to contact dermatitis, treat that condition as necessary. Barrier creams are helpful in preventing this.

How should I manage pruritus vulvae with an unknown cause?

The aim of treatment is to provide symptomatic relief, reduce inflammation, restore the skin barrier and prevent and/or treat secondary infection.

- Offer symptomatic treatment with an emollient.
- Offer self-care advice.
- Consider a mildly anxiolytic antihistamine (such as hydroxyzine) or low dose of a tricyclic antidepressant (such as amitriptyline or doxepin) if sleep is affected.
- Consider prescribing a short trial (1–2 weeks) of low potency topical corticosteroids (hydrocortisone 1% ointment).
- Consider prescribing a combination corticosteroid/antifungal, or corticosteroid/antibacterial if a co-existing infection is suspected.
- If symptoms persist despite treatment with an emollient, an antihistamine or tricyclic antidepressant, and a trial of low potency topical corticosteroids, refer for further investigation to a dermatologist, gynaecologist, or vulval clinic depending on local referral pathways.
- Advise the woman to continue using an emollient while awaiting specialist appointment, and
- an antihistamine or low dose tricyclic antidepressant if these have been helpful.

What symptomatic treatment should I offer a woman with pruritus vulvae?

For all women with pruritus vulvae, consider prescribing:

- A simple emollient — advise the woman to apply the emollient directly to the vulval area regularly as well as using it as a soap substitute.
  - When used regularly (even when there are no symptoms), an emollient protects the skin against local irritants (for example, urine or menstrual blood) and may help to prevent flare-ups.

- A mildly anxiolytic antihistamine or low dose tricyclic antidepressant at night, such as hydroxyzine, amitriptyline (10mg), or doxepin (25mg) to reduce nocturnal itching.
  - Do not prescribe hydroxyzine in the elderly, if the woman has a prolonged QT interval, or risk factors for QT interval prolongation.

What self-care advice should I give to women with pruritus vulvae?

Advise the woman:

- To shower rather than bath, and to clean the vulval area (a maximum of once a day) with a simple, unscented emollient as a soap substitute — over cleaning may aggravate vulval symptoms. Once the vulval area is clean, gently dab the vulval area dry with a soft towel or use a hairdryer on a cool setting held well away from the skin.
- To keep nails short to avoid skin damage from scratching.
- Cooling of the area may be helpful to manage the itching, with a cool flannel or cool gel packs.

Advise women to avoid:

- Washing with water only, or with soap, as these cause dry skin and make itching worse.
- Contact of the vulval skin with:
  - Shampoo.
  - Bubble bath.
  - Over-the-counter preparations used on the vulva.
  - Wet wipes (feminine or baby).
  - Perfumed sanitary towels and panty liners.
  - Sponges or flannels — these may irritate the skin.
  - Antiseptics.
- Tight-fitting garments or synthetic clothes, for example nylon underwear, as these may irritate the vulval area.
- Fabric conditioner or fragranced washing powder when washing underwear.
- Use of spermicidally-lubricated condoms.
- Coloured or scented toilet paper.
- Wearing nail varnish on fingernails if they are scratching.
- Shaving the pubic hair.
- Scratching (Explain the itch-scratch cycle)

TOPIC: BARTHOLIN'S CYST

- Bartholin's cyst develops when the entrance to the Bartholin duct becomes blocked.
  - The gland continues to produce mucus which builds up behind the blockage, eventually leading to the formation of a mass.
  - The initial blockage is most commonly caused by vulvar oedema.
  - These are usually sterile.
- Bartholin's cysts are usually painless and commonly asymptomatic, often being detected at a routine pelvic examination or by the woman herself.
  - If the cyst is particularly large it may cause superficial dyspareunia and could be uncomfortable when sitting.
  - In contrast, Bartholin's abscess is extremely painful, with erythema and often gross deformity of the affected side of the vulva. Bartholin's abscess is three times more common than the cyst in terms of presentations to gynaecology but this is likely to reflect the asymptomatic nature of the cyst in a majority of cases to some extent.
- Bartholin's cysts are usually unilateral and 1-3cm in diameter - the Bartholin's glands should not be palpable in health.
- On examination, a cyst is characterised by the presence of a soft, painless lump in the labium. It is best felt between a finger at the posterior vaginal introitus and a thumb lateral to the labium.
- Risk factors for the development of Bartholin's cyst are poorly understood, but it is thought to increase in incidence with increasing age up to the menopause before decreasing.
- Asymptomatic cysts require no intervention in general, although in older women (over age 40) some gynaecologists advocate incision and drainage with biopsy in order to exclude carcinoma.
  - Cysts that are symptomatic and/or disfiguring should be treated with either incision and drainage (with/without placement of a 'word' catheter to allow continuing drainage) or with a procedure known as marsupialisation.
  - The latter involves the creation of a new orifice through which the glandular secretions may drain, by incising the gland open, everting it and suturing the epithelial lining against the skin.
  - Marsupialisation is thought to be more effective at preventing recurrence, but is a longer operation and is more invasive.
  - There is no place for antibiotic use in the setting of Bartholin's cyst with no evidence of abscess.

BARTHOLIN'S ABSCESS

- Bartholin's glands are a pair of glands located next to the entrance to the vagina.
  - These are normally about the size of a pea, but can become infected and enlarge - forming a Bartholin's abscess.
- This can be treated by antibiotics, by the insertion of a word catheter or by a surgical procedure known as marsupialization.

TOPIC: CERVICAL ECTROPIAN

- On the ectocervix there is a transformation zone where the stratified squamous epithelium meets the columnar epithelium of the cervical canal.
  - Elevated oestrogen levels (ovulatory phase, pregnancy, combined oral contraceptive pill use) result in larger area of columnar epithelium being present on the ectocervix

- The term cervical erosion is used less commonly now.

- This may result in the following features:
  - vaginal discharge
  - post-coital bleeding

- Ablative treatment (for example 'cold coagulation') is only used for troublesome symptoms

TOPIC: CERVICAL CANCER AND HPV

DEFINITION

- The cervix is the lowermost part of the uterus and is composed of the ectocervix which projects into the vagina (lined by squamous epithelium), and the endocervical canal which extends from the internal os to the external os (lined by columnar epithelium).
  - Almost all cases of cervical cancer originate from the ecto- or endocervical mucosa in the 'transformation zone' (the area of the cervix between the old and new squamocolumnar junction)
  - The main cause is persistent infection with a high-risk human papillomavirus (HPV) subtype
  - Low-grade dysplasia (cervical intraepithelial neoplasia 1 [CIN1]) usually regresses, but it can progress to high-grade dysplasia (CIN2 or 3). Cervical cancer occurs when high-grade dysplastic lesions extend beyond the basement membrane of the cervical epithelium
- There are several **types of cervical cancer**, which include:
  - **Squamous cell carcinoma** — accounts for about 70–80% of cases.
  - **Adenocarcinoma** — accounts for over 20–25% of cases.
  - **Other epithelial tumours** — including adenosquamous carcinoma, neuroendocrine tumours, and undifferentiated carcinoma (rare).

PREVALENCE

- Cervical cancer mainly affects sexually active women aged 30–45 years.

CAUSES

- The main cause of cervical cancer is persistent infection with one of about 15 high-risk subtypes of human papillomavirus (HPV), which is predominantly transmitted through sexual intercourse.
- **Human papillomavirus (HPV)**, particularly serotypes 16,18 & 33 is by far the most important factor in the development of cervical cancer.
  - HPV is detected in 99% of cervical cancer, and the high-risk oncogenic subtypes HPV 16 and 18 account for at least two-thirds of cases. HPV subtypes 31, 33, 45, 52, and 58 account for the majority of remaining cases.
- 'Persistent infection' describes the presence of the same type-specific HPV DNA on repeat sampling after 6–12 months.
  - About 90% of incident HPV infections are short-lived and resolve spontaneously within 2 years. Only about 10% of all HPV infections become persistent, which puts the woman at risk of developing precancerous cervical lesions.
- The time interval between HPV infection and development of precancerous lesions is 1–10 years, and progression to invasive carcinoma is usually more than 10 years.

RISK FACTORS

- **Risk factors for acquiring HPV infection include:**
  - Early age of first sexual intercourse.
    - Infection with at least one type of HPV occurs soon after the start of sexual activity, with nearly 40% of women infected within 2 years
  - Multiple sexual partners or a high-risk sexual partner.

- History of sexually transmitted infection (STI).
- Lack of use of barrier methods of contraception, such as condoms.
- History of HPV-related vaginal or vulval dysplasia.
- Conditions causing immunocompromise or immunosuppresion, such as solid organ transplant or HIV infection.

❖ **Factors which increase the risk of progression to cervical cancer include:**
- Co-infection with other STIs.
- Smoking — increases the risk of developing precancerous lesions and squamous cell carcinoma.
- High parity (more than five full-term births) and young age at first birth (less than 17 years of age).
- Family history in a first-degree relative.
- Conditions causing immunocompromise or immunosuppression, such as solid organ transplant or HIV infection — increases the risk of persistent HPV infection, precancerous lesions, and invasive cervical cancer.
- Use of the combined oral contraceptive pill for longer than 5 years — may be related to the effect of sex steroids on oncogene expression.

COMPLICATIONS

❖ Psychosocial
- Coping with the initial shock of the diagnosis
- Reduced quality of life — may include depression, anxiety, low self-esteem, loss of confidence, fear of recurrence, body image issues, embarrassment, and social isolation

❖ Sexual dysfunction
- May be due to radiotherapy treatment, nerve damage to pelvic viscera during surgery, or to invasive spread into the adjacent parametrium (uterine blood vessels, connective tissue, or lymphatic channels), vagina, uterus, and organs such as bladder and rectum
- May include reduced libido, vaginal dryness, dyspareunia, vaginal shortening, or stenosis post-operatively

❖ Early menopause and loss of fertility
- May be due to radiotherapy causing loss of ovarian function in pre-menopausal women or surgery, including bilateral oophorectomy or salpingectomy

❖ Bladder and bowel dysfunction
- May be secondary to pelvic radiotherapy treatment, nerve damage to pelvic viscera during surgery, or due to invasive cancer spread to the bladder and rectum
- Bowel symptoms may include diarrhoea, faecal urgency, and faecal incontinence
- Urinary symptoms may include haematuria, urgency, frequency, urge incontinence, pain on micturition due to chronic interstitial cystitis and bladder pain, detrusor instability, and urinary retention

❖ Advanced disease
- Non-specific
  - Loss of appetite, unexplained weight loss, fatigue
  - May be due to distant metastases to the lungs, liver, or bone by the haematogenous route (often a late phenomenon)
- Pain

        - May be due to enlarged para-aortic or supraclavicular lymph nodes or bony metastases
        - Pelvic pain or back pain may suggest pelvic side wall infiltration
        - Flank or loin pain may be due to hydroureter or hydronephrosis
    - Renal failure
        - May be due to bilateral ureteric obstruction
    - Severe haemorrhage
        - Delayed haemorrhage may occur due to erosion and bleeding from major pelvic vessels, especially after previous pelvic radiotherapy
    - Fistulae
        - May occur as a result of invasive cancer spread to the bladder and rectum, or as a late complication of radiotherapy
        - A vesicovaginal fistula presents with continual passage of urine through the vagina
        - A rectovaginal fistula presents with passage of faeces through the vagina
        - May cause malodour
    - Lymphoedema
        - May be due to invasive cancer spread to lymph nodes or pelvic side wall infiltration, or due to pelvic lymphadenectomy or radiotherapy treatment, causing lower leg swelling.

PROGNOSIS

- The **FIGO stage** (including the extent of disease spread and lymph node involvement) is the most important prognostic factor in cervical cancer survival.
    - Approximately 20% of women with high-grade dysplasia will develop invasive cervical cancer within 5 years if left untreated
- When comparing cervical cancer survival rates at different stages of disease:
    - 96% of women will survive for 1 year or more when diagnosed at stage I (disease confined to the cervix).
    - 50% of women will survive for 1 year or more when diagnosed at stage IV (disease extended beyond the pelvis).
- Survival is highest in women diagnosed under 40 years of age.
    - Around 90% of women diagnosed aged 15–39 years survived for 5 years or more, compared with around 25% of women diagnosed aged 80 years and over.
- Recurrence can occur as local or metastatic disease, usually within 3 years of treatment, and often has a poor prognosis.

DIAGNOSIS

- Cervical cancer is often asymptomatic in its early stages, and many cases are diagnosed through the NHS cervical screening programme.
- **Suspect a diagnosis of cervical cancer if a woman presents with any of the following:**
    - Persistent unexplained abnormal vaginal bleeding, intermenstrual bleeding, and/or postcoital bleeding which is not secondary to infection or other causes.
    - Unexplained persistent vaginal discharge (may be blood-stained) which is not secondary to infection or other causes.
    - Pelvic pain and/or dyspareunia.
    - Postmenopausal bleeding and not taking hormone replacement therapy (HRT).

- Postmenopausal bleeding if there is an increase in heaviness, duration of bleeding or irregular bleeding if taking sequential HRT; or bleeding beyond 6 months or bleeding after a spell of amenorrhoea if taking continuous combined HRT.
- Abnormal appearance of the cervix on examination, such as inflamed or friable appearance with contact bleeding; or a visible ulcerating or necrotic lesion on the cervix.

- **Consider a diagnosis of cervical cancer if a woman presents with a possible complication of advanced disease, depending on clinical judgement.**

Assessment

- If a woman presents with clinical features suggesting a diagnosis of cervical cancer:
- **Ask about:**
  - The presenting symptoms.
  - Previous smear history.
  - Any risk factors including family history.
  - Any comorbidities which increase the risk of progression or invasive cancer.
  - Smoking status.
  - Any previous colposcopy assessments or cervical treatments.
- **Examine the woman:**
  - Offer to perform an abdominal, speculum, and bimanual pelvic examination.
    - Be aware that some early cervical cancers are not easily detected, and the cervix may look normal on speculum examination if the disease is microinvasive or in the endocervical canal.
  - Assess for lymphadenopathy — enlarged, indurated inguinal and supraclavicular lymph nodes may be palpated in advanced disease.
- **Do not arrange an unscheduled cervical smear** if a woman has clinical features of suspected cervical cancer.
  - An urgent referral for colposcopy/gynaecology oncology assessment should be arranged (using a 2-week cancer pathway).

INTERPRETATION OF RESULTS:

- The introduction of HPV testing allowed patients with mild dyskaryosis to be further risk-stratified, i.e. as HPV is such a strong risk factor patients who were HPV negative could be treated as having normal results.
- The NHS has now moved to an HPV first system, i.e. a sample is tested for high-risk strains of human papillomavirus (hrHPV) first and cytological examination is only performed if this is positive.

Management of results

Negative hrHPV

- return to normal recall, unless
  - the test of cure (TOC) pathway: individuals who have been treated for CIN1, CIN2, or CIN3 should be invited 6 months after treatment for a test of cure repeat cervical sample in the community
  - the untreated CIN1 pathway
  - follow-up for incompletely excised cervical glandular intraepithelial neoplasia (CGIN) / stratified mucin producing intraepithelial lesion (SMILE) or cervical cancer
  - follow-up for borderline changes in endocervical cells.

Positive hrHPV

- samples are examined cytologically
- if the cytology is abnormal → colposcopy
  - this includes the following results:
  - borderline changes in squamous or endocervical cells.
  - low-grade dyskaryosis.
  - high-grade dyskaryosis (moderate).
  - high-grade dyskaryosis (severe).
  - invasive squamous cell carcinoma.
  - glandular neoplasia
- if the cytology is normal (i.e. hrHPV +ve but cytologically normal) the test is repeated at 12 months
  - if the repeat test is now hrHPV -ve → return to normal recall
  - if the repeat test is still hrHPV +ve and cytology still normal → further repeat test 12 months later:
  - If hrHPV -ve at 24 months → return to normal recall
  - if hrHPV +ve at 24 months → colposcopy

If the sample is 'inadequate'

- repeat the sample in 3 months
- if two consecutive inadequate samples then → colposcopy

The follow-up of patients who've previously had CIN is complicated but as a first step, individuals who've been treated for CIN1, CIN2, or CIN3 should be invited 6 months after treatment for a test of cure repeat cervical sample in the community.

DIFFERENTIAL DIAGNOSES

- Sexually transmitted infections (STIs)
- Endometrial cancer
- Endometriosis
- Ectropion or cervical polyp
- Hormonal contraception
  - may cause unscheduled bleeding, particularly during the first 3 months of oral preparation use, and first 3–6 months of levonorgestrel intrauterine system [LNG-IUS] or progestogen-only implant use.

MANAGEMENT

Prevention

- Encourage women to participate in the NHS cervical screening programme.
- Advise that for optimum effectiveness, HPV vaccination must be given before the girl or woman becomes sexually active.
- Advise that HPV vaccination with the nine-valent HPV vaccine also provides additional protection against anogenital warts and other vaginal, vulval, and anal HPV-related cancers.
- Advise girls and women who are sexually active about the importance of using additional barrier methods of contraception such as condoms, to reduce the risk of HPV infection.
  - Condom use reduces the risk of HPV infection but does not fully eliminate the risk of sexual transmission.

- Condom use also protects against other sexually transmitted infections (STIs) such as HIV, which is a risk factor for progression of HPV infection to cervical cancer.
- Limiting the number of sexual partners also reduces potential exposure to HPV infection.

Referral

- If a woman has clinical features of suspected cervical cancer:
  - Arrange an urgent referral to colposcopy/gynaecology oncology for specialist assessment (using a 2-week cancer pathway).
    - All women with unexplained postmenopausal bleeding should be referred to gynaecology for specialist assessment.
    - Do not delay referral because a woman has had a previously normal cervical screening result.
    - Do not delay referral until after pregnancy if a woman is pregnant.

Specialist Assessment

- Specialist assessment of suspected cervical cancer may be undertaken by a multidisciplinary team, including gynaecology oncologists, general gynaecologists, clinical oncologists, radiologists, and clinical nurse specialists.
- Following referral to colposcopy or gynaecology oncology, initial assessment may include:
  - Punch or excisional biopsy procedures such as loop electrosurgical excision and conization of suspicious lesions
- If a diagnosis of cervical cancer is confirmed, further specialist investigations may be arranged to determine the extent of disease. Staging using clinical, radiological, or pathological findings (the revised Féderation Internationale de Gynécologie et d'Obstrétrique 2018 FIGO staging system) allows specialist treatment planning and counselling about prognosis
  - Clinical staging is based on the tumour size, lymph node status, vaginal or parametrial involvement (into uterine blood vessels, connective tissue, or lymphatic channels), bladder/rectum extension, and distant metastases.
  - MRI can determine tumour size and extension of disease. Chest X-ray and assessment for hydronephrosis (using renal ultrasound, CT, or MRI) may be arranged pre-treatment to help decide if initial surgery or primary chemoradiotherapy is needed.
  - Examination under anaesthesia (EUA) may be needed in some cases to gain tissue for diagnosis and to assess for vaginal or parametrial extension.
  - Sentinel node biopsy and PET/CT may be needed to check for lymph node involvement and distant metastases (typically to the lungs, liver, or bone).

CONFIRMED CERVICAL CA MANAGEMENT

- If a woman has a confirmed diagnosis of cervical cancer following specialist assessment:
  - Provide advice on sources of information and support
  - Consider testing for HIV if a woman has a diagnosis of cervical dysplasia or cervical cancer.
  - Advise the woman to stop smoking, if needed.
  - Ask about any adverse effects of specialist treatment.
  - Ask about any complications, and manage as appropriate. If there is any uncertainty about management, seek specialist advice or refer to an appropriate specialist.
    - Offer emotional and psychological support.

- Offer advice on analgesia options for the management of chronic pelvic pain.
- Offer advice on management of sexual dysfunction, including use of vaginal lubricants, vaginal dilatation (if vaginal stenosis after chemoradiotherapy), topical oestrogen, and/or referral to psychosexual counselling, depending on clinical judgement.
- Offer advice on management of early menopause, such as use of hormone replacement therapy (HRT).
- Offer management of bladder and bowel symptoms, such as use of laxatives if constipation or loperamide if diarrhoea. Consider arranging referral to gastroenterology, colorectal surgery, urology, gynaecology, and/or a continence nurse, depending on clinical judgement.
- Offer management of lymphoedema, such as referral to a specialist lymphoedema clinic.

➢ Advise about and assess for symptoms suggestive of recurrent disease (such as vaginal discharge, vaginal bleeding, dyspareunia, low back pain, pelvic pain, and unexplained weight loss).

➢ If a woman has advanced disease.

Specialist management

❖ Management options include:

➢ Stage IA1 (microinvasive disease)
- Loop electrosurgical excision and conization with the aim of achieving negative margins to both cancer and dysplasia.
- Simple hysterectomy may be offered if the woman does not wish to preserve fertility.

➢ Stages IA2–IB2 (early-stage disease)
- Radical hysterectomy (resection of the cervix, uterus, parametria, and cuff of upper vagina) and bilateral salpingectomy (if fertility-sparing surgery is appropriate in low-risk disease) and/or bilateral oophorectomy with bilateral pelvic lymphadenectomy. Radical trachelectomy (removal of the cervix) and lymphadenectomy may be considered instead of radical hysterectomy for smaller tumours.
- The woman may be considered for adjuvant chemotherapy or radiotherapy if intermediate or high risk of recurrence, depending on factors such as tumour size, margin involvement, and lymph node involvement.

➢ Stage IB3–IVA (locally advanced disease)
- Treatment with external beam radiotherapy, intracavity brachytherapy (radiation source placed in the uterus and vagina), and concomitant chemotherapy is usually used first-line to reduce recurrence risk, increase chance of cure, and prolong life.
- Surgery is not recommended as it is unlikely to be curative, and the combination of radical surgery and chemoradiotherapy has a high risk of adverse effects and associated morbidity.

➢ Stage IVB (spread to distant organs)
- Systemic chemotherapy is usually offered first-line.

❖ **Management options for recurrent or metastatic disease** are based on the woman's wishes, previous treatment(s), location of disease, the disease-free interval, symptoms,

comorbidities, and performance status. Treatment of recurrent disease confined to the cervix or upper vagina may be curative. Options may include:

- Salvage surgery — pelvic exenteration (removal of female reproductive organs, the lower urinary tract, and a portion of the rectosigmoid bowel) is possible if the relapse is confined to the central pelvis and chemoradiotherapy have failed.
- Chemotherapy and/or radiotherapy if initial surgery has not controlled disease.

ADVANCED CERVICAL CA MANAGEMENT

- If a woman has advanced cervical cancer, offer management in primary care as part of a multidisciplinary team, which may include gynaecology oncologists, general gynaecologists, clinical oncologists, radiologists, palliative care specialists, and clinical nurse specialists.
- Offer management of general issues which may affect a woman at the end-of-life.
- Offer management of complications, taking into account the woman's wishes, previous treatment(s), location of disease, the disease-free interval, symptoms, comorbidities, and performance status.
  - Offer management of pain and consider specialist referral, depending on the underlying cause and clinical judgement.
    - Specialist management may include nerve-blocking procedures, percutaneous cementoplasty for bony metastases, and palliative radiotherapy for pain due to pelvic disease or bony metastases.
  - Offer management of renal complications due to ureteric obstruction.
    - Specialist management may include percutaneous nephrostomy or retrograde stenting.
  - Offer management of problems associated with haemorrhage.
    - Minor vaginal bleeding due to pelvic disease may respond to oral or topical tranexamic acid or short-course palliative radiotherapy.
    - Severe haemorrhage often requires specialist advice and management, which may include post-operative percutaneous embolization. If the woman is receiving palliative care, midazolam may be used for its anxiolytic effect, and/or diamorphine for its hypotensive effect.
  - Offer management of bladder and bowel symptoms, and seek specialist advice if needed.
    - Surgical intervention with diversion stoma and/or stenting may be considered if there is symptomatic obstructive disease. Bilateral percutaneous nephrostomy may be considered if there is fistula-related urinary incontinence.
    - Defunctioning colostomy may be considered if there is fistula-related faecal incontinence.
  - Offer management of lymphoedema and consider specialist referral to a lymphoedema clinic.
    - Management may include lymphatic therapy with compression bandaging, manual lymph drainage and massage, and good skin care.

TOPIC: INFERTILITY

DEFINITION

- Infertility is a disease of the reproductive system in men or women.
- The **National Institute for Health and Care Excellence (NICE)** defines infertility as the period of time people have been trying to conceive without success, after which formal investigation is justified and possible treatment implemented.
- The **World Health Organization (WHO)** defines infertility as the failure to achieve a pregnancy after 12 months or more of regular unprotected sexual intercourse.
- Infertility is classed as:
  - **Primary**
    - in couples who have never conceived.
  - **Secondary**
    - in couples who have conceived at least once before (with the same or different sexual partner).
- Subfertility generally describes any form of reduced fertility that results in a prolonged duration of unwanted lack of conception.

CAUSES

- There is no identifiable cause of infertility in about 25% of couples.
- The presence of disorders in both the man and the woman has been reported in about 40% of infertile couples.

Causes of infertility in women

- **Ovulatory disorders**
  - Anovulation and oligo-ovulation are ovulatory disorders estimated to cause 21% of infertility in women.
  - The World Health Organization (WHO) categorizes ovulation disorders into three groups:
    - Group I ovulation disorders are caused by hypothalamic-pituitary failure. They are present in about 10% of women with infertility and include conditions such as hypothalamic amenorrhea (which commonly develops as a result of low body weight or excessive exercise) and hypogonadotropic hypogonadism (cause is unknown in most cases but may be congenital, for example, when associated with anosmia [Kallmann syndrome]). Women typically present with amenorrhoea (primary or secondary), characterized by low gonadotrophins and oestrogen deficiency.
    - Group II ovulation disorders are defined as dysfunctions of the hypothalamic-pituitary-ovarian axis. They are present in about 85% of women with infertility and include conditions such as hyperprolactinaemic amenorrhoea and polycystic ovary syndrome (PCOS). PCOS is the most commonly encountered type of Group II ovulation disorder and is responsible for most ovulation disorders.

- Group III ovulation disorders are caused by ovarian failure and are characterized by high gonadotrophins, hypogonadism, and a low oestrogen level. They are present in about 4–5% of women with infertility.

➢ Other conditions that can cause ovulatory disorders include:
- Hyperthyroidism and hypothyroidism
  - may lead to menstrual disorders and ovulatory dysfunction. Hypothyroidism for more information.
- Cushing's syndrome and congenital adrenal hyperplasia
  - may cause anovulation.
- Chronic debilitating diseases
  - (such as uncontrolled diabetes, cancer, AIDS, end-stage kidney disease, and malabsorption), may lead to anovulation and amenorrhoea.

❖ **Tubal damage**

➢ Tubal factors account for 14% of the causes of subfertility in women.

➢ The most common cause of tubal factor infertility is pelvic inflammatory disease (PID) and acute salpingitis. However, all pelvic infections, including appendicitis and diverticulitis, can damage the fallopian tubes.

➢ Endometriosis

➢ Other factors affecting tubal patency include:
- Injury to the fallopian tube due to previous tubal surgeries or sterilization, ischemic nodules, polyps or mucus, tubal spasm, and congenitally abnormal tubes.

❖ **Uterine or peritoneal disorders**

➢ Uterine abnormalities (such as adhesions, polyps, submucous leiomyomas, and septae) have been found in 10–15% of women seeking treatment for fertility problems.

➢ Uterine fibroids may be the sole cause of infertility in 2–3% of women. Depending on their location in the uterus, fibroids have been implicated in recurrent pregnancy loss and infertility.

➢ Peritoneal factors, such as peritubular adhesions, endometriosis, altered tubal motility, and fimbrial end blockage, can also affect tubal patency.

❖ **Unexplained cause**

➢ Unexplained infertility is a diagnosis made by exclusion in couples who have not conceived and in whom standard investigations have not detected any abnormality. It accounts for about 40% of infertility in women and 8–28% of infertility in couples.

Causes of infertility in men

❖ **Testicular failure**

➢ Primary testicular failure is the most common cause of infertility in men due to oligozoospermia and is the cause of non-obstructive azoospermia.

➢ Testicular failure may be due to factors such as cryptorchidism (undescended testicles), testicular torsion or trauma, orchitis, chromosome disorders (such as Y chromosome deletions and Klinefelter syndrome [XXY]), systemic disease, radiotherapy, or chemotherapy. However, in most cases (66%) the cause is unknown.

- Varicocele is present in almost 15% of the normal population of men, in 25% of men with abnormal semen analysis, and in 35–40% of men presenting with infertility. Clinically detectable varicoceles can be associated with abnormal gonadotrophin levels, impaired spermatogenesis, and histological changes to sperm.
- Several endocrinopathies are associated with low testosterone levels, including hyperprolactinaemia and genetic conditions (such as Kallmann syndrome). Clinically significant endocrinopathies are found in only 2% of infertile men.
- Risk factors that alter spermatogenesis through low testosterone levels include obesity and exposure to certain drugs or environmental toxins.

❖ **Obstructive azoospermia**

- Obstructive azoospermia is the absence of spermatozoa in the sediment of a centrifuged sample of ejaculate due to obstruction.
- Obstruction may be:
  - Congenital, such as congenital bilateral absence of vas deferens, prostatic cysts (Mullerian cysts), or idiopathic epididymal obstruction.
  - Acquired secondary to epididymal or prostatic infections, vasectomy, or complications of surgical procedures (such as inguinal hernia repair or orchiopexy for undescended testicles).
  - Related frequently to a mutation in the cystic fibrosis transmembrane regulator gene.
- Obstructive azoospermia is less common than non-obstructive azoospermia and occurs in 20–40% of men with azoospermia.

❖ **Ejaculatory and erectile dysfunction**

- Ejaculatory disorders result from conditions that interfere with the ejaculation process, a complex physiological process comprising two phases (emission and expulsion) and is mediated by complex neurological and hormonal pathways. Ejaculatory disorders include:
  - Premature ejaculation.
  - Delayed ejaculation.
  - Retrograde ejaculation (semen passing backwards into the bladder).
  - Anejaculation (no ejaculation).
  - Painful ejaculation.
  - Anorgasmia (perceived absence of orgasm, which can give rise to anejaculation).
  - Haematospermia (blood in the ejaculate, which may indicate underlying pathology).
- Erectile dysfunction is defined as the persistent inability to attain and maintain an erection sufficient to permit satisfactory sexual performance.
- Erectile and ejaculatory dysfunction may be associated with psychological factors, hypogonadism, spinal cord disease, metabolic and vascular conditions (such as diabetes mellitus), lifestyle factors (such as smoking, excessive alcohol intake, and obesity), and the use of certain drugs.

❖ **Abnormal sperm function and quality**

- Conditions or situations that cause abnormal shape (morphology) and movement (motility) of the sperm negatively affect fertility.

- Sperm motility can be reduced in men with Kartagener syndrome (immotile cilia syndrome) and those with antisperm antibodies.
- Urogenital tract infections (such as prostatitis, orchitis, and epididymitis) may result in an inflammatory response leading to obstruction or immunogenic spermatogenic dysfunction.
- Anabolic steroids can cause abnormal semen parameters (such as sperm count and shape).

- **Unexplained cause**
  - Unexplained infertility in men is defined as infertility of unknown origin with normal sperm parameters and partner evaluation. It is now believed that idiopathic infertility may be associated with several previously unidentified pathological factors, including endocrine disruption due to environmental pollution, generation of reactive oxygen species (ROS)/sperm DNA damage, or genetic and epigenetic abnormalities.

RISK FACTORS

WOMEN

- **Age**
  - Natural fertility in women declines with age due to dwindling oocyte numbers and poorer oocyte quality.
  - The fecundity of women decreases gradually, but significantly, beginning approximately at 32 years and decreasing rapidly after 37 years of age.
- **Sexually transmitted infections (STIs)**
  - Chlamydia and gonorrhoea particularly affect fertility by causing damage to the genital anatomy.
- **Obesity**
  - High body fat produces a range of factors, including adipokines and oestrone, that can significantly influence reproductive function.
  - Women with a body mass index (BMI) of 30 or over are likely to take longer to conceive compared with women with a normal BMI.
  - Women with obesity are at increased risk of infertility and of developing maternal and fetal complications during pregnancy.
  - Obesity in women is associated with ovulatory dysfunction, reduced ovarian responsiveness to treatments that induce ovulation, altered oocyte and endometrial function, and lower birth rates after in vitro fertilization.
- **Low body weight**
  - A weight of 10% below normal body weight in most women will cause menstrual irregularities.
  - In women with a BMI of less than 19 who have irregular menstruation or are not menstruating, increasing body weight is likely to improve their chances of conception.
- **Lifestyle factors**
  - Smoking is likely to reduce fertility in women.
  - Stress (in either the man or the woman) may affect the couple's relationship and is likely to reduce libido and the frequency of intercourse, which may contribute to infertility.
- **Occupational and environmental factors**

- Exposure to pesticides, nitrous oxide, metals (lead, cadmium, manganese), solvents, formaldehyde and other occupational and environmental hazards can reduce fertility.

- **Drugs**
  - Some prescription, over-the-counter, and recreational drugs can interfere with fertility. For example:
    - Nonsteroidal anti-inflammatory drugs (NSAIDs) may inhibit ovulation.
    - Chemotherapy can induce ovarian failure, which may be permanent.
    - Metoclopramide, methyldopa, sertraline, fluoxetine, and antipsychotics (such as chlorpromazine, prochlorperazine, haloperidol, and risperidone) can suppress hypothalamic-pituitary function and increase prolactin levels.
    - Recreational drugs (such as marijuana and cocaine) can adversely affect ovulatory and tubal function.

MEN

- **Age**
  - Advanced paternal age has emerged as one of the main risk factors associated with the progressive increase in the prevalence of infertility in men.
  - Increasing age (over 55 years) may affect sperm motility.
- **Sexually transmitted infections (STIs)**
  - STIs can produce scarring and obstruction of the reproductive tract.
- **Obesity**
  - Obesity contributes to infertility in men. This can be attributed to multiple factors, including reduced testosterone levels and increased aromatisation of testosterone to oestrogen (which has a negative feedback effect on follicle-stimulating hormone production).
  - Men with obesity may exhibit impaired reproductive function. However, additional research is needed to understand further the relationship between obesity in men and reproductive function.
  - Obesity is also associated with erectile dysfunction.
- **Lifestyle factors**
  - There is an association between smoking and reduced semen quality, although this impact on fertility is uncertain.
  - Excessive alcohol intake harms semen quality.
  - Stress (in either the man or the woman) may affect the couple's relationship and is likely to reduce libido and the frequency of intercourse, which may contribute to infertility.
- **Occupational and environmental factors**
  - Exposure to pesticides, nitrous oxide, metals (lead, cadmium, manganese), solvents, formaldehyde and other occupational and environmental hazards can reduce fertility.
- **Tight underwear**
  - There is an association between elevated scrotal temperature and reduced semen quality, but it is uncertain whether wearing loose-fitting underwear improves fertility.
- **Drugs**
  - Some prescription, over-the-counter, and recreational drugs can interfere with fertility. For example:

- Sulfasalazine and some antifungal treatments can adversely affect spermatogenesis.
- Certain antipsychotics, antidepressants, and antihypertensives can cause retrograde ejaculation and orgasmic dysfunction.
- Long-term opiate use can cause hormonal dysregulation and impair bulk semen parameters.
- 5-alpha reductase inhibitors can cause sexual dysfunction, Finasteride may impair semen parameters.
- Hormone treatment, particularly testosterone supplementation or anabolic steroid treatment, can inhibit spermatogenesis and impair fertility.
- Recreational drugs (such as anabolic steroids and cocaine) can adversely affect semen quality.

COMPLICATIONS

- Psychological stress (for both the woman and the man) may be caused by infertility in addition to the investigation and treatment.

Complications of assisted conception

- **Ovarian hyperstimulation syndrome (OHSS)**
  - is an iatrogenic and potentially life-threatening complication of superovulation caused by vasoactive products released by hyperstimulated ovaries.
  - The incidence of OHSS varies between different types of fertility treatment, with treatments involving greater degrees of ovarian stimulation being associated with a higher incidence.
  - Risk factors include previous history of OHSS, young age, lean physique, polycystic ovary syndrome, and multiple pregnancies.
  - The onset of OHSS can be early (within 9 days after the ovulatory dose of human chorionic gonadotrophin [hCG]) or late (after 9 days, reflecting endogenous hCG stimulation arising from successful implantation).
- **Ectopic pregnancy**
  - is a possible complication in women undergoing assisted conception.
- **Pelvic infection**
  - is a possible complication of the ovum extraction procedure.
    - Pelvic pain and other signs of infection may develop in the weeks after the procedure.
    - Infection may be introduced into the ovary during the ovum extraction procedure, which involves passing a needle through the vaginal wall into the ovary.
- **Multiple births**
  - are the single most significant risk of fertility treatment.
  - The risk of a multiple birth is largely determined by the number of embryos transferred back during IVF treatment.
  - Multiple births remain a health risk for women undergoing IVF treatment and their babies, but the decreasing multiple birth rate means that IVF is a safer treatment.
- **The evidence on cancer risk in women who have had fertility treatment is limited and of low quality.**

- According to the National Institute for Health and Care Excellence (NICE):
- There is no direct association between ovulation induction or stimulation and invasive cancer.
- No association has been found in the short term and medium term between ovulation induction or stimulation and adverse outcomes (including cancer) in children born from ovulation induction.
- Information on long-term health outcomes in women and children is still awaited.

ASSESSMENT

- All couples concerned about fertility should be offered an initial assessment (history and examination) in primary care to identify those who have modifiable risk factors (such as obesity and smoking) and those who need early investigation or referral.

FOR WOMEN

- **Take a full medical, sexual, and social history. Ask about:**
  - Length of time trying to conceive.
  - Children born to the woman as well as previous pregnancies and miscarriages (with the same or a different partner).
  - Frequency of sexual intercourse, including any difficulties with intercourse — the chances of conception improve with regular (every 2–3 days) sexual intercourse.
  - Length of time since stopping contraception, and type of contraception. Check that there are no problems that may explain the inability to conceive, such as a 'lost' intrauterine contraceptive device.
  - Frequency and regularity of their menstrual cycles.
  - Presence of systemic disease, including thyroid dysfunction, diabetes, and inflammatory bowel disease.
  - History of sexually transmitted infections (STIs) or pelvic inflammatory disease (PID).
  - The timing and result of their most recent cervical smear test.
  - Previous pelvic surgery (for example, appendicitis or ovarian cyst).
  - Symptoms that may indicate ovulatory disorders, such as:
    - Menorrhagia (abnormally heavy menstrual bleeding).
    - Oligomenorrhoea (infrequent or irregular menstrual periods).
    - Amenorrhoea (absence of menstruation).
    - Dysmenorrhoea (painful periods).
    - Galactorrhoea or hirsutism.
    - Excessive exercise, weight loss, or psychological distress.
  - Symptoms that may indicate uterine, cervical, and peritoneal disorders, including:
    - Symptoms of PID or endometriosis, such as dyspareunia (difficult or painful sexual intercourse) and dysmenorrhoea.
    - Intermenstrual or postcoital bleeding.
  - Risk factors for infertility, including age (fertility in women decreases with age), lifestyle factors, drug history, occupational hazards, and stress.
- **Perform a physical examination.**
  - Check body mass index (BMI) to assess for obesity (associated with lower fertility).
  - Look for:
    - Hirsutism and acne (associated with polycystic ovary syndrome).

        - Galactorrhoea (suggestive of hyperprolactinaemia).
    - Perform an abdominal examination to identify any mass, such as an ovarian cyst.
    - Pelvic examination may identify vaginal infection or tenderness, indicating endometriosis or PID. Bimanual examination may reveal fibroids or an ovarian cyst.
    - Vaginal examination may indicate undisclosed sexual difficulties, such as vaginismus.

FOR MEN

- **Take a full medical, sexual, and social history. Ask about:**
    - Length of time trying to conceive.
    - Children born to the man (with the same or a different partner).
    - Frequency of sexual intercourse, including any difficulties with intercourse — the chances of conception improve with regular (every 2–3 days) sexual intercourse.
    - History of systemic diseases (for example, cardiac failure, chronic renal failure, neoplasia, uncontrolled diabetes, liver cirrhosis, or thyrotoxicosis).
    - History that may suggest primary testicular failure or obstructive azoospermia, including:
        - History of mumps, sexually transmitted infections (STIs), or testicular trauma or torsion.
        - Previous urogenital abnormality and treatment (for example, undescended testis or orchidopexy).
        - Previous surgery (for example, hernia repair or orchidopexy).
        - Ejaculatory or erectile dysfunction.
    - Risk factors for infertility, including lifestyle factors, drug history, occupational hazards, and stress.
- **Perform a physical examination to identify factors that may contribute to infertility.**
    - Examine the penis, including a check of the position of the urethral meatus, for structural abnormalities.
    - Examine the scrotum and testicles for lumps (which may indicate varicocele, hernia, or cancer); small, soft testes (which may indicate hypogonadism); or undescended testes.
    - Assess secondary sexual characteristics. In hypogonadism, there may be a decrease in beard and body hair growth and a decrease in muscle mass.
    - Look for gynaecomastia, which may indicate hypogonadism.

MANAGEMENT

How should I advise a couple seeking information on infertility?

- **Involve both partners in all aspects of management.**
    - Provide evidence-based information (verbal and written) to enable them to make an informed decision regarding their care and treatment.
    - Ensure the information provided is in a form that is accessible to people who have additional needs, such as people with physical, cognitive, or sensory disabilities, and people who do not speak or read English.

Give women and men general advice on the following:

- **Chance of conception**
  - Advise that over 80% of couples in the general population will conceive within 1 year if the woman is aged under 40 years and they have regular (every 2–3 days) unprotected sexual intercourse. About half of those who do not conceive in the first year will do so in the second year (cumulative pregnancy rate over 90%).
  - Inform people who are using artificial insemination to conceive and are concerned about their fertility that over 50% of women under 40 years old will conceive within 6 cycles of intrauterine insemination (IUI). About half of those who do not conceive within 6 cycles of IUI will do so with a further 6 cycles (cumulative pregnancy rate over 75%).
  - Advise that fertility in women (and, to a lesser extent) fertility in men declines with age.
- **Smoking**
  - Advise women that smoking (including passive smoking) will likely reduce their fertility and that maternal smoking can harm a developing baby.
  - Advise men that smoking affects semen quality (although it is unknown if this adversely affects fertility).
  - Discuss the benefits of quitting smoking to general health.
  - Offer those who smoke support to help them quit.
- **Alcohol intake**
  - Advise women that alcohol consumption during pregnancy is not advised as it can adversely affect the foetus.
  - Advise men that excessive alcohol consumption may affect semen quality. However, there is no evidence that drinking within recommended safe limits (no more than 14 units per week) has an adverse effect.
  - Offer those who drink excessively support to help them quit.
- **Obesity**
  - Advise women that a body mass index (BMI) of 30 kg/m2 or over may increase their time to conceive.
  - Advise men that a BMI greater than 29 kg/m2 is likely to have reduced fertility.
  - Advise women with a BMI of 30 kg/m2 or over who are not ovulating that losing weight is likely to increase their chance of conception.
  - Explain that women with obesity are at increased risk of infertility and developing maternal and fetal complications during pregnancy.
  - Offer those who are overweight or obese support to help them lose weight.
- **Low body weight**
  - Advise women with a BMI less than 19 kg/m2 plus either amenorrhoea or irregular menstruation that gaining weight is likely to increase their chance of conception.
- **Drugs**
  - Advise that some prescription, over-the-counter, and recreational drugs can interfere with fertility in women and men.
  - Offer users of illicit drugs referral to a specialist drugs and alcohol service.
- **Stress management**
  - Advise that stress in either partner can affect their relationship and is likely to reduce libido and frequency of intercourse, which can contribute to fertility problems.
  - Explain that psychological stress (for both the woman and the man) may be caused by infertility as well as the investigation and treatment.

- Offer counselling before, during, and after investigation and treatment, irrespective of the outcome of these procedures.

- **Occupational risks**
  - Advise anyone concerned about occupational risks to their health, including their fertility, to seek specialist advice from occupational health at their place of work or from the Health and Safety Executive website, which provides detailed information on occupational risks.
- **Caffeinated beverages**
  - Advise that no consistent evidence exists of an association between consumption of caffeinated beverages (tea, coffee, and colas) and fertility problems.
- **Complementary therapies**
  - Advise that the effectiveness of complementary therapies for fertility problems has not been adequately evaluated, so further research is needed before such interventions can be recommended.
- **Tight underwear**
  - Advise men that although an elevated scrotal temperature is associated with reduced semen quality, it is uncertain whether wearing loose-fitting underwear improves semen quality.

Give women additional relevant pre-conception advice.

- **Folic acid**
  - Dietary supplementation with folic acid before conception and up to 12 weeks of gestation reduces the risk of having a baby with neural tube defects.
  - The recommended dose is 0.4 mg per day. For women who have previously had an infant with a neural tube defect or who are receiving anti-epileptic medication, or who have diabetes, a higher dose of 5 mg per day is recommended.
- **Susceptibility to rubella**
  - Offer testing for rubella status.
  - Women who are susceptible to rubella should be offered vaccination and advised not to become pregnant for at least 1 month following vaccination.
- **Cervical cancer screening**
  - Ask about the timing and result of their most recent cervical smear test to avoid delay in fertility treatment.
  - Where applicable, offer cervical screening in accordance with the national cervical screening programme guidance.

Provide sources of additional information and support.

What initial investigations should I arrange in a woman?

- Investigate both partners simultaneously to categorize the cause of infertility. Start investigations in couples who have not conceived after 1 year of regular (every 2–3 days) unprotected sexual intercourse. Offer investigations earlier than 1 year to couples identified as less likely to conceive. Early investigations may be prompted by the same factors that prompt early referral.

- **Measure mid-luteal phase progesterone in all women to confirm ovulation.**
  - Offer a blood test to measure serum progesterone in the mid-luteal phase of the cycle (day 21 of a 28-day cycle) even if the woman has regular menstrual cycles.
- **Screen for chlamydia.**
  - Before undergoing uterine instrumentation, women should be offered screening for Chlamydia trachomatis using an appropriately sensitive technique.
  - If the result is positive, the woman and her sexual partner(s) should be referred for appropriate management with treatment and contact tracing.
  - Prophylactic antibiotics should be considered before uterine instrumentation if screening has not been done.
- **The following additional tests may be needed:**
  - Serum progesterone (in women with prolonged irregular menstrual cycles) — depending on the timing of menstrual periods, serum progesterone may need to be measured later (for example, on day 28 of a 35-day cycle) to confirm ovulation and repeated weekly after that until the next menstrual cycle starts.
  - Gonadotrophin measurement (in women with irregular menstrual cycles) — gonadotrophin (follicle-stimulating hormone and luteinizing hormone) measurements are of value in women with anovulation or oligo-ovulation. They can be used to identify ovulation disorders.
  - Thyroid function tests (in women with symptoms of thyroid disease).
  - Prolactin measurement (in women with symptoms of an ovulatory disorder [for example, polycystic ovary syndrome], galactorrhoea, or a suspected pituitary tumour).
- **The following investigations are not recommended:**
  - Endometrial biopsy — there is no evidence that medical treatment of luteal phase defects improves pregnancy rates.
  - Basal body temperature charts — they do not reliably predict ovulation.
  - Use of ovulation predictor kits — there is no evidence that attempts to time sexual intercourse to the menstrual cycle result in improved conception rates. Furthermore, there is evidence that the use of these kits causes stress.
  - Postcoital testing of cervical mucus — has no predictive value for pregnancy rates.

What initial investigations should I arrange in a man?

- **Arrange for semen analysis.**
  - Give clear written and verbal instructions concerning the collection of the semen sample:
    - The specimen should be collected by masturbation; ejaculated into a clean, wide-mouthed container made of plastic from a batch that has been confirmed to be non-toxic for spermatozoa; and protected from extremes of temperature (below 20°C or above 37°C).
    - The specimen should be collected after a minimum of 2 days and a maximum of 7 days of sexual abstinence.
    - Ideally, the sample should be collected in a private room near the laboratory to avoid exposure of the semen sample to fluctuations in temperature and to control the time between collection and analysis.
    - If the man, for any reason, must collect the sample at another place, the specimen container should be kept close to the body under the clothes (for example, in the armpit) during transport and should be delivered to the

laboratory, preferably within 30 minutes after collection and at least no longer than 50 minutes after collection.

- Emphasize that the semen sample needs to be complete and that the man should report any loss of any fraction of the sample.
- The results should be interpreted using the most recent World Health Organization (WHO) reference values for human semen characteristics.
  - If the result of the first semen sample is normal, there is no need to do a repeat confirmatory test.
  - If the result of the first semen sample is abnormal, order a repeat test.
  - Testing should ideally be repeated 3 months after the initial test to allow time for the cycle of spermatozoa to be completed.
  - It may be appropriate to test earlier, for example, if the man is very anxious about the test result and prefers to have the test done earlier. In these circumstances, or if there is a gross spermatozoa deficiency (azoospermia or severe oligospermia), repeat the test within 2–4 weeks.
  - If the repeat test result is normal, regard the semen as normal; no further testing is required.
  - Refer men with two abnormal semen examination results to secondary care for further assessment.

- **Screen for chlamydia.**
- **The following tests are not recommended:**
  - Screening for antisperm antibodies — there is no evidence of effective treatment to improve fertility.

REFERRAL

Referral criteria for people presenting with infertility may vary between health authorities. Refer to local guidelines.

- **In women younger than 36 years of age:**
  - In general, consider referring the couple for additional investigations and management if history, examination, and investigations are normal in both partners and the couple has not conceived after 1 year.
- **Consider earlier referral if the following factors are present:**
  - In women:
    - Age 36 years and older (refer after 6 months).
    - Amenorrhoea or oligomenorrhoea.
    - Previous abdominal or pelvic surgery.
    - Previous pelvic inflammatory disease.
    - Previous sexually transmitted infection (STI).
    - Abnormal pelvic examination.
    - Known reason for infertility (for example, prior treatment for cancer).
  - In men:
    - Previous genital pathology.
    - Previous urogenital surgery.
    - Previous STI.
    - Varicocele.
    - Significant systemic illness.
    - Abnormal genital examination.

- Two abnormal semen examination results.
- Known reason for infertility (for example, prior treatment for cancer).

❖ **Ensure that the couple is offered counselling before, during, and after investigation and treatment regardless of the outcome of these procedures.**
  - ➢ Psychological stress (for both the woman and the man) may be caused by infertility as well as the investigation and treatment.
  - ➢ Usually, counselling will be arranged by the specialist infertility team.

Additional investigations and management

❖ **Following referral, additional investigations may be undertaken.**
  - ➢ Investigations in women normally include tubal patency tests.
    - Women who are not known to have comorbid conditions (such as pelvic inflammatory disease, endometriosis, or previous ectopic pregnancy) are offered hysterosalpingography or hysterosalpingo-contrast ultrasonography.
    - Women thought to have comorbid conditions are offered diagnostic laparoscopy and dye so that tubal and other pelvic abnormalities can be assessed simultaneously.
  - ➢ Investigations in men may include an assessment of the sperm, starting with a review of results obtained from primary care investigations.
    - In men with abnormal sperm, a more detailed examination is done, which may include microbiological tests, sperm culture, endocrine tests, imaging of the urogenital tract, and testicular biopsy.

❖ **Fertility treatment falls into 3 main types:**
  - ➢ Medical treatment to restore fertility.
    - Clomifene (an anti-oestrogen drug) is an effective treatment for anovulation and may be used in selected women.
    - Gonadotrophins may be offered to women with clomifene-resistant anovulatory infertility. They are also effective in improving fertility in men with hypogonadotropic hypogonadism.
    - Pulsatile gonadotrophin-releasing hormone and dopamine agonists are other treatments that induce ovulation. Dopamine agonists can be considered for women with ovulatory disorders secondary to hyperprolactinaemia.
  - ➢ Surgical treatment to restore fertility. This includes:
    - Tubal microsurgery in women with mild tubal disease — tubal catheterization or cannulation improves the chance of pregnancy in women with proximal tubal obstruction.
    - Surgical ablation, or resection of endometriosis plus laparoscopic adhesiolysis in women with endometriosis.
    - Surgical correction of epididymal blockage in men with obstructive azoospermia — this is likely to restore patency of the duct and improve fertility.
  - ➢ Assisted reproduction techniques (any treatment that deals with means of conception other than vaginal intercourse). These include:
    - Intrauterine insemination (IUI) — in this process, which is timed to coincide with ovulation, sperm is placed in the woman's uterus using a fine plastic tube. Low doses of ovary-stimulating hormones (oral anti-oestrogens or gonadotrophins) might be given (stimulated IUI) to maximize pregnancy rates.
    - In vitro fertilization (IVF) — involves retrieval of one or more ova combined with sperm and incubated for 2–3 days; the resultant embryo is then injected into the

uterus via the cervix. This method is suitable for women who have blocked fallopian tubes, men with a minor degree of subfertility, and couples who have been diagnosed with unexplained infertility or have been unsuccessful with other techniques (such as ovulation induction or IUI).

- Intracytoplasmic sperm injection (ICSI) — involves injecting an individual sperm directly into the ovum to bypass natural barriers that prevent fertilization. The embryo is then transferred into the uterus. This method is suitable when the man has a very low sperm count or problems maintaining an erection and ejaculation (such as diabetes or spinal cord injury).
- Donor insemination — involves insemination of sperm, from a donor, into a woman via her vagina into the cervical canal or into the uterus itself (IUI). This method is considered when the man has no (or very few) sperm on testicular biopsy or surgical extraction, has had a vasectomy and reversal has failed or not been tried, or has an infectious disease (such as HIV), or where there is a high risk of transmitting a genetic disorder to the offspring. It is also considered in couples where there is no male partner.
- Oocyte donation — involves stimulation of the donor's ovaries and collection of ova. The donated ova are then fertilized by the recipient's partner's sperm. After 2–3 days, the embryos are transferred to the uterus of the recipient via the cervix after hormonal preparation of the endometrium. This method is considered for women with ovarian failure (premature or after radiotherapy or chemotherapy); those with bilateral oophorectomy; those with gonadal dysgenesis, including Turner's syndrome; and when the risk of transmitting a genetic disorder is high. It is also used in certain cases of IVF failure. Couples who have had successful IVF or ICSI may decide to donate their spare embryos to help other infertile couples (embryo donation).

How should I manage complications that may occur after assisted conception?

- **Ovarian hyperstimulation syndrome (OHSS)**
  - is a potentially life-threatening complication of superovulation.
  - Symptoms and signs include:
    - Mild — abdominal bloating and mild abdominal pain.
    - Moderate — nausea and vomiting and increased abdominal discomfort.
    - Severe — oliguria, generalized oedema, abdominal pain and/or distension (caused by enlarged ovaries and acute ascites), and hydrothorax (occasionally).
    - Critical — oligo/anuria, tense ascites or large hydrothorax, thromboembolism, and acute respiratory distress syndrome.
  - If a woman presents with these symptoms and signs:
    - Consider alternative diagnoses, such as complications of an ovarian cyst (torsion, haemorrhage), pelvic infection or abscess, intra-abdominal haemorrhage, ectopic pregnancy, bowel perforation, or appendicitis.
    - If OHSS is suspected, seek urgent advice from the specialist unit. The severity of OHSS can worsen over time, and even initially mild presentations should be kept under review.
- **Other possible complications that may occur after assisted conception include:**
  - Ectopic pregnancy
  - Pelvic infection
  - Multiple pregnancy

TOPIC: MENOPAUSE

DEFINITION

- **Menopause** is a biological stage in a woman's life when menstruation stops permanently due to the loss of ovarian follicular activity.
  - It occurs with the final menstrual period and is usually diagnosed clinically after 12 months of amenorrhoea.
- **Perimenopause,** also called the 'menopausal transition' or 'climacteric', is the period before the menopause when the endocrinological, biological, and clinical features of approaching menopause start.
  - It is characterized by irregular cycles of ovulation and menstruation and ends 12 months after the last menstrual period.
- **Postmenopause** is the time after a woman has not had a menstrual period for 12 consecutive months.
- **Early menopause** is the cessation of ovarian function occurring between the ages of 40 and 45 years, in the absence of other causes of secondary amenorrhoea.
- **Premature menopause** describes definitive loss of ovarian function before the age of 40 years, for example following bilateral oophorectomy.
- **Premature ovarian insufficiency** (POI, also known as premature ovarian failure) is a clinical syndrome defined as the transient or permanent loss of ovarian function before the age of 40 years, characterized by menstrual disturbance (amenorrhoea or oligomenorrhoea), and potential spontaneous resumption of ovulation, menstruation, and spontaneous pregnancy.
- **Genitourinary syndrome** of menopause (previously known as vulvovaginal atrophy, atrophic vaginitis, or urogenital atrophy) describes combined vulvovaginal and urinary tract symptoms caused by thinning and shrinking of the tissues of the vulva, vagina, urethra, and bladder caused by oestrogen deficiency

CAUSES

Menopause is a natural stage in a woman's life.

- Unlike in men where sperm production continues into old age, women have a finite number of oocytes (eggs) at birth. The number of oocytes is greatest before birth (between 20- and 28-weeks' gestation) and declines with each menstrual cycle. The menopause is characterized by the eventual depletion of the oocyte store (and the cessation of menstruation).
- During the perimenopause:
  - Ovarian follicular activity begins to fail.
  - Oestrogen and inhibin levels decrease and reduced negative feedback to the pituitary causes follicle stimulating hormone (FSH) and luteinizing hormone (LH) levels to rise.
  - Levels of FSH fluctuate markedly from pre- to postmenopausal values on an almost daily basis during the transition to menopause.
  - Decreasing oestrogen levels begin to disrupt the menstrual cycle and may cause other menopausal symptoms (such as vasomotor symptoms including hot flushes and night sweats). These result from minor increases in core body temperature that trigger an

excessive thermoregulatory reaction and promote heat dissipation by peripheral vasodilatation and sweating.
    - Menstural cycles tend to become anovulatory, and eventually follicular development stops. Estradiol production, which occurs in the granulosa and thecal cells surrounding the oocyte, becomes insufficient to stimulate the endometrium, and amenorrhoea occurs.
    - Eventually, the menopausal pattern of low oestrogen and persistently high FSH and LH levels is established.

The risk of premature or early menopause may be increased if a woman has a history of:

- Early menarche.
- Nulliparity or low parity.
- Smoking (strong dose-response effect).
- Being underweight.

In premature ovarian insufficiency (POI), there is loss of normal ovarian function before the age of 40 years.

- The three main identifiable causes are genetic, autoimmune, and iatrogenic, although in a significant number of women the cause is unknown, and it is termed 'unexplained' or 'idiopathic'.
    - Genetic causes include a strong maternal family history, galactosaemia, and chromosomal abnormalities including X chromosome defects.
    - Women with an autoimmune predisposition may develop autoimmune premature menopause, with or without other autoimmune diseases (such as type 1 diabetes, Addison's disease, and thyroid disorders).
    - Iatrogenic menopause occurs as a result of treatments such as chemotherapy, radiotherapy, treatment with gonadotrophin-releasing hormone analogues (for example for breast cancer), or surgery such as bilateral oophorectomy.
    - In addition, infections (most commonly tuberculosis and mumps) may also cause POI. In most women who develop ovarian failure after mumps, normal ovarian function returns.

SYMPTOMS

The symptoms seen in the climacteric period are caused by reduced levels of female hormones, principally oestrogen.

- Change in periods
    - change in length of menstrual cycles
    - dysfunctional uterine bleeding may occur
- Vasomotor symptoms - affects around 80% of women. Usually occur daily and may continue for up to 5 years
    - hot flushes
    - night sweats
- Urogenital changes - affects around 35% of women
    - vaginal dryness and atrophy
    - urinary frequency

- Psychological
  - anxiety and depression may be seen - around 10% of women
  - short-term memory impairment

COMPLICATIONS

Postmenopausal women are at increased risk of:

- Osteoporosis and fracture.
- Cardiovascular disease.
- Stroke disease.
- Genitourinary syndrome of menopause, due to oestrogen depletion as well as natural ageing processes.

Premature or early menopause may be associated with an increased risk of:

- All-cause mortality.
- Cardiovascular disease.
- Type 2 diabetes mellitus.
- Depression.
- Osteoporosis and fracture.

PROGNOSIS

The duration and severity of symptoms experienced in menopause vary markedly between different women.

- Symptoms typically last for 5–7 years, but some women continue to experience symptoms for at least 10–15 years.
- Perimenopause can last for several years, and menopause can last a few years to more than 13 years.
- Persistent vasomotor symptoms may be associated with:
  - Ethnicity.
  - Younger age at menopause.
  - Current smoking.
  - Weight gain.
  - Lower educational level.

DIAGNOSIS

MENOPAUSE AND PERIMENOPAUSE

Suspect a diagnosis of perimenopause if:

- There is an initial change to the menstrual pattern — the menstrual cycle length may shorten to 2–3 weeks or lengthen to many months. The amount of menstrual blood loss may change, and commonly increases slightly.

In addition, other symptoms associated with the menopause include:

- Hot flushes/night sweats (vasomotor symptoms)
  - A sudden feeling of heat in the upper body (face, neck, and chest) that spreads upwards and downwards. In some cases, this becomes generalized, typically lasting 2–4 minutes, and can be associated with excessive sweating, palpitations, or anxiety.

    - They can be embarrassing and distressing, and triggers may include spicy food and alcohol.
- Cognitive impairment and mood disorders
    - There may be anxiety, mood swings, irritability, sleep disturbance, and reduced quality of life.
    - There may be low mood (mild depressive symptoms that impair quality of life but are usually intermittent and often associated with hormonal fluctuations in the perimenopause), distinct from depression.
    - Low mood may be associated with negative beliefs about the menopause, and stigma about age and reproductive status.
    - There may be poor concentration and memory, and difficulties in multi-tasking, causing social embarrassment.
- Urogenital symptoms (genitourinary syndrome of menopause)
    - These may include vulvovaginal irritation, discomfort, burning, itching, and/or dryness; dyspareunia; reduced libido; dysuria, urinary frequency and urgency, and recurrent lower urinary tract infections.
    - Symptoms of urogenital atrophy may appear for the first time more than 10 years after the last menstrual period.
    - Vaginal dryness tends to increase in severity with time since menopause.
- Altered sexual function
    - Vaginal dryness resulting from urogenital atrophy can lead to dyspareunia, which can impact on libido.
    - Loss of sexual desire and libido may also be a result of declining levels of oestrogen and testosterone as the ovaries fail. The lack of testosterone can be more marked in women who have bilateral oophorectomy.
    - In addition, low mood, loss of confidence, negative self-beliefs about reproductive status, and altered self-image may contribute to psychological causes for altered sexual function.
- Sleep disturbance
    - This may be due to hot flushes and night sweats disrupting sleep, but may also be due to mood disorders or primary sleep disorders.
    - Chronically disturbed sleep may lead to fatigue, irritability, and difficulties with short-term memory and concentration.
- Other
    - Joint and muscle pains, headaches, and fatigue are often reported, and may be related to mood disorders or associated with lack of ovarian hormone production.

Do not routinely use blood tests such as follicle-stimulating hormone (FSH) measurements to diagnose perimenopause or menopause in otherwise healthy women (who are not using hormonal contraception) aged over 45 years, with typical menopausal symptoms.

Diagnose the following without laboratory testing:

- Perimenopause — if the woman has vasomotor symptoms and irregular periods.
- Menopause — if the woman has not had a period for at least 12 months (and is not using hormonal contraception).
- Menopause — based on symptoms in a woman without a uterus.

Consider using serum FSH measurements to diagnose menopause in a woman, provided she is not taking combined hormonal contraception (or hormone replacement therapy), and she is:

- Aged over 45 years with atypical symptoms.
- Aged between 40–45 years with menopausal symptoms, including a change in menstrual cycle.
- Younger than 40 years with a suspected diagnosis of premature ovarian insufficiency (POI).
- Over 50 years of age using progestogen-only contraception, including depot medroxyprogesterone acetate (DMPA).
    - If the FSH level is in the premenopausal range, the woman should continue contraception and the FSH level should be rechecked in 1 year.

Do not routinely perform a pelvic examination to diagnose menopause.

- It should only be performed if clinically indicated and to exclude other possible causes of symptoms.

DIAGNOSIS OF PREMATURE OVARIAN INSUFFICIENCY

Suspect a diagnosis of premature ovarian insufficiency (POI), taking into account the woman's clinical history (for example previous medical or surgical treatment) and family history.

- **Suspect a diagnosis of POI in a woman younger than 40 years, provided she is not taking combined hormonal contraception, if she has**:
    - Menopausal symptoms, including no or infrequent periods (taking into account whether the woman has a uterus) and
    - Elevated serum follicle-stimulating hormone (FSH) levels (more than 30 IU/L) on two blood samples taken 4–6 weeks apart.
        - Note: additional testing of serum luteinizing hormone (LH), oestradiol, prolactin, testosterone, and thyroid-stimulating hormone (TSH) levels may be helpful for the diagnostic work-up.
        - Note: do not routinely use anti-Müllerian hormone (AMH) testing to diagnose suspected POI.

Do not routinely perform a pelvic examination to diagnose POI.

- It should only be performed if clinically indicated and to exclude other possible causes of symptoms.

ASSESSMENT

If a woman has suspected menopause, perimenopause, or premature ovarian insufficiency (POI):

Ask about:

- The symptoms of menopause, including nature, frequency, duration, time of day, and their severity and impact on quality of life.
- Other potential causes of amenorrhoea, including pregnancy.
- Lifestyle factors, such as smoking, alcohol, exercise, and nutrition.
- The need for ongoing or future contraception.
- Smear status and engagement with national screening programmes (if appropriate).
- Family history including premature menopause or POI, venous thromboembolism, or hormone-dependent cancer including breast cancer.
- Her treatment goals, including non-pharmacological, non-hormonal, or hormonal drug options.

- Previous treatments tried, effectiveness, and tolerability (if any).
- Current medications that may affect the choice of treatments available.
- Co-morbid conditions that may affect the choice and safety of hormonal treatment options, depending on the woman's wishes, such as:
  - History of, or increased risk of, cardiovascular disease, hypertension, diabetes mellitus, venous thromboembolism, or stroke disease.
  - History of, or increased risk of, hormone-dependent cancer including breast, endometrial, and ovarian cancer.
  - Previous medical or surgical treatment, including chemotherapy, radiotherapy, hysterectomy, and/or bilateral oophorectomy.
- Bone health and assess risk of osteoporosis.

Examine the woman:

- Check blood pressure and body mass index (BMI).
- If there is a sudden change in menstrual pattern, intermenstrual bleeding, postcoital bleeding, or postmenopausal bleeding — assess appropriately and arrange an urgent 2-week referral if a gynaecological cancer is suspected.

DIFFERENTIAL DIAGNOSES

The following menopausal symptoms may also be caused by other conditions:

- Secondary amenorrhoea
- Irregular vaginal bleeding
  - During the perimenopause, possible causes include endometrial polyps; uterine fibroids; adenomyosis; endometrial hyperplasia or cancer; and vulval, vaginal, or cervical lesions.
- Hot flushes
  - Endocrine causes such as hyperthyroidism and phaeochromocytoma (typically causes hypertension, flushing, and profuse sweating).
  - Tumours such as carcinoid syndrome (typically causes flushing without sweating), pancreatic cancer, medullary thyroid cancer, renal cell cancer, lymphoma, mastocytoma and mast cell disorders (usually with gastrointestinal symptoms), and paraneoplastic syndrome.
  - Excess alcohol consumption, spicy food, food additives (such as monosodium glutamate and sulphites).
  - Dumping syndrome (such as post-weight loss surgery).
  - Anxiety and panic disorders.
  - Tuberculosis.
  - Drugs such as opiates, nitrates, selective serotonin reuptake inhibitors (SSRIs), calcium-channel blockers, levodopa, gonadotrophin-releasing hormone agonists, and anti-oestrogens or selective estrogen receptor modulators (SERMs).
- Vaginal atrophy — trauma, infection, or lichen sclerosis may cause similar symptoms.

The following symptoms associated with the menopause may also be caused by other conditions:

- Urinary incontinence — this is more likely to be due to mechanical factors, such as obesity, gynaecological surgery, or multiparity, than the menopause.

- Mood changes (including anxiety, irritability, and depression) — these symptoms may not be due to the menopause alone. General population studies suggest that most women do not experience major changes in mood during the menopause transition.
- Sleep disturbance — may be associated with the normal ageing process.
- Cognitive impairment (such as memory problems or difficulty concentrating) — cross-sectional studies suggest that these symptoms are unlikely to be related to the menopause.
- Loss of libido — this may be attributed to androgen deficiency, however other non-hormonal factors, such as insomnia, inadequate sexual stimulation, life stresses, and depression, may also contribute to symptoms.
- Muscle and joint pains — pain and swelling resulting in restriction of mobility most often affects the small joints of the hands and feet as well as the knees, elbows, and cervical spine. These symptoms have been linked to a decrease in oestrogen levels, but other musculoskeletal causes (such as osteoarthritis and rheumatoid arthritis) are possible.
- Skin changes — it is difficult to separate skin changes due to ageing, smoking, and sun exposure, from skin changes due to declining hormonal secretion and menopause. Soon after the menopause, skin collagen content and skin thickness decrease, resulting in decreased skin elasticity.
- Weight gain — this is unlikely to be solely due to the perimenopause or menopause. Body weight in women tends to increase with age, especially beginning at or near the menopause (the average weight gain ranges from 2.25–4.19 kg). Body fat redistribution to the abdomen also occurs with age (independently of weight gain).

MANAGEMENT

What information and advice should I give a woman diagnosed with menopause?

Provide evidence-based information about menopause and treatment options to help women make informed, individualized decisions about management, including:

- The stages of menopause, common symptoms, and short- and long-term health implications.
- Treatment options, such as lifestyle measures, non-hormonal treatments, and hormonal drug treatments.
- **Give information and advice on lifestyle measures for menopause symptom relief.**
- Hot flushes and night sweats
  - regular exercise, weight loss (if applicable), wearing lighter clothing/layers of clothing, turning down central heating, sleeping in a cooler room, using fans, reducing stress, and avoiding possible triggers (such as spicy foods, caffeine, smoking, and alcohol).
- Sleep disturbances
  - avoiding exercise late in the day and maintaining a regular bedtime.
- Low mood and anxiety
  - adequate sleep, regular physical activity, and relaxation exercises.
- Cognitive symptoms
  - exercise and good sleep hygiene.

Give information and advice on general health measures that may improve quality of life.

- Advise the woman to seek occupational health advice if support is needed in the workplace.
- Advise on measures to manage cardiovascular risk factors if a woman has, or is at increased risk of, cardiovascular disease.
- Encourage engagement with national screening programmes (if appropriate).
- Advise on bone health, especially if a woman is diagnosed with premature menopause or primary ovarian insufficiency.

- Advise on the need for contraception.
  - It is recommended to use effective contraception until the following time:
    - 12 months after the last period in women > 50 years
    - 24 months after the last period in women < 50 years
- Advise that hormone replacement therapy (HRT) does not provide contraception.
  - Advise that in general all women can stop contraception at the age of 55 years.
  - Be aware that all progestogen-only methods of contraception are safe to use alongside cyclical HRT.
  - Be aware that combined hormonal contraception can be used in eligible women under 50 years of age as an alternative to HRT for relief of menopausal symptoms and prevention of loss of bone mineral density. Women should be advised to switch to a progestogen-only method of contraception at 50 years of age, if needed

How should I manage menopausal symptoms with HRT?

If a woman wishes to consider the use of hormone replacement therapy (HRT) for management of menopausal symptoms:

- Advise on lifestyle measures for menopause symptom relief.
- Offer an individualized choice of HRT preparation, based on her age, symptoms, and co-morbidities, after discussing potential risks, benefits, adverse effects, and contraindications. Aim to prescribe the lowest dose for the shortest possible duration.
  - **For vasomotor symptoms:**
    - In women with a uterus, offer an oral or transdermal combined (oestradiol plus progestogen) preparation.
    - In women without a uterus, offer an oral or transdermal oestrogen-only preparation.
  - **For mood disorders,** offer a choice of oral or transdermal HRT preparations.
  - **For urogenital symptoms** (genitourinary syndrome of menopause):
    - Offer low-dose vaginal oestrogen first-line and continue treatment for as long as needed to relieve symptoms. Note: some women on systemic HRT may also benefit from additional low-dose vaginal oestrogen.
    - If a low-dose preparation does not relieve symptoms sufficiently, consider seeking specialist advice about increasing the dose.
    - If low-dose vaginal oestrogen is not tolerated or is contraindicated, consider a trial of oral ospemifene (a selective oestrogen receptor modulator) if there are moderate-to-severe symptoms and no contraindications, depending on local prescribing guidelines.
    - Advise that vaginal moisturizers and lubricants can be used alone or in addition to vaginal oestrogen preparations.
- **For altered sexual function**, seek specialist advice on the appropriateness of testosterone supplementation (off-label indication).
  - Be aware that women with premature menopause or premature ovarian insufficiency (POI) may need high oestrogen doses for symptom relief, whereas older women may need to start or be maintained on a lower oestrogen dose.
  - If there is any uncertainty about the safety or appropriateness of prescribing HRT, liaise with the woman's relevant specialist, or seek advice from a health professional with expertise in menopause.
- Do not recommend the use of bioidentical hormone replacement therapy.

- **Arrange to review the woman after 3 months if HRT has been started or changed, then at least annually thereafter**, unless there are clinical indications for an earlier review (such as treatment ineffectiveness or adverse effects). At each review:
  - Reinforce information and lifestyle advice.
  - Assess the efficacy and tolerability of treatment(s).
  - Assess for bothersome adverse effects or persistent symptoms, and offer to adjust the HRT dose or preparation if appropriate. Options include:
    - Reduce the dose of oestrogen.
    - Change the dose or type of progestogen.
    - Alter the route of administration, for example switch from oral to transdermal.
    - Switch to a combined oestrogen/bazedoxifene acetate preparation (a selective oestrogen receptor modulator), if progestogen-containing therapy is not appropriate.
  - If there is a sudden change in menstrual pattern, intermenstrual bleeding, postcoital bleeding, or postmenopausal bleeding — assess appropriately and arrange an urgent 2-week referral if a gynaecological cancer is suspected.
  - If there are persistent symptoms despite adjustment of the HRT dose or preparation, consider an alternative cause for symptoms.
  - Review the duration of HRT treatment.
    - If HRT was started in the perimenopause, discuss the option of changing the treatment regimen and/or reducing the dose of oestrogen.
- Support the woman to make an individual decision on when and how to stop HRT. Advise that:
  - HRT should be continued for as long as benefits of symptom control and improved quality of life outweigh any risks, and there is no arbitrary limit for duration of HRT use.
    - For vasomotor symptoms, most women require 2–5 years of treatment, but some women may need longer.
    - Women with premature menopause or POI should take HRT up to the average age of the natural menopause (51 years in the UK), after which the need for ongoing HRT should be reassessed.
  - HRT may be gradually reduced over 3–6 months, or stopped suddenly, depending on the woman's preferences.
    - Symptoms may recur in the short term after stopping treatment, particularly if HRT is stopped suddenly.
    - If troublesome symptoms recur, options include restarting HRT at a low dose, or considering alternative non-hormonal treatments.
- Vaginal oestrogen preparations may be required long term, but regular attempts to stop treatment, such as annually, can be made.

HRT RISKS:

- **Venous thromboembolism (VTE)**
  - The risk of VTE associated with HRT is greater for oral than transdermal preparations.
  - The risk associated with transdermal HRT given at standard therapeutic doses is no greater than baseline risk.
- **Coronary heart disease (CHD) and stroke**
  - The baseline risk of CHD and stroke for women around menopausal age varies from one woman to another, depending on the presence of cardiovascular risk factors.

- HRT with oestrogen alone is associated with no, or reduced, risk of CHD. Combined HRT with oestrogen and progestogen is associated with little or no increase in the risk of CHD.
- The baseline risk of stroke in women younger than 60 years is very low.
- Oral (but not transdermal) oestrogen is associated with a small increase in the risk of stroke.
- HRT does not increase CVD risk when started in women younger than 60 years.
- HRT does not affect the risk of dying from CVD.

- **Type 2 diabetes**
  - HRT (either orally or transdermally) is not associated with an increased risk of developing type 2 diabetes.
- **Dementia**
  - The likelihood of HRT affecting the risk of dementia is unknown.
- **Breast cancer**
  - HRT with oestrogen alone is associated with little or no increase in the risk of breast cancer.
  - Combined HRT with oestrogen and progestogen is associated with an increased risk of breast cancer that is dependent on duration of treatment, which reduces after stopping HRT.
  - HRT does not affect the risk of dying from breast cancer.

HRT BENEFITS

- **Fragility fractures**
  - The baseline risk of fragility fracture for women around menopausal age in the UK is low and varies from one woman to another.
  - The risk of fragility fracture is decreased while taking HRT, and this benefit is maintained during treatment but decreases once treatment stops; this benefit may continue for longer in women who take HRT for longer.
- **Muscle mass and strength**
  - There is limited evidence that HRT may improve muscle mass and strength.
- NICE reviewed the evidence for women diagnosed with premature menopause/premature ovarian insufficiency [NICE, 2019]:
  - Starting hormonal treatment (either with HRT or a combined hormonal contraceptive) and continuing treatment until at least the age of natural menopause (unless contraindicated) reduces the risk of chronic diseases, including cardiovascular disease and osteoporosis.
  - HRT may have a beneficial effect on blood pressure when compared with a combined hormonal contraceptive.
  - Both HRT and combined hormonal contraceptives offer bone protection.

What non-hormonal treatments can I offer?

If a woman chooses not to take hormone replacement therapy (HRT), or it is not tolerated or is contraindicated:

- Advise on lifestyle measures for menopause symptom relief.
- If lifestyle measures are insufficient, advise on non-hormonal treatments and non-drug treatment options for symptom relief, including risks and benefits.
- **For vasomotor symptoms,** consider a trial of:

- Selective serotonin reuptake inhibitors (SSRIs, off-label) or serotonin and norepinephrine reuptake inhibitors (SNRIs, off-label) for 2 weeks initially.
    - Options include fluoxetine (20 mg daily), citalopram (20 mg daily), paroxetine (10 mg daily), or venlafaxine modified-release (37.5 mg daily for 1 week then increased to 75 mg once daily if needed).
- Clonidine (an alpha-2 adrenergic receptor agonist).
    - Initially 50 micrograms twice daily for 2 weeks, then increased to 75 micrograms twice daily, if needed.
- Gabapentin (off-label), depending on local prescribing guidelines.
    - Up to 300 mg three times a day, initial dose should be lower and titrated up over 3 days.
- Cognitive behavioural therapy (CBT), depending on local service provision.

- **For mood disorders,** consider:
    - Self-help resources and a trial of CBT for low mood and/or anxiety.
    - Antidepressant treatment for a confirmed diagnosis of depression and/or anxiety, if indicated.
- **For urogenital symptoms** (genitourinary syndrome of menopause), consider a trial of:
    - Vaginal moisturizers, such as Replens MD®, to be used at least twice weekly.
    - Vaginal lubricants if there are insufficient vaginal secretions for comfortable sexual activity.
    - Note: these preparations may be used alone or in addition to vaginal oestrogen preparations.
- **Advise women considering complementary therapies:**
    - There is some evidence that isoflavones and black cohosh may relieve vasomotor symptoms, but the quality, purity, constituents, and safety of these products may be unknown, and different preparations may vary.
- **Arrange to review the woman after 3 months, then at least annually thereafter,** unless there are clinical indications for an earlier review (such as treatment ineffectiveness or adverse effects). At each review:
    - Reinforce information and lifestyle advice.
    - Assess the efficacy and tolerability of treatment(s).
    - If the woman is symptom free on antidepressant treatment, consider a gradual withdrawal of treatment after 1–2 years. Advise that symptoms may recur once treatment is stopped.
    - Advise that the use of vaginal moisturizers and lubricants may be continued indefinitely.
- **Offer to refer the woman to a healthcare professional with expertise in menopause, if treatments do not improve menopausal symptoms.**

How should I manage women with comorbidities?

- **Women with, or at high risk of, breast cancer:**
    - Stop systemic hormone replacement therapy (HRT) in women who are diagnosed with breast cancer.
    - Do not offer HRT routinely to women with menopausal symptoms and a history of breast cancer.
        - Seek specialist advice if a woman wishes to consider the use of hormonal therapy, such as treatment with low-dose vaginal oestrogen.
    - Advise on lifestyle measures, non-hormonal, and non-drug treatment options for symptom relief.

        - Advise that women taking tamoxifen should not use fluoxetine or paroxetine, as they may inhibit the effect of tamoxifen.
    - Do not recommend the use of isoflavones, red clover, black cohosh, vitamin E, or magnetic devices to treat menopausal symptoms in women with breast cancer.
        - Advise that the herbal preparation St John's wort may interact with other drugs such as tamoxifen, anticoagulants, and anticonvulsants. In addition, there is uncertainty about the appropriate dose, and possible variation of potency of over-the-counter preparations.
- **Women with increased risk of venous thromboembolism (VTE):**
    - Consider the use of transdermal rather than oral HRT for women at increased risk of VTE, including women with a BMI over 30 kg/m2.
    - Consider referring women at high risk of VTE (for example, with a strong family history of VTE or a hereditary thrombophilia) to a haematologist for assessment before considering the use of HRT.
- **Women with increased risk of cardiovascular disease (CVD):**
    - Manage any cardiovascular risk factors before considering the use of HRT.
        - Be aware that the presence of cardiovascular risk factors is not a contraindication to taking HRT as long as they are optimally managed.
    - Consider the use of transdermal rather than oral HRT for women at increased risk of CVD.
- **Women with type 2 diabetes:**
    - Consider the use of HRT after taking any other co-morbidities into account.
        - Advise that HRT is not associated with an adverse effect on blood glucose control.
- **Women with hypothyroidism:**
    - Advise that thyroid-stimulating hormone (TSH) levels should be monitored regularly (for example, 6–12 weeks after starting oral HRT), to ensure that levels remain in the acceptable range, as the dose of levothyroxine (LT4) medication may need to be increased.

When should I refer a woman with menopausal symptoms?

If a woman has menopausal symptoms, consider arranging referral to a healthcare professional with expertise in menopause if:

- The woman has ongoing symptoms and lifestyle measures, hormonal, non-hormonal, or non-drug treatments are ineffective.
- The woman has persistent, troublesome adverse effects from treatment.
- There is uncertainty about the most suitable treatment option, for example if the woman has co-morbidities and/or contraindications to treatment.
- The woman has persistent altered sexual function and hormonal and/or non-hormonal, or non-drug treatments are ineffective:
    - Seek specialist advice regarding the use of testosterone supplementation (off-label use).
    - Consider referral for psychosexual counselling, depending on the woman's wishes.
- There is uncertainty about diagnosing premature ovarian insufficiency, or specialist advice is needed to manage the condition.

TOPIC: OVARIAN ENLARGEMENT

- The initial imaging modality for suspected ovarian cysts/tumours is **ultrasound**.
- The report will usually report that the cyst is either:
  - **simple:** unilocular, more likely to be physiological or benign
  - **complex**: multilocular, more likely to be malignant
- Management depends on the age of the patient and whether the patient is symptomatic.
  - It should be remembered that the diagnosis of ovarian cancer is often delayed due to a vague presentation.

Premenopausal women

- A conservative approach may be taken for younger women (especially if < 35 years) as malignancy is less common. If the cyst is small (e.g. < 5 cm) and reported as 'simple' then it is highly likely to be benign. A repeat ultrasound should be arranged for 8-12 weeks and referral considered if it persists.

Postmenopausal women

- By definition physiological cysts are unlikely
- Any postmenopausal woman with an ovarian cyst regardless of nature or size should be referred to gynaecology for assessment

TOPIC: OVARIAN CYSTS:

- Benign ovarian cysts are extremely common.
- They may be divided into:
  - physiological cysts
  - benign germ cell tumours
  - benign epithelial tumours
  - benign sex cord stromal tumours
- **Complex** (i.e. multi-loculated) ovarian cysts should be biopsied to exclude malignancy.

Physiological cysts (functional cysts)

Follicular cysts

- commonest type of ovarian cyst
- due to non-rupture of the dominant follicle or failure of atresia in a non-dominant follicle
- commonly regress after several menstrual cycles

Corpus luteum cyst

- during the menstrual cycle if pregnancy doesn't occur the corpus luteum usually breaks down and disappears. If this doesn't occur the corpus luteum may fill with blood or fluid and form a corpus luteal cyst
- more likely to present with intraperitoneal bleeding than follicular cysts

Benign germ cell tumours

Dermoid cyst

- also called mature cystic teratomas. Usually lined with epithelial tissue and hence may contain skin appendages, hair and teeth
- most common benign ovarian tumour in woman under the age of 30 years
- median age of diagnosis is 30 years old
- bilateral in 10-20%
- usually, asymptomatic. Torsion is more likely than with other ovarian tumours

Benign epithelial tumours

Arise from the ovarian surface epithelium

Serous cystadenoma

- the most common benign epithelial tumour which bears a resemblance to the most common type of ovarian cancer (serous carcinoma)
- bilateral in around 20%

Mucinous cystadenoma

- second most common benign epithelial tumour
- they are typically large and may become massive
- if ruptures may cause pseudomyxoma peritonei

TOPIC: OVARIAN TORSION

- Ovarian torsion may be defined as the partial or complete torsion of the ovary on their supporting ligaments that may in turn compromise the blood supply.
  - If the fallopian tube is also involved then it is referred to as adnexal torsion.

Risk factors

- ovarian mass: present in around 90% of cases of torsion
- being of a reproductive age
- pregnancy
- ovarian hyperstimulation syndrome

Features

- Usually the sudden onset of deep-seated colicky abdominal pain.
- Associated with vomiting and distress
- fever may be seen in a minority (possibly secondary to adnexal necrosis)
- Vaginal examination may reveal adnexial tenderness

Ultrasound may show free fluid or a whirlpool sign.

Laparoscopy is usually both diagnostic and therapeutic.

TOPIC: OVARIAN CANCER

DEFINITION

Ovarian cancer arises from the ovary.

- **Epithelial carcinomas**
  - Account for approximately 90% of primary ovarian cancers.
  - Most are thought to arise from the fimbria of the fallopian tube. Primary ovarian-type tumours may also develop within the peritoneum.
  - The World Health Organization (WHO) classification of ovarian tumours recognizes six major histotypes: serous (70–85% of cases), mucinous, endometrioid, clear cell, transitional cell, and other carcinomas.
  - Tumours of each type are further subdivided into benign, malignant, and intermediate (or borderline malignancy).
- **Non-epithelial cancers**
  - account for approximately 10% of primary ovarian cancers.
  - They include germ cell tumours, and sex cord/stromal cell tumours.

Secondary metastatic tumours can occur in the ovary, most notably from cancers of the:

- Endometrium.
- Breast.
- Gastrointestinal tract.

Ovarian cancer may spread to:

- Intraperitoneal structures and organs, causing intestinal obstruction and cachexia.
- The liver.
- Para-aortic lymph nodes.
- The lung, causing pleural effusions.

RISK FACTORS

The risk of ovarian CA is increased by:

- **Increasing age**
  - the incidence of ovarian cancer increases with age and peaks in the eighth decade.
- **Genetic factors, including:**
  - Gene mutations — BRCA1 and BRCA2 are tumour suppressor genes involved in DNA (deoxyribonucleic acid) repair, and BRCA-related cancers result from mutation or alteration of these genes.
  - Family history of ovarian cancer — about 3% of ovarian cancer cases occur in women with a family history of ovarian cancer. The risk of ovarian cancer risk is 2.7–3.5 times higher in women whose mother or sister has (or has had) ovarian cancer compared with women with no such family history; the risk may be higher if the affected relative was diagnosed at a younger age.
  - Family history of cancer — the risk of ovarian cancer is higher in families with a history of breast cancer compared with the general population, at least partly due to BRCA1 or BRCA2 mutations. Ovarian cancer has been associated with genetic syndromes that can also cause bowel, uterine, stomach, liver, pancreatic, prostate, gallbladder, and urinary tract cancers.

- **Reproductive and hormonal factors, including:**
  - Conditions that increase the number of ovulatory cycles, such as nulliparity, early menarche, and late menopause
    - evidence from epidemiological studies shows that the number of ovulatory cycles a woman has in her lifetime is proportional to her risk of developing ovarian cancer.
  - Use of hormone replacement therapy (HRT)
    - there appears to be a small increased risk of ovarian cancer associated with the use of HRT (both oestrogen-only and combined HRT).
- **Medical conditions, including:**
  - Cancer
    - a personal history of ovarian, breast, or bowel cancer is associated with an increased risk of ovarian cancer.
  - Endometriosis
    - endometriosis is sometimes found next to endometrioid or clear cell ovarian cancers, and evidence has shown an increased incidence of ovarian cancer in women with documented endometriosis.
  - Diabetes
    - particularly in women who use insulin.
- **Lifestyle factors, including:**
  - Tobacco smoking.
  - Being overweight or obese.
  - Occupational exposure to asbestos.
- Note: there is conflicting evidence on the perineal use of talc as a risk factor for developing ovarian cancer.

The risk of ovarian CA is reduced by:

- Conditions that decrease the number of ovulatory cycles, including:
- A higher number of pregnancies.
- Breastfeeding.
- Use of the combined oral contraceptive pill.
- Tubal ligation and hysterectomy.

PROGNOSIS:

- The survival rate for ovarian cancer is strongly related to the stage of the disease at the time of diagnosis.
  - Women diagnosed with more localised (stage I or II) ovarian cancer have a better one-year survival than those diagnosed with advanced or disseminated (stage IV) cancer.

DIAGNOSIS

Suspect ovarian cancer and carry out tests:

- In any woman (particularly if over 50 years of age) if any of the following symptoms are persistent or frequent (particularly more than 12 times per month):
  - Abdominal distension (bloating).
  - Feeling full (early satiety) and/or loss of appetite.
  - Pelvic or abdominal pain.
  - Increased urinary urgency and/or frequency.

- In any woman over 50 years of age, if she has had:
  - Symptoms suggestive of irritable bowel syndrome (IBS) within the last 12 months.
- Consider the possibility of ovarian cancer and consider carrying out tests in any woman who reports any of the following unexplained symptoms:
  - Weight loss.
  - Malaise or fatigue.
  - Change in bowel habit.
- Other symptoms of ovarian cancer that may be present include:
  - Abnormal or postmenopausal bleeding.
  - Gastrointestinal symptoms, such as dyspepsia, nausea, or bowel obstruction.
  - Shortness of breath (due to pleural effusion).

DIFFERENTIAL DIAGNOSIS

Other conditions that could cause symptoms similar to those of ovarian cancer include:

- Other cancers, including cancer of the cervix, uterus, rectum, and bladder.
- Other causes of abdominal distension or bloating such as:
  - Uterine fibroids.
  - Ascites secondary to cirrhosis of the liver, or heart failure.
  - Adenomyosis.
- Other causes of early satiety, such as gastric cancer.
- Other causes of urinary frequency or urgency, such as recurrent urinary tract infections (UTIs).
- Other causes of altered bowel habit, such as:
  - Irritable bowel syndrome.
  - Constipation (functional or drug-induced).
  - Coeliac disease.
  - Inflammatory bowel disease.
  - Gastrointestinal infection.
  - Antibiotic-associated diarrhoea.
- Other causes of abdominal pain or discomfort, such as:
  - Adhesions.
  - Pelvic inflammatory disease (PID).
  - Diverticular disease.
  - Chronic pancreatitis.
  - Gallstones.

Other causes of a raised serum CA125 include:

- Peritoneal trauma, disease, or irritation.
- Other cancers, such as primary peritoneal cancer, lung cancer, and pancreatic cancer.
- Endometriosis.
- PID.
- Ovarian cyst torsion, rupture, or haemorrhage.
- Pregnancy.
- Heart failure.

MANAGEMENT:

- **Carry out an abdominal and pelvic examination.**

- If there is ascites, or a pelvic or abdominal mass (which is not caused by known uterine fibroids), refer the woman urgently to a gynaecological cancer service (to be seen within 2 weeks) for further investigation.

- **If the examination is normal, measure serum CA125 concentration.**
  - If serum CA125 concentration is raised (35 IU/mL or greater), arrange an urgent ultrasound scan of the abdomen and pelvis.
    - If the ultrasound is suggestive of ovarian cancer, refer the woman urgently to a gynaecological cancer service (to be seen within 2 weeks) for further investigation.
  - If serum CA125 is normal (less than 35 IU/mL), or raised (35 IU/mL or greater) but with a normal ultrasound:
    - Consider other causes for symptoms or a raised CA125.
    - If no other clinical cause is apparent, advise the woman to return for review if her symptoms become more frequent and/or persistent.
- **NOTE: A CA125 should not be used for screening for ovarian cancer in asymptomatic women.**

TOPIC: ENDOMETRIAL HYPERPLASIA

- Endometrial hyperplasia may be defined as an abnormal proliferation of the endometrium in excess of the normal proliferation that occurs during the menstrual cycle.
- A minority of patients with endometrial hyperplasia may develop endometrial cancer

TYPES

- simple
- complex
- simple atypical
- complex atypical

FEATURES

- abnormal vaginal bleeding e.g. intermenstrual

MANAGEMENT

- simple endometrial hyperplasia without atypia: high dose progestogens with repeat sampling in 3-4 months. The levonorgestrel intra-uterine system may be used
- atypia: hysterectomy is usually advised

TOPIC: ENDOMETRIAL CANCER

- Endometrial cancer is classically seen in post-menopausal women but around 25% of cases occur before the menopause. It usually carries a good prognosis due to early detection.

RISK FACTORS

- excess oestrogen
  - nulliparity, early menarche, late menopause
  - unopposed oestrogen.
    - The addition of a progestogen to oestrogen reduces this risk (e.g. In HRT). The BNF states that the additional risk is eliminated if a progestogen is given continuously
- metabolic syndrome
  - obesity, DM, PCOS
- tamoxifen
- hereditary non-polyposis colorectal carcinoma

PROTECTIVE FACTORS

- multiparity
- combined oral contraceptive pill
- smoking (the reasons for this are unclear)

FEATURES

- The classic symptom is **postmenopausal bleeding**
  - usually slight and intermittent initially before becoming heavier

Other features:

- premenopausal women may develop menorrhagia or intermenstrual bleeding
- pain is not common and typically signifies extensive disease
- vaginal discharge is unusual

INVESTIGATIONS

- All women >= 55 years who present with postmenopausal bleeding should be referred using the suspected cancer pathway
  - first-line investigation is trans-vaginal ultrasound - a normal endometrial thickness (< 4 mm) has a high negative predictive value
  - hysteroscopy with endometrial biopsy

MANAGEMENT

- The mainstay of management for endometrial cancer is **surgery.**
  - localised disease is treated with total abdominal hysterectomy with bilateral salpingo-oophorectomy
  - patients with high-risk disease may have postoperative radiotherapy
- Progestogen therapy is sometimes used in frail elderly women not considered suitable for surgery.